

Goblin Shadow Theory

A research notebook on open-neighborhood filtrations, blocker barcodes, and dynamic repair

Research notebook

April 29, 2026

Status note. This document packages the findings. Several statements are given as theorems with proof sketches; several are explicitly marked as conjectures where the exploration produced strong evidence but no complete proof. The names “goblin”, “shadow”, “torch”, and related terms are local terminology from this notebook. I have not performed a full literature-priority search; the novelty claims should be read as mathematical discoveries within this exploration, not as certified publication-priority claims.

Contents

1	Executive summary	3
2	Notation and first principles	4
3	The passive shadow spectrum	4
3.1	Lower hull and curvature measure	4
3.2	Reciprocity	5
3.3	Arithmetic normal form	5
3.4	Farey phase budget	6
3.5	Flow and matching interpretations	6
4	Repair, torching, and safe cores	6
4.1	Exposure and edge hardening	6
4.2	Torch sets and safe cores	7
4.3	Fractional and reciprocal thresholds	7
4.4	Star-Cap and first breakaway laws	7
5	Blocker filtration and barcode	8
5.1	Minimal bad caves	8
5.2	Barcode theorem	8
5.3	Cave-kernel theorem	9
6	Shadow-LYM theory and blocker entropy	9
6.1	Cross-shadow theorem	9
6.2	Shadow-LYM theorem	9
6.3	Complete-bipartite entropy theorem	10
6.4	Barcode area and count	10

7	Exact families	11
7.1	Complete bipartite graphs	11
7.2	Parallel cave trees	11
7.3	Three-step spiders	11
8	Dynamic edge edits	12
8.1	Unmasking equation	12
8.2	Endpoint mutation and trident law	12
8.3	Rooted Shadow-LYM	12
8.4	Churn laws	13
8.5	Double-star mutation	13
9	Root pressure, support, and endpoint capture	13
9.1	Root pressure	13
9.2	Support-core dichotomy	14
9.3	Endpoint capture	14
10	Computational corroboration	14
10.1	Graph-atlas blocker maxima and Shadow-LYM checks	14
10.2	Barcode count maxima	15
10.3	Spider blocker law check	15
10.4	Double-star mutation check	15
10.5	Core script excerpt	16
11	Corrections and conjectures	16
12	Applications and interpretation	17
12.1	Network vulnerability and hardening	17
12.2	Audit complexity	17
12.3	Design principles	17
13	Named laws index	17
14	Closing view	18

1 Executive summary

Let $G = (V, E)$ be a finite simple graph, $n = |V|$, and let $N(S)$ denote the open neighborhood of S :

$$N(S) = \{v \in V : \exists s \in S \text{ with } vs \in E\}.$$

The central passive invariant is the *goblin shadow function*

$$g_G(\lambda) = \max_{S \subseteq V} \{\lambda|S| - |N(S)|\}, \quad \lambda \geq 0.$$

It measures the best profit of choosing vertices while paying for the watchers they wake. The theory develops four layers:

1. **Passive spectrum.** The function g_G is a convex rational spline with a reciprocal curvature measure

$$g_G(\lambda) = \int_0^\infty (\lambda - t)_+ d\mu_G(t).$$

The measure μ_G has an arithmetic normal form built from reciprocal atoms.

2. **Repair and safe cores.** For a target exposure α , an α -safe core is a set Q such that every nonempty $S \subseteq Q$ satisfies $|N(S)| \geq \alpha|S|$. The minimum number of vertices that must be “torched” is

$$\tau_\alpha(G) = n - \sigma_\alpha(G),$$

where $\sigma_\alpha(G)$ is the largest safe-core size.

3. **Blocker filtration and barcode.** The active obstructions are inclusion-minimal α -bad sets

$$\mathfrak{B}_\alpha(G) = \min_{\subseteq} \{S \subseteq V : |N(S)| < \alpha|S|\}.$$

Each set has a lifetime interval

$$I_G(S) = \left(\frac{|N(S)|}{|S|}, \min \left(1, \min_{\emptyset \neq T \subsetneq S} \frac{|N(T)|}{|T|} \right) \right],$$

and $S \in \mathfrak{B}_\alpha(G)$ exactly when $\alpha \in I_G(S)$.

4. **Dynamic theory.** Edge edits are local in endpoint support but can cause exponential blocker churn. At fixed threshold the churn exponent is

$$c(\alpha) = -\log(1 - x_\alpha), \quad x_\alpha = (1 - x_\alpha)^{1+\alpha}.$$

At $\alpha = 1$, this is $\log \varphi$, where $\varphi = (1 + \sqrt{5})/2$.

The most useful equations are:

$$\begin{aligned} g_G(\lambda) &= \int (\lambda - t)_+ d\mu_G(t), \\ \tau_\alpha(G) &= \tau(\mathfrak{B}_\alpha(G)) = n - \sigma_\alpha(G), \\ \mathfrak{B}_\alpha(G) &= \{S : \alpha \in I_G(S)\}, \\ \sum_{C \in \mathfrak{B}_\alpha(G)} \frac{1}{\binom{n - |N(C)|}{|C|}} &\leq 1. \end{aligned}$$

The last inequality is the *Shadow-LYM theorem*. It yields the exact maximum possible number of minimal bad caves:

$$\max_{|V(G)|=n} |\mathfrak{B}_\alpha(G)| = \max_{1 \leq p \leq n-1} \binom{n-p}{\lfloor p/\alpha \rfloor + 1},$$

attained by complete bipartite graphs.

2 Notation and first principles

Definition 2.1 (Goblin profit and shadow function). *For $\lambda \geq 0$ and $S \subseteq V$, define*

$$\Phi_\lambda(S) = \lambda|S| - |N(S)|.$$

The goblin shadow function is

$$g_G(\lambda) = \max_{S \subseteq V} \Phi_\lambda(S).$$

A set maximizing Φ_λ is called a λ -lair.

The function $S \mapsto |N(S)|$ is a coverage function, hence submodular:

$$|N(A)| + |N(B)| \geq |N(A \cup B)| + |N(A \cap B)|.$$

Therefore $S \mapsto \Phi_\lambda(S)$ is supermodular.

Theorem 2.2 (Lair lattice and monotone chain). *For fixed λ , the family of λ -lairs is closed under union and intersection. Hence there is a unique smallest and a unique largest λ -lair. Moreover, as λ increases, the largest canonical lairs form an increasing chain.*

Proof sketch. If A and B are optimal, supermodularity gives

$$\Phi_\lambda(A \cup B) + \Phi_\lambda(A \cap B) \geq \Phi_\lambda(A) + \Phi_\lambda(B) = 2g_G(\lambda).$$

Neither term on the left can exceed $g_G(\lambda)$, so both are optimal. Monotonicity follows by comparing optimality inequalities at $\lambda < \mu$ and using supermodularity to force the lower- λ canonical lair inside the higher- λ canonical lair. $\square$

Lemma 2.3 (Purification below one mana). *If $0 \leq \lambda < 1$, every λ -lair is independent. At $\lambda = 1$, every optimal lair can be replaced by an independent optimal lair.*

Proof sketch. Let $S = I \sqcup T$, where I consists of vertices with no neighbor inside S and T consists of internally adjacent vertices. Let $B = N(S) \setminus S$. Then $N(S) = T \cup B$ and $N(I) \subseteq B$. Hence

$$\Phi_\lambda(I) - \Phi_\lambda(S) \geq (1 - \lambda)|T|.$$

For $\lambda < 1$, internal adjacency strictly hurts. $\square$

3 The passive shadow spectrum

3.1 Lower hull and curvature measure

Define

$$m_G(k) = \min_{|S|=k} |N(S)|.$$

Then

$$g_G(\lambda) = \max_{0 \leq k \leq n} \{\lambda k - m_G(k)\}.$$

The generic phases are the exposed vertices of the lower convex hull of the points $(k, m_G(k))$. If consecutive hull points are (k_{i-1}, w_{i-1}) and (k_i, w_i) , put

$$a_i = k_i - k_{i-1}, \quad b_i = w_i - w_{i-1}, \quad \theta_i = \frac{b_i}{a_i}.$$

Definition 3.1 (Shadow curvature measure).

$$\mu_G = \sum_i a_i \delta_{\theta_i}.$$

Theorem 3.2 (Spline equation).

$$g_G(\lambda) = \int_0^\infty (\lambda - t)_+ d\mu_G(t) = \sum_i a_i (\lambda - \theta_i)_+.$$

This is the first compact form of the theory: the whole passive optimization curve is a rational curvature spectrum.

3.2 Reciprocity

For $S \subseteq V$ define the polar set

$$S^\star = V \setminus N(S).$$

Theorem 3.3 (Reciprocity). *For $\lambda > 0$,*

$$g_G(\lambda) = n(\lambda - 1) + \lambda g_G(1/\lambda).$$

If S is optimal at λ , then $S^\star$ is optimal at $1/\lambda$. On the hull, points are paired by

$$(k, w) \longleftrightarrow (n - w, n - k).$$

Therefore slopes are paired by $\theta \leftrightarrow 1/\theta$.

Proof sketch. For $T = S^\star$, $|T| = n - |N(S)|$ and $N(T) \subseteq V \setminus S$, so $|N(T)| \leq n - |S|$. Hence

$$\lambda \Phi_{1/\lambda}(T) = |T| - \lambda |N(T)| \geq n - |N(S)| - \lambda(n - |S|) = \Phi_\lambda(S) - n(\lambda - 1).$$

Maximizing over S gives one inequality, and applying it to $1/\lambda$ gives the reverse. $\square$

3.3 Arithmetic normal form

The curvature measure has a unique atom decomposition.

Theorem 3.4 (Goblin normal form). *For a finite simple graph,*

$$\mu_G = z\delta_0 + c\delta_1 + \sum_{0 < p < q, (p,q)=1} t_{p,q} (q\delta_{p/q} + p\delta_{q/p}),$$

where $z \geq 0$ is the number of isolated vertices, $t_{p,q} \in \mathbb{Z}_{\geq 0}$, and

$$c \in \{0, 2, 3, 4, \dots\}.$$

The value $c = 1$ is impossible in a loopless graph. For trees with more than one vertex, $z = 0$ and c is even.

Proof sketch. If an atom lies at p/q in lowest terms, its horizontal mass must be a multiple of q , because the vertical jump must be integral. Reciprocity forces the reciprocal mass to be the corresponding multiple of p . A central mass-one jump would have vector $(1, 1)$, requiring one vertex to wake exactly itself, impossible without loops. The tree parity follows from bipartition symmetry. $\square$

Conversely, every such measure is realized by disjoint unions of complete bipartite atoms $K_{q,p}$ and central cliques K_c . For trees, the complete bipartite atom can be replaced by a tree atom; fusing atoms through watcher-side vertices preserves the shadow hull.

3.4 Farey phase budget

A subunit slope $p/q < 1$ costs at least $p + q$ vertices, because its reciprocal pair contributes a transition vector at least (q, p) plus (p, q) . Thus the number of possible phases is controlled by a Farey knapsack.

Let

$$\rho(N) = \max \left\{ |\mathcal{A}| : \mathcal{A} \subseteq \{p/q : 0 < p < q, (p, q) = 1\}, \sum_{p/q \in \mathcal{A}} (p + q) \leq N \right\}.$$

Then the maximum number $M(n)$ of positive generic phases satisfies

$$M(n) = \max_{\varepsilon \in \{0,1\}} (1 + \varepsilon + 2\rho(n - 2\varepsilon)),$$

and trees realize this bound. Asymptotically,

$$M(n) \sim \frac{3}{\pi^{2/3}} n^{2/3}.$$

3.5 Flow and matching interpretations

Build a bipartite network with a left and right copy of each vertex. Put arcs $v_L \rightarrow u_R$ when $v \sim u$, source capacity λ into each v_L , and sink capacity 1 out of each u_R . If $F_G(\lambda)$ is the maximum flow value, then

$$g_G(\lambda) = \lambda n - F_G(\lambda).$$

At one mana,

$$g_G(1) = n - 2\nu_f(G),$$

where $\nu_f(G)$ is the maximum fractional matching number. Thus $g_G(1) = 0$ iff G has a fractional perfect matching.

4 Repair, torching, and safe cores

4.1 Exposure and edge hardening

Define the worst exposure ratio

$$\eta(G) = \min_{\emptyset \neq S \subseteq V} \frac{|N(S)|}{|S|}.$$

If r new edges may be added, define

$$\eta_r(G) = \max_{|F| \leq r} \eta(G + F).$$

The passive spectrum gives an optimistic ceiling:

$$A_r(G) = \sup\{\alpha : g_G(\alpha) \leq 2r\alpha\}.$$

But examples showed this ceiling can be unattainable when hiddenness is split across parallel caves.

4.2 Torch sets and safe cores

Definition 4.1 (Torch set). *A set $T \subseteq V$ is an α -torch set if*

$$|N(S)| \geq \alpha|S \setminus T| \quad \forall S \subseteq V.$$

Let $\tau_\alpha(G)$ be the minimum size of such a T .

Definition 4.2 (Safe core). *A set $Q \subseteq V$ is α -safe if*

$$|N(S)| \geq \alpha|S| \quad \forall \emptyset \neq S \subseteq Q.$$

Let $\sigma_\alpha(G)$ be the maximum size of an α -safe core.

Theorem 4.3 (Safe-core equation).

$$\tau_\alpha(G) = n - \sigma_\alpha(G).$$

Proof sketch. A torch set T works iff $V \setminus T$ contains no surviving α -bad set, i.e. iff $V \setminus T$ is an α -safe core. $\square$

For rational $\alpha = a/b$, safety is a grouped Hall condition: Q is a/b -safe iff a demand clones of each vertex of Q can be matched into b capacity clones of its neighbors.

4.3 Fractional and reciprocal thresholds

The passive spectrum gives the fractional torch budget:

$$\tau_\alpha^*(G) = \frac{g_G(\alpha)}{\alpha}.$$

The integral torch number can be larger. Define the blocker defect

$$\text{bid}_\alpha(G) = \tau_\alpha(G) - \left\lceil \frac{g_G(\alpha)}{\alpha} \right\rceil.$$

Theorem 4.4 (Reciprocal integrality). *For every integer $q \geq 1$,*

$$\tau_{1/q}(G) = qg_G(1/q) = \max_S (|S| - q|N(S)|)_+.$$

Thus $\text{bid}_{1/q}(G) = 0$ for every graph.

For non-reciprocal thresholds, the defect can grow linearly. Parallel cave trees $C_{m,q}$, formed by attaching q leaves to each of m centers, show that away from $1/q$ the spectrum can undercount active repair by $\Theta(n)$.

4.4 Star-Cap and first breakaway laws

Theorem 4.5 (Star-Cap law). *For every connected n -vertex graph,*

$$\tau_\alpha(G) \leq \max(0, n - \lfloor 1/\alpha \rfloor - 1).$$

The star $K_{1,n-1}$ attains equality.

Equivalently, the minimax survivor curve is

$$\inf_{G \text{ connected}} \Sigma_b(G) = \begin{cases} \frac{1}{n-b-1}, & 0 \leq b \leq n-2, \\ 1, & b \geq n-2, \end{cases}$$

where $\Sigma_b(G) = \sup\{\alpha : \tau_\alpha(G) \leq b\}$.

Let $a(G)$ be the number of inclusion-minimal sets attaining $\eta(G)$.

Theorem 4.6 (Goblin Breakaway law).

$$\min\{r : \eta_r(G) > \eta(G)\} = \lceil a(G)/2 \rceil.$$

Moreover $a(G) \leq \lfloor n/3 \rfloor$ for connected non-balanced graphs, and this is sharp up to rounding.

Conjecture 4.7 (Torch Pairing conjecture). For all finite graphs and $0 < \alpha \leq 1$,

$$r_\alpha(G) := \min\{r : \eta_r(G) \geq \alpha\} = \lceil \tau_\alpha(G)/2 \rceil.$$

This is proved in the first-breakaway case and for several exact families. Exhaustive small-graph tests found no counterexample.

5 Blocker filtration and barcode

5.1 Minimal bad caves

Definition 5.1 (Blocker family).

$$\mathfrak{B}_\alpha(G) = \min_{\subseteq} \{S \subseteq V : |N(S)| < \alpha|S|\}.$$

Members of $\mathfrak{B}_\alpha(G)$ are called minimal α -bad caves.

Theorem 5.2 (Shadow blocker theorem). A set T is an α -torch set iff it hits every cave in $\mathfrak{B}_\alpha(G)$. Hence

$$\tau_\alpha(G) = \tau(\mathfrak{B}_\alpha(G)),$$

where the right side is the transversal number of the hypergraph $\mathfrak{B}_\alpha(G)$.

Proof sketch. If T misses a minimal bad cave C , then $C \setminus T = C$ is still bad. Conversely, if T is not a torch set, some S violates $|N(S)| \geq \alpha|S \setminus T|$. Then $S \setminus T$ contains a minimal bad cave disjoint from T . $\square$

5.2 Barcode theorem

For nonempty S , define

$$\rho(S) = \frac{|N(S)|}{|S|},$$

$$\bar{\rho}(S) = \min \left(1, \min_{\emptyset \neq T \subsetneq S} \rho(T) \right),$$

with the inner minimum treated as $+\infty$ for singletons. Define the interval

$$I_G(S) = (\rho(S), \bar{\rho}(S)].$$

Theorem 5.3 (Blocker Barcode theorem).

$$S \in \mathfrak{B}_\alpha(G) \iff \alpha \in I_G(S).$$

Consequently,

$$\mathfrak{B}_\alpha(G) = \{S : \alpha \in I_G(S)\}.$$

This theorem corrects a tempting local-only view. A set can satisfy local cave conditions and still never be minimal, because a smaller subcave suppresses it. The barcode interval is nonempty exactly when the set is minimally responsible for some threshold.

5.3 Cave-kernel theorem

Theorem 5.4 (Cave kernel). *If $C \in \mathfrak{B}_\alpha(G)$ and $p = |N(C)|$, $k = |C|$, then:*

1. C is independent;
2. $N(C \setminus \{v\}) = N(C)$ for every $v \in C$;
3. $k = \lfloor p/\alpha \rfloor + 1$.

Proof sketch. If C has an internal edge, purification gives a proper bad subset. If some vertex has a private watcher, deleting it lowers the shadow by at least one, which still leaves a bad subset because $\alpha \leq 1$. Finally, C is bad but every one-vertex deletion is not, so $p < \alpha k$ and $p \geq \alpha(k - 1)$. $\square$

6 Shadow-LYM theory and blocker entropy

6.1 Cross-shadow theorem

For a cave D , define its free region

$$F(D) = V \setminus (D \cup N(D)).$$

Theorem 6.1 (Cross-shadow theorem). *If $C, D \in \mathfrak{B}_\alpha(G)$ are distinct, then*

$$C \cap F(D) \neq \emptyset.$$

Proof sketch. Assume $C \subseteq D \cup N(D)$. Let $X = C \cap D$ and $U = C \cup D$. Since X is a proper subset of minimal bad caves, it is not bad. Submodularity gives

$$|N(U)| + |N(X)| \leq |N(C)| + |N(D)| < \alpha|C| + \alpha|D|,$$

so $|N(U)| < \alpha|U|$: U is bad. The assumption $C \subseteq D \cup N(D)$ forces an internal edge between $C \setminus D$ and $D \setminus C$. Purification produces a proper bad subset inside D , contradicting the minimality of D . $\square$

6.2 Shadow-LYM theorem

Theorem 6.2 (Shadow-LYM theorem). *For every finite graph and $0 < \alpha \leq 1$,*

$$\sum_{C \in \mathfrak{B}_\alpha(G)} \frac{1}{\binom{n - |N(C)|}{|C|}} \leq 1.$$

Proof sketch. Set $A_C = C$ and $B_C = F(C)$. Then $A_C \cap B_C = \emptyset$, and the cross-shadow theorem gives $A_C \cap B_D \neq \emptyset$ for $C \neq D$. Bollobas's set-pairs inequality gives

$$\sum_C \binom{|A_C| + |B_C|}{|A_C|}^{-1} \leq 1.$$

Since $|A_C| + |B_C| = n - |N(C)|$, the result follows. $\square$

6.3 Complete-bipartite entropy theorem

The cave-kernel theorem gives $|C| = \lfloor |N(C)|/\alpha \rfloor + 1$. Thus the Shadow-LYM theorem implies:

Theorem 6.3 (Complete-bipartite blocker entropy). *For connected n -vertex graphs,*

$$\max_G |\mathfrak{B}_\alpha(G)| = \max_{1 \leq p \leq n-1} \binom{n-p}{\lfloor p/\alpha \rfloor + 1}.$$

The maximum is attained by the complete bipartite graph $K_{n-p,p}$.

At $\alpha = 1$, this becomes

$$\max_G |\mathfrak{B}_1(G)| = \max_p \binom{n-p}{p+1}.$$

Asymptotically,

$$\log \max_G |\mathfrak{B}_\alpha(G)| = c(\alpha)n + O(\log n),$$

where $x_\alpha \in (0, 1)$ solves

$$x_\alpha = (1 - x_\alpha)^{1+\alpha},$$

and

$$c(\alpha) = -\log(1 - x_\alpha).$$

At $\alpha = 1$, $c(1) = \log \varphi$.

α	$c(\alpha)$
1	0.481212 = $\log \varphi$
3/4	0.517729
2/3	0.531573
1/2	0.562399
1/4	0.618787
$\alpha \rightarrow 0$	$\log 2$

6.4 Barcode area and count

The Shadow-LYM inequality can be integrated over all thresholds:

$$\mathcal{A}(G) = \sum_{S: I_G(S) \neq \emptyset} \frac{|I_G(S)|}{\binom{n-|N(S)|}{|S|}} \leq 1.$$

For connected graphs, no bad cave exists below $1/(n-1)$, so

$$\mathcal{A}(G) \leq \frac{n-2}{n-1},$$

with equality for the star.

The total number of cave bars satisfies

$$\text{Bar}(G) := |\{S : I_G(S) \neq \emptyset\}| \leq 2^{n-1} - n,$$

again with equality for the star.

Thus complete bipartite graphs maximize fixed-threshold blocker count, while stars maximize total barcode persistence.

7 Exact families

7.1 Complete bipartite graphs

For $K_{M,P}$ with $M \geq P$, every $S \subseteq A$ on the large side has $N(S) = B$. Hence S is α -bad iff $P < \alpha|S|$. Let

$$q = \lfloor P/\alpha \rfloor + 1.$$

Then

$$\mathfrak{B}_\alpha(K_{M,P}) = \binom{A}{q}, \quad |\mathfrak{B}_\alpha(K_{M,P})| = \binom{M}{q}.$$

The torch number is

$$\tau_\alpha(K_{M,P}) = M - q + 1,$$

and

$$\eta_r(K_{M,P}) = \min\left(1, \frac{P}{M - 2r}\right).$$

7.2 Parallel cave trees

Let $C_{m,q}$ be a tree with m centers, each carrying q leaves. It has the same passive spectrum as $K_{mq,m}$:

$$mq\delta_{1/q} + m\delta_q.$$

But its active repair is staircase-like:

$$\eta_r(C_{m,q}) = \frac{1}{q - \min(q - 1, \lfloor 2r/m \rfloor)}.$$

At non-reciprocal thresholds, this family gives linear blocker defect.

7.3 Three-step spiders

Let Sp_m be the tree with one center and m arms $o - a_i - b_i - c_i$. Put $A_i = \{a_i, c_i\}$. For $J \subseteq [m]$,

$$A_J = \bigcup_{i \in J} A_i,$$

so

$$|A_J| = 2|J|, \quad |N(A_J)| = |J| + 1.$$

Let

$$\rho_q = \frac{q + 1}{2q}.$$

If $\rho_q < \alpha \leq \rho_{q-1}$, then

$$\mathfrak{B}_\alpha(Sp_m) \cong \binom{[m]}{q},$$

so

$$|\mathfrak{B}_\alpha(Sp_m)| = \binom{m}{q}, \quad \tau_\alpha(Sp_m) = m - q + 1.$$

Yet the passive spectrum is only

$$\mu_{Sp_m} = 2m\delta_{(m+1)/(2m)} + (m+1)\delta_{2m/(m+1)}.$$

This gives an exponential blocker explosion from a two-atom passive spectrum.

The exact edge-hardening curve is

$$\eta_r(Sp_m) = \begin{cases} \frac{m-2r+1}{2(m-2r)}, & 2r < m, \\ 1, & 2r \geq m. \end{cases}$$

8 Dynamic edge edits

8.1 Unmasking equation

Let $H = G + F$ and

$$\delta_F(S) = |N_H(S) \setminus N_G(S)|.$$

Then

$$\mathfrak{B}_\alpha(H) = \min_{\subseteq} \{S \in \mathcal{D}_\alpha(G) : |N_G(S)| + \delta_F(S) < \alpha|S|\},$$

where $\mathcal{D}_\alpha(G) = \{S : |N_G(S)| < \alpha|S|\}$. Thus adding edges deletes some old bad sets, and the new blockers are the minimal surviving bad sets. This operation can increase the number of minimal blockers.

8.2 Endpoint mutation and trident law

If $H = G + uv$ and $C \in \mathfrak{B}_\alpha(H) \setminus \mathfrak{B}_\alpha(G)$, then

$$|C \cap \{u, v\}| = 1.$$

For edge deletion, new blockers may contain one endpoint or both. Thus an edge edit has a trident: the u -cone, the v -cone, and the uv bridge cone.

More generally, if G and H differ only on edges whose endpoints lie in a support set R , then every changed blocker intersects R .

8.3 Rooted Shadow-LYM

For $R \subseteq V$, define

$$\mathfrak{B}_\alpha(G; R) = \{C \in \mathfrak{B}_\alpha(G) : R \subseteq C\}.$$

Then

$$\sum_{C \in \mathfrak{B}_\alpha(G; R)} \frac{1}{\binom{n-|N(C)|-|R|}{|C|-|R|}} \leq 1.$$

Consequently

$$|\mathfrak{B}_\alpha(G; R)| \leq M_{|R|}(n, \alpha),$$

where

$$M_s(n, \alpha) = \max_p \binom{n - p - s}{\lfloor p/\alpha \rfloor + 1 - s}.$$

8.4 Churn laws

At fixed threshold,

$$\log \max_{G,e} |\mathfrak{B}_\alpha(G \triangle e) \triangle \mathfrak{B}_\alpha(G)| = c(\alpha)n + O(\log n).$$

At $\alpha = 1$ this gives golden-ratio churn:

$$\max_{G,e} |\mathfrak{B}_1(G \triangle e) \triangle \mathfrak{B}_1(G)| = \varphi^{n+o(n)}.$$

Across all thresholds, one edge can reveal or change nearly Boolean-many bars:

$$\max_{G,e} \text{Churn}_{\text{bar}}(G, e) = 2^{n+O(\log n)}.$$

8.5 Double-star mutation

Let $D_{a,b}$ be a double-star with centers x, y , leaf sets A, B , and add one edge from y to a special leaf $a_1 \in A$. At $\alpha = 3/4$:

$$|\mathfrak{B}_{3/4}(D_{a,b})| = \binom{a}{2} + \binom{b}{2},$$

while after mutation

$$|\mathfrak{B}_{3/4}(D'_{a,b})| = \binom{a-1}{2} + \binom{b}{2} + (a-1)b.$$

The gross new blockers are $(a-1)b$, the erased blockers are $(a-1)$, and the net increase is $(a-1)(b-1)$.

Across all thresholds, the gross new barcode bars are

$$\sum_{r \geq 1} \binom{a-1}{r} \binom{b}{r} = \binom{a+b-1}{b} - 1.$$

For balanced a, b , this is exponential.

9 Root pressure, support, and endpoint capture

9.1 Root pressure

If $|R| \approx \beta n$ and every blocker is forced to contain R , then

$$\log M_{\lfloor \beta n \rfloor}(n, \alpha) = (1 - (1 + \alpha)\beta)_+ c(\alpha)n + O(\log n).$$

At one mana,

$$\log M_{\lfloor \beta n \rfloor}(n, 1) = (1 - 2\beta)_+ (\log \varphi)n + O(\log n).$$

A common core of size at least $n/(1 + \alpha)$ kills exponential fixed-threshold blocker entropy.

9.2 Support-core dichotomy

A support condition $C \cap R \neq \emptyset$ behaves differently from a core condition $R \subseteq C$. For every fixed nonempty support set R ,

$$\log \max_G |\{C \in \mathfrak{B}_\alpha(G) : C \cap R \neq \emptyset\}| = c(\alpha)n + O(\log n).$$

Thus touching a support does not reduce exponential rate. Containing a core does.

A finite support-hit extremal formula, verified in small tests but still conjectural, is

$$H_b(n, \alpha) = \max_p \left[\binom{n-p}{\lfloor p/\alpha \rfloor + 1} - \binom{n-p - \min(b, n-p)}{\lfloor p/\alpha \rfloor + 1} \right].$$

9.3 Endpoint capture

In the entropy-extremal complete-bipartite universe, let $x_\alpha = q/M$ where q is the minimal bad-cave size on the large side. Then

$$x_\alpha = (1 - x_\alpha)^{1+\alpha}.$$

A fixed support set of b vertices touches asymptotically

$$1 - (1 - x_\alpha)^b$$

of all blockers. A fixed core of s vertices is contained in asymptotically

$$x_\alpha^s$$

of all blockers.

At $\alpha = 1$, $x_1 = \varphi^{-2}$, so one endpoint captures $\varphi^{-2} \approx 0.382$ of extremal blockers, and two endpoints capture $\varphi^{-1} \approx 0.618$.

To touch a fraction θ of extremal blockers, one needs approximately

$$b_\theta = \left\lceil \frac{\log(1 - \theta)}{\log(1 - x_\alpha)} \right\rceil$$

high-capture endpoints.

10 Computational corroboration

All computations in this section used exact rational arithmetic and exhaustive subset enumeration for the stated finite graphs. The Python code that generated the main tables was saved as a supplementary script.

10.1 Graph-atlas blocker maxima and Shadow-LYM checks

Connected nonisomorphic graphs from the NetworkX graph atlas were checked through 7 vertices. For selected thresholds, the maximum observed blocker count matched the complete-bipartite formula. The maximum Shadow-LYM mass observed was 1.

α	n	graphs	max $ \mathfrak{B}_\alpha $	formula	max LYM
1	3	2	1	1	1
1	4	6	3	3	1
1	5	21	6	6	1
1	6	112	10	10	1
1	7	853	15	15	1
1/2	4	6	1	1	1
1/2	5	21	4	4	1
1/2	6	112	10	10	1
1/2	7	853	20	20	1
2/3	5	21	6	6	1
2/3	6	112	10	10	1
2/3	7	853	15	15	1

10.2 Barcode count maxima

The maximum total number of cave bars among connected graph-atlas graphs matched the star formula $2^{n-1} - n$.

n	max bars found	$2^{n-1} - n$
2	0	0
3	1	1
4	4	4
5	11	11
6	26	26
7	57	57

10.3 Spider blocker law check

For Sp_6 , exact enumeration matched $|\mathfrak{B}_\alpha| = \binom{6}{q}$ at the indicated thresholds.

q	sample α	computed caves	predicted $\binom{6}{q}$
2	7/8	15	15
3	17/24	20	20
4	31/48	15	15
5	49/80	6	6
6	71/120	1	1

10.4 Double-star mutation check

At $\alpha = 3/4$, exact enumeration gave the following before/after blocker counts and barcode counts.

a	b	blockers before	blockers after	gross new	erased	bars before	bars after
3	2	4	6	4	2	5	7
4	3	9	15	9	3	15	27
5	4	16	28	16	4	37	91
5	5	20	36	20	4	52	162

10.5 Core script excerpt

The core exact enumeration routine was as follows.

```
def blockers_pre(N, pc, npc, a, b):
    # alpha = a/b, exact integer comparison
    bad = [False] * len(N)
    mins = []
    for S in range(1, len(N)):
        if npc[S] * b < a * pc[S]:
            bad[S] = True
    for S in range(1, len(N)):
        if not bad[S]:
            continue
        ok = True
        sub = (S - 1) & S
        while sub:
            if bad[sub]:
                ok = False
                break
            sub = (sub - 1) & S
        if ok:
            mins.append(S)
    return mins
```

11 Corrections and conjectures

The theory evolved through several corrections. The important corrections are:

- A proposed disjoint torch-packing formula was false. The correct object is the blocker hypergraph $\mathfrak{B}_\alpha(G)$ and its transversal number.
- Local cave-kernel conditions are necessary but not sufficient. The barcode theorem captures subcave suppression exactly.
- The passive spectrum is complete for passive vulnerability but not for active repair. Same-spectrum graphs can have different blocker filtrations and hardening curves.

The main remaining conjectures are:

Conjecture 11.1 (Torch Pairing). *For all graphs and $0 < \alpha \leq 1$,*

$$r_\alpha(G) = \lceil \tau_\alpha(G)/2 \rceil.$$

Conjecture 11.2 (Finite support-hit extremal formula). *For $b \geq 1$,*

$$H_b(n, \alpha) = \max_p \left[\binom{n-p}{\lfloor p/\alpha \rfloor + 1} - \binom{n-p - \min(b, n-p)}{\lfloor p/\alpha \rfloor + 1} \right].$$

12 Applications and interpretation

12.1 Network vulnerability and hardening

A set S with small $|N(S)|/|S|$ is a hidden cluster: many internal vertices, few exposed neighbors. The passive spectrum measures how hidden such clusters can be. The blocker filtration identifies which minimal clusters must be hit to reach a target exposure. The torch number tells how many vertices must be touched. The hardening curve tells how much an edge budget can improve the worst exposure.

12.2 Audit complexity

The number of minimal explanations can be exponential. At $\alpha = 1$, the worst-case blocker count is $\varphi^{n+o(n)}$. Across all thresholds, the total number of cave bars can be $2^{n-1} - n$. A single edge edit can cause golden-ratio fixed-threshold churn and nearly Boolean barcode churn.

Thus explicit blocker enumeration is inherently expensive in the worst case. Practical algorithms should use:

- separation oracles rather than full lists;
- safe-core certificates;
- rooted LYM mass as a completeness certificate;
- endpoint-capture summaries before cave enumeration.

12.3 Design principles

The support-core dichotomy suggests a practical rule:

touching a monitored set is not enough; forcing containment in a common core compresses complexity.

For fixed-threshold blockers, a common core of density β reduces entropy to

$$(1 - (1 + \alpha)\beta) + c(\alpha)n.$$

At one mana, a core larger than $n/2$ kills golden blocker entropy.

13 Named laws index

Law / object	Main statement
Goblin shadow spline	$g_G(\lambda) = \int (\lambda - t)_+ d\mu_G(t).$
Reciprocity	$g_G(\lambda) = n(\lambda - 1) + \lambda g_G(1/\lambda).$
Normal form	$\mu_G = z\delta_0 + c\delta_1 + \sum t_{p,q}(q\delta_{p/q} + p\delta_{q/p}).$
Farey budget	Positive phase count is controlled by costs $p + q$ and grows like $(3/\pi^{2/3})n^{2/3}.$
Shadow skeleton	Every passive spectrum is realized by weighted edges.
Safe-core equation	$\tau_\alpha(G) = n - \sigma_\alpha(G).$
Reciprocal integrality	$\tau_{1/q}(G) = qg_G(1/q).$

Star-Cap law	$\tau_\alpha(G) \leq n - \lfloor 1/\alpha \rfloor - 1$, sharp for stars.
Breakaway law	First improvement budget is $\lceil a(G)/2 \rceil$.
Blocker theorem	$\tau_\alpha(G)$ is the transversal number of $\mathfrak{B}_\alpha(G)$.
Barcode theorem	$S \in \mathfrak{B}_\alpha(G)$ iff $\alpha \in I_G(S)$.
Shadow-LYM theorem	$\sum_{C \in \mathfrak{B}_\alpha} \binom{n - N(C) }{ C }^{-1} \leq 1$.
Complete-bipartite entropy	$\max \mathfrak{B}_\alpha = \max_p \binom{n-p}{\lfloor p/\alpha \rfloor + 1}$.
Golden Blocker law	At $\alpha = 1$, max blocker count is $\varphi^{n+o(n)}$.
Spider Blocker law	$\mathfrak{B}_\alpha(Sp_m) \cong \binom{[m]}{q}$ in threshold bands.
Unmasking law	Adding edges deletes bad sets but can reveal new minimal survivors.
Edge-edit trident	Changed blockers under an edge toggle live in endpoint cones / bridge cone.
Root pressure	Core-root entropy is $(1 - (1 + \alpha)\beta)_+ c(\alpha)n$.
Support-core dichotomy	Support-hit keeps full entropy rate; common core reduces it.
Endpoint capture	Extremal support fraction is $1 - (1 - x_\alpha)^b$.

14 Closing view

The theory that emerged can be summarized as follows.

The passive shadow spectrum is a compact, rational, reciprocal spline measuring hiddenness. It is algebraically elegant and often small. But active repair is governed by minimal bad caves, whose blocker hypergraph can be exponentially large even when the passive spectrum has only two atoms. The Shadow-LYM theorem gives a global entropy budget for these caves, complete bipartite graphs maximize fixed-threshold cave multiplicity, stars maximize total barcode persistence and torch cost, and spiders expose overlapping blocker geometry.

Dynamically, edge edits are endpoint-local but entropy-global: all changed blockers must touch the edited support, yet a single endpoint cone can contain a full exponential blocker universe. The practical lesson is to separate passive risk, active blocker geometry, and dynamic audit complexity. The goblin metaphor condenses this into one sentence:

The spectrum tells how dark the kingdom is; the blockers tell which caves must be burned; the barcode tells how long their shadows persist.

15 Round II: spark parity, ignition numbers, and a correction

This continuation studies the edge-addition side of the theory more sharply. The previous notebook isolated the universal lower bound

$$r_\alpha(G) \geq \lceil \tau_\alpha(G)/2 \rceil,$$

because one new edge has only two endpoints. The new finding is that the lower bound is not always realizable: an odd, completely unsupported island at high exposure must be closed by a triangle rather than merely paired by a path. This produces a clean parity law and a corrected view of the Torch Pairing conjecture.

Throughout this round, $0 < \alpha \leq 1$, $\eta(G) = \min_{\emptyset \neq S \subseteq V} |N(S)|/|S|$, and

$$r_\alpha(G) = \min\{|F| : \eta(G + F) \geq \alpha\}.$$

15.1 Endpoint support and the goblin pairing gap

Definition 15.1 (Endpoint support and pairing gap). *For an added edge set F , write*

$$R(F) = \{v \in V : v \text{ is incident to an edge of } F\}.$$

The goblin pairing gap is

$$\gamma_\alpha(G) = r_\alpha(G) - \lceil \tau_\alpha(G)/2 \rceil.$$

Theorem 15.2 (Endpoint support lower bound). *If $\eta(G + F) \geq \alpha$, then $R(F)$ is an α -torch set in G . Consequently*

$$r_\alpha(G) \geq \lceil \tau_\alpha(G)/2 \rceil \quad \text{and} \quad \gamma_\alpha(G) \geq 0.$$

Proof. Let $R = R(F)$ and $Q = V \setminus R$. No added edge touches a vertex of Q . Hence for every $S \subseteq Q$,

$$N_{G+F}(S) = N_G(S).$$

Since $G + F$ is α -safe, $|N_G(S)| \geq \alpha|S|$ for every $S \subseteq Q$. Thus Q is an α -safe core in G . By the safe-core equation from the previous notebook, R is an α -torch set. Therefore $|R| \geq \tau_\alpha(G)$. Since $|R| \leq 2|F|$, every successful repair satisfies $|F| \geq \lceil \tau_\alpha(G)/2 \rceil$. $\square$

The lower bound is exact in every connected graph tested so far, but it is false as an equality for arbitrary finite graphs. The obstruction already appears in the edgeless graph on an odd number of vertices.

15.2 The Null-Ignition Law

Definition 15.3 (Ignition number). *Let*

$$I_\alpha(n) = \min\{|E(H)| : |V(H)| = n, \eta(H) \geq \alpha\}.$$

Equivalently, $I_\alpha(n) = r_\alpha(E_n)$, where E_n is the edgeless graph on n vertices.

Theorem 15.4 (Null-Ignition Law). *For $n \geq 2$ and $0 < \alpha \leq 1$,*

$$I_\alpha(n) = \begin{cases} n/2, & n \text{ even,} \\ (n+1)/2, & n \text{ odd and } 0 < \alpha \leq 1/2, \\ (n+3)/2, & n \text{ odd and } 1/2 < \alpha \leq 1. \end{cases}$$

Thus

$$I_\alpha(n) = \lceil n/2 \rceil + \mathbf{1}_{\{n \text{ odd}, \alpha > 1/2\}}.$$

Since $\tau_\alpha(E_n) = n$, the odd high-exposure edgeless graphs have

$$\gamma_\alpha(E_n) = 1 \quad (n \text{ odd}, \alpha > 1/2).$$

Proof. Every α -safe graph with $\alpha > 0$ has no isolated vertex, because a singleton $\{v\}$ would otherwise satisfy $|N(\{v\})| = 0 < \alpha$. Hence at least $\lceil n/2 \rceil$ edges are needed to cover all vertices.

If n is even, a perfect matching is 1-safe: the matching involution maps every set S bijectively to a set of neighbors of the same size. Hence $I_\alpha(n) = n/2$.

If n is odd and $0 < \alpha \leq 1/2$, take a matching on $n - 3$ vertices plus one path P_3 . The only deficient-looking set in P_3 is the pair of leaves, whose neighborhood has size 1, so P_3 is exactly $1/2$ -safe. This gives $(n + 1)/2$ edges.

If n is odd and $\alpha > 1/2$, equality in the no-isolates lower bound would force the graph to be a disjoint union of K_2 components and exactly one P_3 component. The two leaves of that P_3 have neighborhood size 1, so the ratio is $1/2$, which is too small. Thus one more edge is necessary. A triangle plus a matching on the remaining $n - 3$ vertices is 1-safe and uses $(n + 3)/2$ edges. $\square$

n	$\tau_\alpha(E_n)$	$r_\alpha(E_n)$ for $\alpha \leq 1/2$	$r_\alpha(E_n)$ for $\alpha > 1/2$	high- α gap
3	3	2	3	1
5	5	3	4	1
7	7	4	5	1
9	9	5	6	1

This is the first discovered correction to the unrestricted Torch Pairing conjecture: the conjecture needs either a connectedness hypothesis or a spark-parity correction.

15.3 The Connected Spark Law

The same parity mechanism has a connected version. Connectivity lets a tree absorb the odd defect until the target exposure exceeds the imbalance of an odd path.

Definition 15.5 (Connected ignition number). *Let*

$$J_\alpha(n) = \min\{|E(H)| : |V(H)| = n, H \text{ connected}, \eta(H) \geq \alpha\}.$$

Lemma 15.6 (Fractional load certificate). *Suppose a graph H has nonnegative edge weights x_e such that each vertex load*

$$\ell(v) = \sum_{e \ni v} x_e$$

satisfies $\alpha \leq \ell(v) \leq 1$. Then H is α -safe.

Proof. For any $S \subseteq V$,

$$\alpha|S| \leq \sum_{v \in S} \ell(v).$$

Each weighted incidence counted on the right can be charged to the opposite endpoint, which lies in $N(S)$. Internal edges of S are charged twice, once to each endpoint, and crossing edges are charged to the outside endpoint. Therefore

$$\sum_{v \in S} \ell(v) \leq \sum_{u \in N(S)} \ell(u) \leq |N(S)|.$$

So $|N(S)| \geq \alpha|S|$. □

Theorem 15.7 (Connected Spark Law). *For $n \geq 2$ and $0 < \alpha \leq 1$,*

$$J_\alpha(n) = \begin{cases} n-1, & n \text{ even}, \\ n-1, & n = 2m+1 \text{ odd and } \alpha \leq m/(m+1), \\ n, & n = 2m+1 \text{ odd and } \alpha > m/(m+1). \end{cases}$$

Equivalently, for odd n the one-extra-edge threshold is

$$\alpha > \frac{n-1}{n+1}.$$

Proof. Every connected graph has at least $n-1$ edges.

If n is even, the path P_n has a perfect matching, hence is 1-safe by the matching-involution argument. Thus $J_\alpha(n) = n-1$.

Let $n = 2m+1$. The path P_{2m+1} is $m/(m+1)$ -safe. One certificate is given by the following edge weights on consecutive edges $e_i = (i, i+1)$:

$$x_{2j-1} = \frac{m+1-j}{m+1}, \quad x_{2j} = \frac{j}{m+1}, \quad j = 1, \dots, m.$$

The vertex loads alternate between $m/(m+1)$ and 1, so the fractional load certificate applies.

No tree on $2m+1$ vertices can be α -safe for $\alpha > m/(m+1)$. Indeed, any tree is bipartite. Its larger bipartition class A has size at least $m+1$, while $N(A)$ lies in the other class and has size at most m . Thus

$$\eta(T) \leq \frac{|N(A)|}{|A|} \leq \frac{m}{m+1}.$$

So above this threshold a connected safe graph needs at least n edges.

For the upper bound with n edges, take a triangle and $m-1$ disjoint copies of K_2 . This disjoint graph is 1-safe. Add $m-1$ extra edges to connect the components in a chain. Adding edges cannot decrease $|N(S)|$, so the resulting connected graph is still 1-safe and has

$$3 + (m-1) + (m-1) = 2m+1 = n$$

edges. □

15.4 Refined conjectures after the spark correction

The unrestricted Torch Pairing conjecture should be replaced by a connected version and a spark-corrected version.

Conjecture 15.8 (Connected Torch Pairing). *For every connected finite graph G and every $0 < \alpha \leq 1$,*

$$r_\alpha(G) = \lceil \tau_\alpha(G)/2 \rceil.$$

Conjecture 15.9 (Spark-corrected pairing). *The gap $\gamma_\alpha(G)$ is caused by unsupported odd spark islands. In particular, after contracting an already safe core, each all-torch component that has odd order and target $\alpha > 1/2$ contributes one parity spark unless it can attach to old neighborhood support or to another odd island.*

The second conjecture is intentionally less formal. It is meant as a design rule for the next round: the half-torch lower bound counts endpoints, while the spark correction counts whether those endpoints can be wired as safe local templates.

15.5 Exact computational continuation

A second exact enumeration was run to test the connected version of Torch Pairing. The method was different from direct edge-subset search. For each n and α , all labeled safe supergraphs were precomputed, and the following superset dynamic program was used:

$$h(M) = \min\{|E(H)| : H \supseteq M, \eta(H) \geq \alpha\}, \quad r_\alpha(M) = h(M) - |E(M)|.$$

Initialize $h(M) = |E(M)|$ for safe masks M and $h(M) = \infty$ otherwise, then sweep over missing edges e using

$$h(M) \leftarrow \min\{h(M), h(M \cup \{e\})\}.$$

The torch number was computed independently by subset zeta propagation of bad sets, i.e. by finding the largest safe core.

α	connected atlas graphs checked, $2 \leq n \leq 7$	counterexamples	max γ_α
1/2	994	0	0
2/3	994	0	0
3/4	994	0	0
4/5	994	0	0
1	994	0	0

Thus the new facts coexist cleanly: unrestricted Torch Pairing is false, but connected Torch Pairing survived all exact graph-atlas tests through seven vertices.

15.6 Applications of the spark laws

Sparse safe design. A high-exposure network does not have to be dense. For even n , a perfect matching is already 1-safe. For odd n , a triangle plus a matching is a minimal high-exposure nonconnected design. If connectedness is required, a tree is enough for even n , and for odd n a single cycle beyond a tree is enough exactly when the target exceeds $(n-1)/(n+1)$.

Odd-batch onboarding. When a batch of previously isolated devices is onboarded, pairwise links are optimal unless the batch is odd and the target is above $1/2$. In that case, one triadic seed is unavoidable. Operationally: do not leave an odd unsupported batch as a path P_3 when the desired exposure is above $1/2$; close the triangle.

Repair planning. The endpoint support lower bound remains the first audit: every successful edge repair must expose at least $\tau_\alpha(G)$ endpoints. The new spark correction is the second audit: those endpoints must themselves be wired into local templates that meet the target exposure after old support is discounted.

Algorithmic use. Before enumerating the blocker hypergraph, search for unsupported odd spark islands. They are small certificates that the naive half-torch budget is short by one. Conversely, if every torch island has an even pairing or can be absorbed into old support, the connected atlas evidence suggests that the half-torch budget is sharp.

16 Round II summary

The exceptional discovery in this round is a parity wall hidden inside edge repair. A new edge supplies two endpoints, so $\lceil \tau_\alpha/2 \rceil$ is always a lower bound; however, endpoints must also ignite into a safe local graph. The exact sparse ignition laws are

$$I_\alpha(n) = \lceil n/2 \rceil + \mathbf{1}_{\{n \text{ odd}, \alpha > 1/2\}},$$

and

$$J_\alpha(2m) = 2m - 1, \quad J_\alpha(2m + 1) = \begin{cases} 2m, & \alpha \leq m/(m + 1), \\ 2m + 1, & \alpha > m/(m + 1). \end{cases}$$

These equations correct the unrestricted Torch Pairing conjecture while strengthening the design theory: isolated odd high-exposure systems need a triangle; connected odd systems need one extra cycle only above the path imbalance threshold. Exact enumeration found no connected counterexample through seven vertices at the tested rational thresholds.

The goblin sentence for this round is:

Two torches can make a pair, but three lonely goblins above half-exposure must form a triangle.

17 Round III: reciprocal matroids and packet cliffs

This round turned the safe-core problem sideways. Instead of asking only which vertex sets are safe, clone the demand and supply. A reciprocal exposure $1/q$ becomes an ordinary transversal matroid. A general rational exposure a/b becomes a transversal matroid with vertices bundled into packets of size a . The packet viewpoint explains why reciprocal thresholds are algorithmic and why non-reciprocal thresholds can break greedy exchange.

The exceptional discovery in this round is a family of connected graphs showing that, for every $2/3 < \alpha < 1$, the α -safe sets need not form a matroid. Thus the reciprocal point $\alpha = 1$ is a sharp endpoint: exactly at one mana the blockers are matroid circuits, but just below one mana packet geometry can already violate circuit elimination.

17.1 The packet-lift law

Let $\alpha = a/b$ with a, b positive coprime integers and $0 < a \leq b$. Build a bipartite clone graph $\mathcal{B}_{a,b}(G)$ as follows. The left side contains a demand clones (v, i) of each vertex $v \in V(G)$, and the right side contains b capacity clones (u, j) of each vertex $u \in V(G)$. Connect (v, i) to (u, j) exactly when $vu \in E(G)$. Let $M_{a,b}(G)$ be the transversal matroid on the left clones. For $Q \subseteq V(G)$, write

$$P_a(Q) = Q \times [a]$$

for the union of the full demand packets over Q .

Theorem 17.1 (Packet-lift law). *For every finite graph G and every rational $\alpha = a/b$,*

$$Q \text{ is } \alpha\text{-safe in } G \iff P_a(Q) \text{ is independent in } M_{a,b}(G).$$

Equivalently, the safe-core problem is the problem of choosing as many whole a -packets as possible inside a transversal matroid.

Proof. By Hall's theorem, $P_a(Q)$ is matchable into the b neighbor-clones if and only if every clone subset $X \subseteq P_a(Q)$ has at least $|X|$ right-neighbors. If $U \subseteq Q$ is the support of X , then its right-neighbor set has size $b|N_G(U)|$ and $|X| \leq a|U|$. Hence the worst Hall test for a fixed support U is the full packet $P_a(U)$. Therefore Hall's condition is exactly

$$a|U| \leq b|N_G(U)| \quad \text{for all } U \subseteq Q,$$

which is the definition of a/b -safety. □

This is the cleanest structural equation found so far:

$$\boxed{\mathcal{I}_{a/b}(G) = \{Q \subseteq V(G) : P_a(Q) \in M_{a,b}(G)\}.$$

Here $\mathcal{I}_{a/b}(G)$ denotes the hereditary family of a/b -safe sets. The previous blocker hypergraph is the family of minimal packet-dependencies after projecting clone packets back down to vertices.

17.2 The reciprocal matroid corridor

When $a = 1$, packets have size one, so the projection disappears. This produces a true matroid on the original vertex set.

Theorem 17.2 (Reciprocal matroid law). *For every integer $q \geq 1$, the $1/q$ -safe sets of G are the independent sets of a transversal matroid $M_q(G)$. Its rank on $X \subseteq V(G)$ is*

$$r_q(X) = \min_{Y \subseteq X} (|X \setminus Y| + q|N(Y)|).$$

Consequently,

$$\sigma_{1/q}(G) = r_q(V), \quad \tau_{1/q}(G) = n - r_q(V) = \max_{Y \subseteq V} (|Y| - q|N(Y)|) = qg_G(1/q).$$

Moreover, $\mathcal{B}_{1/q}(G)$ is precisely the circuit set of $M_q(G)$.

Proof. The first statement is the packet-lift law with $a = 1$. The rank formula is Rado's rank formula for a transversal matroid with q copies of each right vertex. Substituting $X = V$ gives

$$r_q(V) = \min_{Y \subseteq V} (n - |Y| + q|N(Y)|) = n - \max_{Y \subseteq V} (|Y| - q|N(Y)|).$$

The final maximum equals $qg_G(1/q)$. Minimal dependent sets of this matroid are exactly minimal sets with $|C| > q|N(C)|$, i.e. the minimal $1/q$ -bad caves. $\square$

Thus the reciprocal integrality theorem from the earlier notebook is not a numerical accident; it is matroid nullity. At reciprocal thresholds the goblin can use all the standard matroid spells: rank certificates, circuit elimination, augmentation, greedy bases with weights, and matching oracles.

Corollary 17.3 (Reciprocal circuit elimination). *If $C, D \in \mathcal{B}_{1/q}(G)$ and $e \in C \cap D$, then there exists $E \in \mathcal{B}_{1/q}(G)$ such that*

$$E \subseteq (C \cup D) \setminus \{e\}.$$

This corollary is false for general non-reciprocal packet projections.

17.3 The hinge-packet cliff above two thirds

For $p \geq 2$, define a connected graph H_p with vertices

$$z, w_0, X = \{x_1, \dots, x_p\}, \quad Y = \{y_1, \dots, y_p\}, L = \{\ell_1, \dots, \ell_{p-1}\}, \quad R = \{r_1, \dots, r_{p-1}\}.$$

Put $W_L = \{w_0\} \cup L$ and $W_R = \{w_0\} \cup R$. Add the edge zw_0 , join every x_i to every vertex of W_L , and join every y_i to every vertex of W_R . The two intended caves are

$$C_X = \{z\} \cup X, \quad C_Y = \{z\} \cup Y.$$

Theorem 17.4 (Hinge-packet cliff). *Let $p \geq 2$ and*

$$\frac{p}{p+1} < \alpha \leq \frac{2p-1}{2p}.$$

Then the α -safe sets of H_p do not form a matroid. More precisely, C_X and C_Y are minimal α -bad caves sharing the vertex z , but $(C_X \cup C_Y) \setminus \{z\} = X \cup Y$ is α -safe. Hence blocker circuit elimination fails.

Proof. We have $|C_X| = |C_Y| = p + 1$ and $|N(C_X)| = |N(C_Y)| = p$. Since $p/(p + 1) < \alpha$, both caves are α -bad.

Every proper subset of C_X is safe. The singleton $\{z\}$ has one neighbor. Any nonempty subset meeting X sees all p vertices of W_L , while its size is at most p . Since $\alpha \leq 1$, its neighbor count is at least α times its size. The same argument applies to C_Y .

Now set $U = X \cup Y$. If a subset of U lies only in X or only in Y , it has at most p vertices and sees p watchers. If it meets both X and Y , it sees $W_L \cup W_R$, of size $2p - 1$, while its size is at most $2p$. The upper bound on α gives

$$2p - 1 \geq \alpha(2p),$$

so every subset of U is safe. Thus no blocker lies inside $(C_X \cup C_Y) \setminus \{z\}$, contradicting the circuit-elimination axiom required of matroid circuits. $\square$

The intervals

$$\left(\frac{p}{p+1}, \frac{2p-1}{2p} \right], \quad p = 2, 3, 4, \dots,$$

cover $(2/3, 1)$. Therefore every non-endpoint exposure $2/3 < \alpha < 1$ has a connected graph whose safe-core family is not a matroid. This is the packet cliff.

17.4 A sparse first witness at $\alpha = 3/4$

The general graph H_2 proves the first cliff interval $(2/3, 3/4]$. An exact search also found a sparser tree witness on eight vertices. Let T_8 be the tree with edge set

$$\{02, 06, 16, 25, 27, 34, 45\}.$$

Equivalently, it is the path $1 - 6 - 0 - 2 - 5 - 4 - 3$ with one extra leaf 7 attached to 2. At $\alpha = 3/4$ the only minimal bad caves are

$$C_1 = \{0, 1, 7\}, \quad C_2 = \{3, 5, 7\},$$

with

$$|N(C_1)| = |N(C_2)| = 2, \quad |C_1| = |C_2| = 3.$$

They share 7, but

$$(C_1 \cup C_2) \setminus \{7\} = \{0, 1, 3, 5\}$$

sees exactly three neighbors and is therefore $3/4$ -safe. The exchange failure can be written as

$$A = \{0, 3, 7\}, \quad B = \{0, 1, 3, 5\}.$$

Both A and B are safe, $|A| < |B|$, and $B \setminus A = \{1, 5\}$, but $A \cup \{1\}$ contains the bad cave C_1 and $A \cup \{5\}$ contains the bad cave C_2 . No augmentation from B is legal.

This tree has $\sigma_{3/4}(T_8) = 7$ and $\tau_{3/4}(T_8) = 1$, but it also has maximal non-optimal safe cores of size 6. Thus a naive greedy safe-core builder can get stuck even though one more safe vertex exists elsewhere.

17.5 Exact computational checks

The computations in this round reused the exact bitmask style of the supplementary script: open neighborhood masks are precomputed, bad sets are compared by exact integer inequalities, and safe cores are checked by subset propagation.

q	graphs	exch. fail	rank mismatch	nullity mismatch	max τ
1	996	0	0	0	5
2	996	0	0	0	4
3	996	0	0	0	3
4	996	0	0	0	2

For comparison, the non-reciprocal exchange test found no connected graph-atlas failure through seven vertices at $\alpha \in \{2/3, 3/4, 4/5, 5/6\}$. The sparse T_8 witness above is therefore a minimal connected 3/4-failure relative to exhaustive enumeration up to seven vertices.

17.6 Applications

Algorithm selection. At reciprocal thresholds, the maximum safe-core problem is a matching/matroid-rank problem. One can compute $\sigma_{1/q}$ by matching into q neighbor copies and obtain a dual obstruction Y from the rank formula. This gives short certificates for both safety and nonsafety.

Greedy warning. Away from reciprocal thresholds, whole demand packets must be selected. The projection of a matroid through packets can break augmentation. The T_8 tree shows that even a tree at 3/4 can have maximal safe cores that are not maximum. Greedy repair should therefore be trusted at $1/q$ levels, but audited by blocker or packet certificates at non-reciprocal levels.

Threshold engineering. If an application can choose its exposure standard, reciprocal levels are unusually auditable: blockers are circuits and torch number is passive nullity. If the standard must lie in $(2/3, 1)$, the hinge-packet theorem says matroid exchange cannot be assumed in all networks.

Circuit-elimination diagnostics. In a live audit, two blockers C, D with a common vertex but with $(C \cup D) \setminus \{e\}$ safe form a packet-cliff certificate. Such a certificate warns that local circuit-style explanations are incomplete and that packet-level optimization is needed.

18 Round III summary

The main new law is the packet-lift equation

$$\mathcal{I}_{a/b}(G) = \{Q \subseteq V(G) : P_a(Q) \in M_{a,b}(G)\}.$$

It says that every rational exposure is secretly a transversal matroid problem on demand clones, but only reciprocal exposures have packet size one and therefore descend to an honest matroid on vertices. At those reciprocal levels,

$$r_q(X) = \min_{Y \subseteq X} (|X \setminus Y| + q|N(Y)|), \quad \tau_{1/q}(G) = qg_G(1/q),$$

and the blockers are matroid circuits.

The exceptional new obstruction is the hinge-packet cliff: for every $2/3 < \alpha < 1$ there is a connected graph whose α -safe cores are not a matroid. The failure is not merely aesthetic; it creates

greedy traps and destroys circuit elimination. The first sparse witness is the eight vertex tree T_8 at $\alpha = 3/4$.

The remaining mystery is the interval $1/2 < \alpha \leq 2/3$. Exact graph-atlas tests through seven vertices found no exchange failure at $2/3$, but the packet-lift law does not yet prove a matroid there. This suggests a new target problem: determine the true lower edge of the packet cliff.

The goblin sentence for this round is:

At reciprocal mana every goblin is a matroid point; above two-thirds mana, goblins travel in packets, and packets can break exchange.

Round IV - Universal Matroid Thresholds and Packet Exchange

Status note. Round III showed that reciprocal thresholds $\alpha = 1/q$ have a matroid structure, while a small tree at $\alpha = 3/4$ already breaks circuit elimination. The new target in this round was to decide whether this is an accident of high exposure, or whether non-reciprocal thresholds are fundamentally nonmatroidal. The answer is sharp: reciprocal thresholds are exactly the universal matroid thresholds. I have not performed a literature-priority search; the novelty claim here means “new within this notebook.”

15.1 The universal matroid threshold law

For a graph $G = (V, E)$ and $0 < \alpha \leq 1$, recall the safe-core independence system

$$\mathcal{I}_\alpha(G) = \{Q \subseteq V : |N(S)| \geq \alpha|S| \text{ for every nonempty } S \subseteq Q\}.$$

It is always hereditary. The question is when it is always a matroid.

Theorem 1 (Universal Matroid Threshold Law). *For real thresholds $0 < \alpha \leq 1$,*

$$\mathcal{M}_{\text{univ}} := \{\alpha : \mathcal{I}_\alpha(G) \text{ is a matroid for every finite graph } G\} = \left\{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots\right\}.$$

Equivalently, exact greedy exchange is universal precisely at reciprocal exposure levels.

Proof idea. The inclusion $\{1/q\} \subseteq \mathcal{M}_{\text{univ}}$ is the Reciprocal Matroid Law from Round III: at $\alpha = 1/q$, the safe sets are independent sets of a transversal matroid with q neighbor-capacity clones. For every non-reciprocal threshold, the core-hinge construction below gives two minimal bad caves whose circuit elimination fails. Thus no non-reciprocal threshold can be universally matroidal.

15.2 The core-hinge equation

The following four integers are the new small machine for producing nonmatroid witnesses. Choose integers

$$k \geq 2, \quad 1 \leq s < k, \quad p \geq 1, \quad 1 \leq t \leq p,$$

satisfying the hinge feasibility equation

$$\boxed{\alpha(k-1) \leq p < \alpha k, \quad \alpha s \leq t \leq \min\{p, 2p - \alpha(2k - s - 1)\}.} \quad (\text{HFE})$$

Build a graph $X(k, p, s, t)$ as follows.

- A shared core H has size s .
- Two petal sets A and B each have size $k - s$.
- Common watchers W have size t .
- Side watchers U_A, U_B each have size $p - t$.
- Every vertex of H is adjacent to every vertex of W .
- Every vertex of A is adjacent to every vertex of $W \cup U_A$.
- Every vertex of B is adjacent to every vertex of $W \cup U_B$.

Let

$$C_A = H \cup A, \quad C_B = H \cup B.$$

Then $|C_A| = |C_B| = k$ and $|N(C_A)| = |N(C_B)| = p$, so the inequality $p < \alpha k$ makes both caves bad. Every proper subcave of C_A or C_B is safe because it either has neighborhood p and at most $k - 1$ vertices, or it lies in H and has neighborhood $t \geq \alpha s$. Hence C_A and C_B are minimal bad caves.

Now choose any $e \in H$. The set

$$(C_A \cup C_B) \setminus \{e\} = A \cup B \cup (H \setminus \{e\})$$

has neighborhood $2p - t$ and size $2k - s - 1$. The last inequality in (HFE) makes it safe, and the same bound makes all of its subsets safe. Therefore no minimal bad cave is contained in $(C_A \cup C_B) \setminus \{e\}$. The circuit-elimination axiom fails.

The witness has

$$n_H(k, p, s, t) = 2(k - s) + s + 2p - t = 2k - s + 2p - t \quad (1)$$

vertices. This gives a new target invariant:

$$h(\alpha) = \min\{2k - s + 2p - t : (k, p, s, t) \text{ satisfies (HFE)}\},$$

the smallest core-hinge certificate for nonmatroidality at threshold α .

15.3 Rational and irrational existence

Rational non-reciprocals. Let $\alpha = a/b$ in lowest terms with $a \geq 2$. Choose $r \geq b$ satisfying

$$ar \equiv b - a + 1 \pmod{b},$$

and set

$$k = r + 1, \quad p = \left\lceil \frac{ar}{b} \right\rceil, \quad s = b, \quad t = a.$$

Then

$$p - \frac{a}{b}(k - 1) = \frac{a - 1}{b},$$

$a/b \cdot s = t$, and

$$2p - t - \frac{a}{b}(2k - s - 1) = \frac{a - 2}{b} \geq 0.$$

Thus (HFE) holds. This proves that every rational threshold with numerator $a > 1$ has a connected finite graph whose α -safe sets are not a matroid.

Irrational thresholds. If α is irrational, the fractional parts of αr are dense. Choose r with

$$\delta = \lceil \alpha r \rceil - \alpha r \in (\alpha/2, \alpha),$$

put $p = \lceil \alpha r \rceil$ and $k = r + 1$. Then choose $s < r$ with $\lceil \alpha s \rceil - \alpha s < 2\delta - \alpha$, possible again by density, and set $t = \lceil \alpha s \rceil$. For r large enough, $t \leq p$, and (HFE) holds. Hence irrational thresholds also fail universal matroidality.

15.4 Exact computation: compact hinge witnesses

The following table was generated by exact bitmask enumeration. For each threshold, I searched the core-hinge template for a small witness, then enumerated all subsets to verify the number of minimal bad caves and the circuit-elimination failure. The column δ is the slack $p - \alpha(k - 1)$.

α	n_H	k	p	s	t	δ	caves	CE fail
2/7	19	11	3	7	2	1/7	2	yes
2/5	15	8	3	5	2	1/5	2	yes
3/5	14	7	4	5	3	2/5	2	yes
2/3	11	5	3	3	2	1/3	2	yes
3/4	8	3	2	1	1	1/2	2	yes
4/5	10	4	3	2	2	3/5	2	yes
5/6	12	4	3	1	1	1/2	2	yes

The exceptional point is the row $\alpha = 2/3$. The earlier high-exposure hinge explained $3/4 < \alpha < 1$ behavior. The core-hinge equation shows that even lower non-reciprocal thresholds such as $2/7$ are nonmatroidal; the obstruction merely requires a larger shared core. This is a denominator mirage: small graphs can make a non-reciprocal threshold look matroidal because its first hinge may be too large to appear.

15.5 Packet extendibility: the useful replacement for matroids

The matroid cliff is not the end of algorithmic structure. Rational thresholds still have a bounded exchange law.

Definition 1. A hereditary family $\mathcal{I}$ is c -extendible if whenever $A \subseteq B \in \mathcal{I}$ and $A \cup \{x\} \in \mathcal{I}$, there is a set $Y \subseteq B \setminus A$ with $|Y| \leq c$ such that

$$(B \setminus Y) \cup \{x\} \in \mathcal{I}.$$

Theorem 2 (Packet Extendibility Law). If $\alpha = a/b$ is rational in lowest terms, then $\mathcal{I}_{a/b}(G)$ is a -extendible for every finite graph G .

Proof sketch. Round III's Packet-Lift Law says that Q is a/b -safe exactly when the full packet $P_a(Q)$ of a clones per vertex is independent in a transversal matroid with b capacity clones per neighbor. Suppose $A \subseteq B$ are safe and $A \cup \{x\}$ is safe. In the clone matroid, $P_a(B)$ and $P_a(A) \cup P_a(x)$ are independent. The matroid exchange lemma inserts the a clones of x into $P_a(B)$ after deleting at most a clones from $P_a(B) \setminus P_a(A)$. Deleting the original vertices that own those clones removes at most a vertices, and independence is preserved under further deletion. Projecting back to vertices gives the claim.

At reciprocal thresholds $a = 1$, this is exactly the matroid exchange axiom. At non-reciprocal rational thresholds, exact matroid exchange fails, but packet exchange remains controlled by the numerator a .

15.6 Applications and new targets

Threshold engineering. If an application needs exact greedy optimization, exact rank oracles, or clean circuit elimination, choose $\alpha = 1/q$. If a non-reciprocal threshold is required, the packet numerator a gives a conservative exchange-loss certificate: adding one safe vertex may force the removal of at most a older vertices.

Audit test benches. The core-hinge equation is a compact adversarial generator. Given a proposed safety threshold, the tester either recognizes a reciprocal matroid threshold or emits a small connected graph where greedy exchange can fail. This is useful for stress-testing safe-core algorithms.

New target 1: the true matroidality radius. Define

$$m(\alpha) = \min\{|V(G)| : \mathcal{I}_\alpha(G) \text{ is not a matroid}\}.$$

The hinge number $h(\alpha)$ is an upper bound for $m(\alpha)$. Determining when $h(\alpha) = m(\alpha)$ is now a natural finite extremal problem. The previous atlas checks through seven vertices explain why the first 3/4 failure at eight vertices was invisible to smaller searches.

New target 2: exact extendibility. Let

$$e(\alpha) = \sup_G \{\text{minimum extendibility parameter of } \mathcal{I}_\alpha(G)\}.$$

The Packet Extendibility Law gives $e(a/b) \leq a$, while the Universal Matroid Threshold Law gives $e(1/q) = 1$ and $e(\alpha) > 1$ for non-reciprocal α . The sharp value of $e(a/b)$ is open inside this notebook and may be the next most useful invariant for approximation algorithms.

New target 3: irrational threshold compilers. For irrational α , the hinge proof uses Diophantine approximations. A practical compiler should return the smallest found tuple (k, p, s, t) satisfying (HFE), not merely prove existence. This would turn a real-valued safety policy into an explicit stress-test graph.

15.7 Round IV summary

The new structural picture is:

$\begin{aligned} \alpha = 1/q &\implies \text{safe cores are matroids;} \\ \alpha = a/b, a > 1 &\implies \text{not universally matroidal, but } a\text{-extendible;} \\ \alpha \notin \mathbb{Q} &\implies \text{not universally matroidal by Diophantine hinges.} \end{aligned}$

Thus the useful law of Round IV is: reciprocal thresholds give exact exchange, rational thresholds give packet-bounded exchange, and every other threshold admits a finite core-hinge trap.

16 Round V - Exact Exchange Loss and Defect Atoms

Status note. Round IV left three targets: the true matroidality radius, the exact extendibility parameter, and practical compilers for non-reciprocal thresholds. This round resolves the second target, adds a finite exhaustive check around the first target, and separates two different ways non-reciprocal thresholds fail: exchange failure and passive-active repair defect. The word “new” again means new inside this notebook; I have not performed a literature-priority search.

16.1 The exact extendibility law

Recall that a hereditary family $\mathcal{I}$ is c -extendible if whenever $A \subseteq B \in \mathcal{I}$ and $A \cup \{x\} \in \mathcal{I}$, there is a set $Y \subseteq B \setminus A$ with $|Y| \leq c$ such that

$$(B \setminus Y) \cup \{x\} \in \mathcal{I}.$$

Round IV proved the upper bound $e(a/b) \leq a$ for rational thresholds by lifting vertices to a -clone demand packets. The missing question was whether this packet numerator is merely a proof artifact or the true worst-case exchange loss.

Law 1 (Exact Extendibility Law). *For $0 < \alpha \leq 1$, let*

$$e(\alpha) = \sup_G \{ \text{minimum extendibility parameter of } \mathcal{I}_\alpha(G) \}.$$

Then

$$e(\alpha) = \begin{cases} 1, & \alpha = 1/q \text{ for an integer } q \geq 1, \\ a, & \alpha = a/b \text{ in lowest terms with } a \geq 2, \\ \infty, & \alpha \notin \mathbb{Q}. \end{cases}$$

Thus the numerator of the exposure threshold is not cosmetic: it is the exact worst-case number of old safe vertices that may have to be sacrificed to insert one new safe vertex.

The reciprocal case is the Round III matroid law. The rational upper bound is the Packet Extendibility Law from Round IV. It remains to build lower-bound examples.

16.2 The exchange fan

Fix integers $m \geq 1$, $k \geq 2$, $1 \leq s < k$, $p \geq 1$, and $1 \leq t \leq p$. The m -fan $F_m(k, p, s, t)$ is the graph with:

- a core H of size s ;
- petals $P_1, \dots, P_m$, each of size $k - s$;
- common watchers W of size t ;
- side watchers $U_1, \dots, U_m$, each of size $p - t$;
- every core vertex adjacent to every vertex of W ;
- every vertex of petal P_i adjacent to every vertex of $W \cup U_i$.

Put $C_i = H \cup P_i$. The fan inequalities are

$$p < \alpha k, \tag{F1}$$

$$p \geq \alpha(k-1), \tag{F2}$$

$$t \geq \alpha s, \tag{F3}$$

$$qp - (q-1)t \geq \alpha(q(k-s) + s-1) \quad (1 \leq q \leq m). \tag{F4}$$

Then each C_i is a minimal bad cave, but every union of q petals with one core vertex removed is safe. Choose a protected core vertex $x \in H$ and set

$$A = H \setminus \{x\}, \quad B = A \cup P_1 \cup \dots \cup P_m.$$

The sets A , B , and $A \cup \{x\} = H$ are safe. But if we insert x into B , every intact petal P_i completes the bad cave C_i . Since the protected set A cannot be deleted, restoring safety requires deleting at least one vertex from each petal. Hence this fan forces exchange loss at least m .

For rational $\alpha = a/b$ in lowest terms with $a \geq 2$, choose an integer $r \geq b$ satisfying

$$ar \equiv b - a + 1 \pmod{b},$$

and define

$$k = r + 1, \quad p = \left\lceil \frac{ar}{b} \right\rceil, \quad s = b, \quad t = a, \quad m = a.$$

The arithmetic collapses to the fan slack equation

$$qp - (q-1)t - \frac{a}{b}(q(k-s) + s-1) = \frac{a-q}{b}, \quad 1 \leq q \leq a. \tag{1}$$

Also

$$\frac{a}{b}k - p = \frac{1}{b}, \quad p - \frac{a}{b}(k-1) = \frac{a-1}{b}. \tag{2}$$

Equations (1) and (2) prove all fan inequalities and show that the last fan layer, $q = a$, is exactly tight. Therefore $e(a/b) \geq a$. Together with the packet upper bound, this proves $e(a/b) = a$.

For irrational α , the same fan idea has no finite ceiling. Given any m , choose r so that

$$\delta = \lceil \alpha r \rceil - \alpha r \in ((1 - 1/m)\alpha, \alpha),$$

and choose s so that

$$0 < \varepsilon = \lceil \alpha s \rceil - \alpha s < \min_{2 \leq q \leq m} \left(\frac{q\delta}{q-1} - \alpha \right).$$

This is possible by density of irrational fractional parts. Put $k = r + 1$, $p = \lceil \alpha r \rceil$, and $t = \lceil \alpha s \rceil$, with r large enough that $s < k$ and $t \leq p$. Then

$$q\delta - (q-1)\varepsilon \geq (q-1)\alpha \implies qp - (q-1)t \geq \alpha(q(k-s) + s-1),$$

so an m -fan exists. Since m is arbitrary, $e(\alpha) = \infty$ for irrational thresholds.

16.3 Exact checks and the seven-vertex mirage

The supplementary script for this round used exact rational arithmetic. It checked every connected NetworkX graph-atlas graph through seven vertices, and for each such graph it checked all threshold intervals that can be distinguished by ratios p/k with $k \leq 7$. There are 36 endpoint/midpoint samples and 35,856 graph-threshold checks. Every safe-core family was matroidal. Thus the smallest possible nonmatroid witness has at least eight vertices, and the Round III/Round IV 3/4 hinge on eight vertices is genuinely the first obstruction visible to atlas-scale search.

n	graphs	n	graphs	n	graphs	n	graphs
1	1	2	1	3	2	4	6
5	21	6	112	7	853		

The fan table below gives compact rational lower-bound witnesses for $e(\alpha)$. The column n_F is the number of vertices in the fan generated by the congruence construction above.

α	$e(\alpha)$	n_F	k	p	s	t
2/3	2	11	5	3	3	2
3/5	3	17	7	4	5	3
3/4	3	22	7	5	4	3
4/5	4	37	9	7	5	4
5/7	5	37	10	7	7	5
5/8	5	53	13	8	8	5
7/8	7	106	15	13	8	7

The exceptional conclusion is that denominator size is the wrong first-order measure of greedy fragility. The numerator controls exchange. For instance, 5/7 and 5/8 both force five-for-one exchange in the worst case, while 7/8 forces seven-for-one exchange.

16.4 Residue atoms and passive-active defect

The exchange fan solves the exact exchange target, but it also reveals a second, independent failure mode. A graph may have perfect or near-perfect exchange behavior and still fool the passive fractional torch bound.

Let $\alpha = a/b$ be rational in lowest terms. Choose integers k, p with

$$p < \alpha k, \quad p \geq \alpha(k-1),$$

and put

$$r = \alpha k - bp, \quad 1 \leq r \leq a.$$

In the complete bipartite atom $K_{k,p}$, the k -side is the unique minimal bad cave. Fuse m such atoms into one connected graph by adding edges only between watcher-side vertices. These added edges do not change the cave neighborhoods. The resulting graph has m disjoint bad caves, so

$$\tau_\alpha = m, \quad g_G(\alpha) = m \frac{r}{b}, \quad \left\lceil \frac{g_G(\alpha)}{\alpha} \right\rceil = \left\lceil \frac{mr}{a} \right\rceil.$$

Therefore the blocker defect is

$$\text{bid}_\alpha(G) = m - \left\lceil \frac{mr}{a} \right\rceil. \quad (3)$$

If $r < a$, the defect grows linearly with m . If $a = 1$, then necessarily $r = a = 1$, so (3) gives zero defect: this is another view of reciprocal integrality.

This gives a residue-density target:

$$D(\alpha) = \sup_G \limsup_{|V(G)| \rightarrow \infty} \frac{\text{bid}_\alpha(G)}{|V(G)|},$$

with the constructive lower bound

$$D(a/b) \geq \max_{k,p} \frac{1 - (ak - bp)/a}{k + p}, \quad (4)$$

where the maximum ranges over valid atoms with $1 \leq ak - bp < a$. Determining whether (4) is always sharp is a new extremal problem.

α	k	p	residue r	defect per atom	efficiency
2/3	2	1	1	1/2	1/6
3/5	2	1	1	2/3	2/9
3/4	3	2	1	2/3	2/15
4/5	4	3	1	3/4	3/28
5/8	2	1	2	3/5	1/5
5/7	3	2	1	4/5	4/25

16.5 The two-failure principle

The round’s main conceptual update is that non-reciprocal thresholds have two orthogonal demons.

Exchange demon. Fan graphs force the exact extendibility loss $e(a/b) = a$ but have no passive-active torch defect: one protected core vertex hits all fan caves, and the passive lower bound rounds up to one.

Residue demon. Fused residue atoms have a matroid-like partition structure, but the passive lower bound can undercount active repair by a linear amount.

Thus the passive spectrum, the exchange parameter, and the blocker-defect density are genuinely separate invariants. A useful policy should test all three.

16.6 Applications and new targets

Threshold engineering. If bounded greedy repair is required, use rational thresholds with small numerator. The denominator affects arithmetic resolution, but the numerator is the exact worst-case exchange cost. Irrational thresholds should be rounded to small-numerator rationals if a uniform exchange guarantee is needed.

Algorithm stress tests. Two compact adversarial generators are now available. Use fans to test greedy augmentation and exchange routines. Use residue atoms to test whether a passive fractional risk score underestimates the number of active interventions.

New target A: defect density. Decide whether the residue lower bound (4) equals the true asymptotic defect density $D(a/b)$.

New target B: mixed fan-atom composites. Combine fans and residue atoms in one graph and study the Pareto frontier between exchange loss, torch defect, and blocker entropy.

New target C: matroidality radius. The seven-vertex mirage check and the eight-vertex $3/4$ hinge suggest a finite phase diagram for

$$m(\alpha) = \min\{|V(G)| : \mathcal{I}_\alpha(G) \text{ is not a matroid}\}.$$

The Round IV hinge number is an upper bound; proving exactness remains open inside this notebook.

16.7 Round V summary

The most useful new law is

$$e(\alpha) = \begin{cases} 1, & \alpha = 1/q, \\ a, & \alpha = a/b \in \mathbb{Q}, (a, b) = 1, a \geq 2, \\ \infty, & \alpha \notin \mathbb{Q}. \end{cases}$$

The goblin interpretation is: reciprocal thresholds are true matroid gates; rational thresholds are packet gates whose numerator is the exact exchange toll; irrational thresholds admit arbitrarily large exchange traps. Separately, residue atoms show that even matroid-like systems can have a linear passive-active repair defect. The next theory should treat exchange loss and torch defect as a two-coordinate phase space rather than a single notion of difficulty.

Round VI — Defect Spectrum, Residue Knapsacks, and Reciprocal Cliffs

Status. This round continues the Round V target “exact defect density.” I separate two levels: a proved atom-generated lower envelope, and a conjectural full extremal envelope. All computations below use exact integer comparisons; no floating point threshold decisions are used.

The active-passive defect is the amount by which the passive shadow spline underpays the actual number of torches required:

$$\Delta_\alpha(G) = \tau_\alpha(G) - \text{ceil}(g_G(\alpha)/\alpha).$$

For asymptotic design, define the defect density target

$$D(\alpha) = \limsup_{n \rightarrow \infty} (1/n) \max\{|\Delta_\alpha(G)| : |V(G)|=n\}.$$

The new object of this round is the atom defect spectrum. It predicts where the passive theory and active repair theory separate most violently.

Complete-bipartite atom. Let $A_{\{k,p\}}=K_{\{k,p\}}$, and regard the k -side as the hidden cave. It is a one-cave α -atom exactly when

$$p/k < \alpha \leq p/(k-1).$$

In that interval one copy has $\tau_\alpha=1$, while its passive payment is

$$x_{\{k,p\}}(\alpha) = g(\alpha)/\alpha = k - p/\alpha, \quad 0 < x_{\{k,p\}}(\alpha) \leq 1.$$

Thus m fused copies have the exact rounding defect

$$\Delta_\alpha(m A_{\{k,p\}}) = m - \text{ceil}(m x_{\{k,p\}}(\alpha)),$$

and the asymptotic defect density contributed by this one atom type is

$$\begin{aligned} d_{\{k,p\}}(\alpha) &= (1 - x_{\{k,p\}}(\alpha))/(k+p) \\ &= (p/\alpha - k + 1)/(k+p). \end{aligned}$$

Atom Defect Spectrum Law. The atom-generated lower envelope is

$$D_{\text{atom}}(\alpha) = \max_{\{k,p \geq 1, \ p/k < \alpha \leq p/(k-1)\}} (p/\alpha - k + 1)/(k+p).$$

The maximum exists because every candidate has value at most $1/(k+p)$. Therefore large atoms become irrelevant unless all smaller atoms vanish. This is the first compact “defect spectrum” equation.

20.1 Rational residue form

For $\alpha=a/b$ in lowest terms, the atom condition is equivalent to a positive integer residue

$$r = a k - b p, \quad 1 \leq r \leq a.$$

The case $r=a$ pays exactly one full torch passively and creates no asymptotic defect. The useful residues are $1 \leq r < a$. Put

$$c_r(a,b) = \min \{ k+p : k,p \geq 1, \ a k - b p = r \}.$$

Then the atom spectrum becomes a finite residue maximization:

$$D_{\text{atom}}(a/b) = \max_{\{1 \leq r < a\}} (a-r)/(a \ c_r(a,b)).$$

This is the Residue Density Equation. The numerator a is no longer just the exchange-loss parameter from Round V; it is also the denominator that converts residue gaps into active-passive defect density.

The finite-size version is a small unbounded knapsack. If m_r copies of residue r are used, then

$$\begin{aligned} M &= \sum_r m_r, & R &= \sum_r r \ m_r, \\ \Delta &= M - \text{ceil}(R/a), & N &= \sum_r c_r \ m_r. \end{aligned}$$

Equivalently, positive defect appears as soon as the accumulated residue gap reaches a :

$$n_0(a/b) = \min \{ \sum_r c_r \ m_r : \sum_r (a-r)m_r \geq a \}.$$

Interpretation. Residue is passive credit. Gap $a-r$ is unpaid active demand. The first-awakening number n_0 tells how large a graph must be before passive and active repair can provably disagree by at least one torch through residue atoms.

This also refines the Round V residue-atom law. Instead of merely saying that fused atoms can have linear defect, the knapsack identifies the cheapest first failure and the best asymptotic failure rate.

The construction can be made connected without changing the accounting: connect the watcher-side vertices of consecutive atoms by extra watcher-watcher edges. Exact enumeration in the script verifies that the resulting connected witnesses retain the predicted defect.

20.2 Exact experiment table

The following table was generated by exact bitmask enumeration. “passive” means $\text{ceil}(g_G(\alpha)/\alpha)$. Each witness is a connected watcher-fused atom graph at the first-awakening size.

α	best density atom	D_atom	n _o	connected witness invariants
2/3	$K_{\{2,1\}}$, $r=1$	1/6	6	$n=6$, $\tau=2$, passive=1, $\Delta=1$, blockers=3
3/4	$K_{\{3,2\}}$, $r=1$	2/15	8	$n=8$, $\tau=2$, passive=1, $\Delta=1$, blockers=3
3/5	$K_{\{2,1\}}$, $r=1$	2/9	6	$n=6$, $\tau=2$, passive=1, $\Delta=1$, blockers=3
4/5	$K_{\{4,3\}}$, $r=1$	3/28	10	$n=10$, $\tau=2$, passive=1, $\Delta=1$, blockers=3
5/8	$K_{\{2,1\}}$, $r=2$	1/5	6	$n=6$, $\tau=2$, passive=1, $\Delta=1$, blockers=3
5/7	$K_{\{3,2\}}$, $r=1$	4/25	8	$n=8$, $\tau=2$, passive=1, $\Delta=1$, blockers=3
3/7	$K_{\{5,2\}}$, $r=1$	2/21	11	$n=11$, $\tau=2$, passive=1, $\Delta=1$, blockers=3
7/10	$K_{\{3,2\}}$, $r=1$	6/35	8	$n=8$, $\tau=2$, passive=1, $\Delta=1$, blockers=3

Small surprise. At $\alpha=3/4$, the best asymptotic atom $K_{\{3,2\}}$ alone does not create the first defect most cheaply. The first defect uses a mixed residue pair: $K_{\{3,2\}}$ with residue 1 and $K_{\{2,1\}}$ with residue 2. Their gaps 2 and 1 sum to $a=3$, so $\tau=2$ while the passive ceiling is only 1. The first defect therefore appears at 8 vertices.

This mirrors the earlier 8-vertex nonmatroid witness, but it is a different failure mode: here the issue is not circuit elimination; it is passive-active rounding across residues.

Experimental reading. In the tested witnesses, connecting atoms through watcher-side edges creates one extra mixed blocker but does not alter τ , passive payment, or defect. This suggests that the connected and disconnected asymptotic defect densities may coincide.

20.3 Reciprocal Cliff Law

The exceptional discovery of this round is a sharp one-sided instability at reciprocal thresholds. From Rounds III-V, $\alpha=1/q$ is matroidal and has zero active-passive defect. But immediately above it, the atom $K_{\{q,1\}}$ wakes up.

Reciprocal Cliff Law. For every integer $q \geq 1$,

$$\begin{aligned} D_{\text{atom}}(1/q) &= 0, \\ \lim_{\{\varepsilon \downarrow 0\}} D_{\text{atom}}(1/q + \varepsilon) &= 1/(q+1), \\ \lim_{\{\varepsilon \downarrow 0\}} D_{\text{atom}}(1/q - \varepsilon) &= 0. \end{aligned}$$

Proof sketch. At $\alpha=1/q$, every candidate atom satisfies $k=q+1$ and has $d_{\{k,p\}}=0$, reproducing reciprocal integrality. For α just above $1/q$, $K_{\{q,1\}}$ is admissible and has density $(1/\alpha - q + 1)/(q+1)$, which tends to $1/(q+1)$. No atom with smaller cost can be bad near $1/q$, giving the upper bound. From below, the nearest $p=1$ atom is $K_{\{q+1,1\}}$, whose density tends to zero.

A concrete rational approach from above is

$$\alpha_a = a/(qa-1).$$

For this sequence, the best atom is $K_{\{q,1\}}$, with residue 1, and

$$D_{\text{atom}}(\alpha_a) = (a-1)/(a(q+1)), \quad e(\alpha_a)=a.$$

Thus an arbitrarily tiny upward perturbation of $1/q$ can simultaneously create large exchange loss and a defect density approaching $1/(q+1)$. At the exact reciprocal, both collapse to the matroidal, zero-defect case.

q	sample $\alpha=a/(qa-1)$	D_{atom} samples	right-limit
2	2/3, 3/5, 10/19, 25/49	1/6, 2/9, 3/10, 8/25	1/3
3	2/5, 3/8, 10/29, 25/74	1/8, 1/6, 9/40, 6/25	1/4
4	2/7, 3/11, 10/39, 25/99	1/10, 2/15, 9/50, 24/125	1/5
5	2/9, 3/14, 10/49, 25/124	1/12, 1/9, 3/20, 4/25	1/6

20.4 Defect Spectrum Conjecture and applications

Defect Spectrum Conjecture. For every $0 < \alpha \leq 1$,

$$D(\alpha) = D_{\text{atom}}(\alpha).$$

In words: the worst asymptotic passive-active defect is already generated by complete-bipartite residue atoms, and arbitrary blocker hypergraph geometry cannot beat the best residue gap per vertex. I do not regard this as proved. The exact experiments and the Shadow-LYM budget make it plausible, but a full proof would need a charging theorem that assigns every unpaid torch to a private residue gap.

Why this is useful. The equation $D_{\text{atom}}(\alpha)$ is cheap to compute. For rational $\alpha = a/b$ it is a finite modular scan over $r = 1, \dots, a-1$; the first visible failure n_0 is an unbounded knapsack with at most $a-1$ item types. This gives a policy stress-test compiler: input α and graph-size budget N , output the cheapest connected graph where passive repair underpays active repair.

Applications.

- Threshold engineering: exact reciprocal thresholds $1/q$ are algorithmically safe; non-reciprocal thresholds just above $1/q$ are dangerous even when numerically close.
- Audit calibration: $n_0(\alpha)$ tells when a testbed is too small to reveal passive-active defect. For $\alpha = 3/4$, the first atom-generated defect appears at 8 vertices, not at 5.
- Greedy repair warning: $e(a/b) = a$ and $D_{\text{atom}}(a/b) > 0$ together mean a rational policy with large numerator can have both large exchange traps and large passive underpayment.
- Adversarial benchmark design: residue knapsacks generate minimal connected witnesses for stress-testing safe-core solvers.

New targets.

- Prove or refute the full Defect Spectrum Conjecture $D(\alpha) = D_{\text{atom}}(\alpha)$. The missing tool is a residue-gap charging theorem for arbitrary blocker hypergraphs.
- Compute the finite connected correction $D_n(\alpha)$ and the true first-defect radius. Compare it with the matroidality radius from Rounds IV-V.
- Build mixed fan-atom composites that simultaneously maximize exchange loss $e(\alpha)$ and passive-active defect. Current constructions show the demons separately; the tensor product may expose a sharper combined law.
- Analyze the real-valued graph of $D_{\text{atom}}(\alpha)$. It appears as a Farey field of hyperbolic teeth with one-sided reciprocal cliffs.
- Turn the residue knapsack into a design oracle: choose α under constraints on exchange loss, defect density, and first-awakening size.

Round VI summary. The numerator a of $\alpha = a/b$ now has three roles: it is the exact exchange toll $e(\alpha)$, the residue denominator in passive-active rounding, and the gap budget needed before defect awakens. Reciprocal thresholds are not merely matroidal; they are isolated zero-defect ledges. Just above them lies a cliff of active repair cost invisible to the passive shadow spline.

Round VII: exchange barcodes and sparse goblin trees

After the defect-spectrum round, I returned to the matroidality-radius target left by the exchange-loss theory. This continuation moves one level above blocker bars. The new object is an exchange barcode: a finite list of threshold intervals on which the safe-core family fails the matroid circuit-elimination axiom.

Exceptional discovery. Matroid failure has its own barcode. A graph can be safe-core matroidal at some thresholds and nonmatroidal at others, and the transition is exactly governed by exchange bars computable from shadow ratios. This turns the vague problem "when does greedy exchange break?" into an interval arithmetic problem.

21.1 The exchange-bar formula

For a nonempty vertex set S , keep the Round I notation

$$\rho(S) = |N(S)| / |S|.$$

Let the blocker upper endpoint be

$$\rho_{\text{bar}}(S) = \min(1, \min_{\{\emptyset \neq T \text{ proper subset } S\}} \rho(T)).$$

Thus S is a minimal α -bad cave exactly for α in the blocker bar

$$I_G(S) = (\rho(S), \rho_{\text{bar}}(S)].$$

For any U , define the internal exposure floor

$$\eta(U) = \min_{\{\emptyset \neq T \subset U\}} \rho(T).$$

Now take two distinct candidate circuits A and B and a shared element x in $A \cap B$. Put $U = (A \cup B) \setminus \{x\}$. The new exchange bar is

$$J_G(A, B, x) = (\max\{\rho(A), \rho(B)\}, \min\{\rho_{\text{bar}}(A), \rho_{\text{bar}}(B), \eta(U)\}].$$

The lower endpoint says both A and B are bad. The first two upper endpoints say they are minimal. The $\eta(U)$ endpoint says the circuit-elimination container U is safe, so no smaller bad cave can rescue the matroid axiom.

21.2 Exchange Barcode Theorem

Theorem. For every finite graph G and every $0 < \alpha \leq 1$, the safe-core independence system $I_\alpha(G)$ is nonmatroidal if and only if α lies in at least one nonempty exchange bar $J_G(A, B, x)$.

Proof sketch. In a hereditary independence system, the minimal dependent sets are the circuits. Here those circuits are exactly the active blockers $B_\alpha(G)$. Matroidality is equivalent to circuit elimination: for all active blockers A and B and all x in $A \cap B$, some active blocker must lie inside $(A \cup B) \setminus \{x\}$.

Fix A , B , and x . The conditions that A and B are active blockers are $\alpha \in I_G(A)$ and $\alpha \in I_G(B)$. The condition that no active blocker lies in $U = (A \cup B) \setminus \{x\}$ is exactly that U is α -safe, namely $\alpha \leq \eta(U)$. Intersecting these three requirements gives $J_G(A, B, x)$. Taking the union over all triples gives the theorem.

This theorem is the exchange-level analogue of the blocker barcode theorem: blocker bars say when individual caves exist; exchange bars say when pairs of caves break greedy augmentation.

21.3 Theta intervals are exchange bars

Round IV introduced a theta signature (k, p, s, r) : two caves of size k , each with p watchers, sharing s cave vertices and r watchers. The exchange-bar theorem condenses the previous construction to one line.

$$J_{\text{theta}}(k,p,s,r) = (p/k, \min\{ p/(k-1), r/s, (2p-r)/(2k-s-1) \}].$$

The three upper gates have direct meanings: deleting one vertex from either cave must be safe; the shared cave core must be safe; and the union with one shared cave vertex removed must be safe. The reciprocal moat proof from Round V becomes a barcode statement: reciprocal thresholds $1/q$ never lie in any theta exchange bar.

This also explains why the seven-vertex atlas checks saw no failure. A nonmatroid witness needs an exchange bar, not merely two bad sets. Up through seven vertices the exact search found no such bar; the first $3/4$ bar appears at eight vertices.

21.4 Sparse tree witnesses

A surprise of this round is that exchange failure is not a dense phenomenon. By taking spanning trees inside theta-like incidence graphs and testing them exactly, I found sparse tree witnesses whose exchange bars match the expected theta bands.

alpha	n	blockers at alpha	active exchange bar	certificate shape
3/4	8	2	(2/3, 3/4]	two 3-caves share one vertex
2/3	11	3	(3/5, 2/3]	three 5-caves; pair failure
3/5	14	3	(4/7, 3/5]	three 7-caves; pair failure
2/5	15	3	(3/8, 2/5]	three 8-caves; pair failure
3/10	14	2	(2/7, 3/10]	two 7-caves share a 3-core
4/5	10	2	(3/4, 4/5]	asymmetric 3/4-cave pair

Exact certificate example at $\alpha = 2/5$. The tree has three active blockers. Two of them are

$$A = \{0, 1, 2, 3, 4, 5, 6, 10\},$$

$$B = \{0, 1, 2, 6, 7, 8, 9, 10\}.$$

They share $x = 6$. Removing x gives $U = \{0, 1, 2, 3, 4, 5, 7, 8, 9, 10\}$, which is safe through $\alpha = 2/5$. The exchange bar is exactly

$$(\max\{\rho(A), \rho(B)\}, \min\{\rho_{\text{bar}}(A), \rho_{\text{bar}}(B), \eta(U)\}]$$

$$= (3/8, 2/5].$$

Thus the failure is not an artifact of a dense complete-bipartite gadget: it survives in a tree with 15 vertices.

21.5 Arboreal universality target

Conjecture: Arboreal Universality. Every non-reciprocal threshold α in $(0, 1]$ has a finite tree T for which $I_{\alpha}(T)$ is not a matroid. Equivalently, reciprocal thresholds are the only thresholds that are universally matroidal even after restricting to trees.

The data above support the conjecture across high, middle, and low non-reciprocal bands. The mechanism appears to be a sparse version of theta: distribute cave leaves across watcher paths so that individual caves remain minimal while the union-minus-one container remains safe.

A proof would sharpen Round IV in a useful way. Dense gadgets prove existence, but trees are minimal infrastructure. If trees already contain every exchange demon, then audits for greedy failure can be stress-tested using acyclic networks alone.

21.6 Useful applications

Threshold engineering. For a fixed system graph, compute exchange bars once. Thresholds outside the union of exchange bars are matroid-safe: greedy augmentation and rank-style reasoning cannot fail by circuit elimination. Thresholds inside an exchange bar require packet-level or blocker-level auditing.

Sparse stress tests. The tree witnesses give small acyclic adversarial examples. They can be used to test safe-core optimizers, onboarding rules, network-hardening heuristics, and any algorithm that assumes exchange-like behavior.

Matroidality radius. The new barcode reframes $m(\alpha)$ as the shortest graph carrying an exchange bar that contains α . The tree version $m_T(\alpha)$ is now a concrete target; the table gives upper bounds $m_T(3/4) \leq 8$, $m_T(2/3) \leq 11$, $m_T(3/5) \leq 14$, $m_T(2/5) \leq 15$, $m_T(3/10) \leq 14$, and $m_T(4/5) \leq 10$.

21.7 New targets after Round VI

The exchange barcode opens a new layer of the theory. The most promising next targets are:

$$\begin{aligned} m(\alpha) &= \min\{|V(G)| : \text{some exchange bar of } G \text{ contains } \alpha\}, \\ m_T(\alpha) &= \text{same minimum restricted to trees}, \\ X_n(\alpha) &= \max\{|\{\text{exchange bars active at } \alpha\}| : |V(G)|=n\}, \\ \text{Pers}X(G) &= \text{sum over exchange bars } J \text{ of } |J|. \end{aligned}$$

Target 1: exact matroidality radius. Round V proved $m(\alpha) \geq 8$ for all non-reciprocal α visible through seven vertices, and $m(3/4)=8$. The exchange-bar formula should make the full radius problem an integer optimization over cave sizes and overlaps.

Target 2: exchange entropy. Blocker entropy can be golden-ratio large at $\alpha=1$. Exchange entropy asks how many simultaneous circuit-elimination traps can coexist. This is the next audit-complexity invariant after blocker count.

Target 3: reciprocal moats. For a size bound n , there should be intervals around each reciprocal $1/q$ where no exchange bar of any n -vertex graph can enter. These moats quantify how close a non-reciprocal policy can be to a reciprocal policy before finite systems notice the difference.

Target 4: tree sparsification theorem. Prove that every theta exchange bar has a tree sub-witness, or characterize exactly which theta bars do. The experiments suggest that dense watcher sharing can be replaced by balanced acyclic watcher paths.

21.8 Round VI summary

1. New exact theorem: matroid failure is governed by exchange bars $J_G(A, B, x)$.
2. The theta obstruction interval is exactly one exchange bar.
3. Sparse tree witnesses were found for $\alpha = 3/4, 2/3, 3/5, 2/5, 3/10$, and $4/5$.
4. New conjecture: every non-reciprocal threshold already fails universal matroidality on trees.
5. Main application: compute exchange bars to locate thresholds where greedy/rank-style safe-core algorithms are trustworthy.

The goblin metaphor now has a fourth layer: the spectrum tells how dark the kingdom is; blockers tell which caves must be burned; packet fans tell the exchange toll; and exchange bars tell exactly when two caves conspire to break the greedy laws.

Round VIII — Arboreal Exchange Compiler and Möbius Leaf Law

This round attacks the tree-sparsification target from Round VII. The previous evidence suggested that every non-reciprocal exposure threshold should already fail matroidality on a tree. The new result is a constructive compiler: input a non-reciprocal alpha, output a finite tree with an exchange bar containing alpha.

Exceptional discovery. The Arboreal Universality conjecture from Round VII becomes a theorem, subject to the local exchange-bar proof given below: reciprocal thresholds are the only universally matroidal thresholds even when the universe is restricted to trees.

22.1 Arboreal Universality Theorem

Theorem. For every alpha in (0,1] that is not of the form $1/q$, there is a finite tree T such that the safe-core independence system $I_{\alpha}(T)$ is not a matroid. Equivalently,

$\{ \alpha : I_{\alpha}(T) \text{ is a matroid for every finite tree } T \}$
 $= \{1, 1/2, 1/3, 1/4, \dots\}.$

The reciprocal inclusion is inherited from the Reciprocal Matroid Law: at $\alpha=1/q$, safe cores are independent sets of a transversal matroid. The non-reciprocal direction is new here and is constructive.

The proof has three moving parts: two base ladders covering the interval $(1/2,1)$, and a watcher-leaf Möbius transform that maps any exchange bar down into a chosen interval between consecutive reciprocals.

22.2 Two tree ladders covering $(1/2,1)$

Plus ladder. For every integer $t \geq 3$ there is a tree P_t with $2t+2$ vertices and two caves A, B sharing one vertex such that

$$J(P_t; A, B, x) = ((t-1)/t, \quad t/(t+1)].$$

Construction: take a central watcher c , a shared cave vertex x adjacent to c , a short arm $c-a_1-wA-a_2$, and a long alternating arm $c-b_1-w_1-b_2-\dots-w_{t-2}-b_{t-1}$. Let $A=\{x, a_1, a_2\}$ and $B=\{x, b_1, \dots, b_{t-1}\}$. Then $\rho(B)=(t-1)/t$ and the union with x removed has ratio $t/(t+1)$. The proper-subset gates are no smaller.

Minus ladder. For every integer $r \geq 2$ there is a tree M_r with $3r+5$ vertices and two caves A, B such that

$$J(M_r; A, B, x) = ((r+1)/(2r+1), \quad r/(2r-1)].$$

Construction: make a watcher chain $W_0, \dots, W_{r-1}$ connected by cave vertices, then attach one cave leaf to every watcher. This shared core has $2r-1$ cave vertices and r watchers. Add one length-three extension path at W_0 and another at W_{r-1} ; A takes the left extension and B takes the right extension. The shared core supplies the upper endpoint $r/(2r-1)$.

The plus intervals cover $(2/3,1)$. The minus intervals cover $(1/2, 2/3]$. Thus the two ladders cover every non-reciprocal beta in $(1/2,1)$.

$$\begin{aligned} \text{Union}_{\{t \geq 3\}} ((t-1)/t, \quad t/(t+1)] &= (2/3, 1), \\ \text{Union}_{\{r \geq 2\}} ((r+1)/(2r+1), \quad r/(2r-1)] &= (1/2, 2/3]. \end{aligned}$$

22.3 Watcher-Leaf Möbius Transform

Let a tree certificate have exchange bar $J=[L,U]$. Fix an integer $m \geq 0$. Attach m new leaves to every watcher in the local watcher union

$$W = N(A) \cup N(B) \cup N((A \cup B) - \{x\}).$$

Inflate each certified cave S by adding all new leaves attached to $N(S)$. If $\rho(S)=p/k$, the inflated cave has the same p watchers but $k+mp$ cave vertices, so its ratio becomes

$$T_m(\rho) = \rho / (1 + m \rho).$$

Möbius Leaf Law. The transformed tree has an exchange bar

$$T_m(J) = (T_m(L), T_m(U)], \quad T_m(z) = z / (1 + mz).$$

Proof sketch. The certified sets A , B , and U transform exactly because their watcher sets are unchanged while each watcher contributes m cave leaves. For proper subsets, the added leaf packets cannot introduce a lower exposure gate inside the transformed interval: a packet attached to h watchers contributes at most mh leaves and therefore has ratio at least $1/m$, while every transformed target lies below $1/m$ for $m > 0$. Projecting any mixed subset back to the original certificate gives the old proper-subset lower bound, then applying T_m gives the new bound. The graph remains a tree because only pendant leaves are added.

This is the goblin version of a continued-fraction move. It sends the open interval $(1/2, 1)$ onto

$$T_m((1/2, 1)) = (1/(m+2), 1/(m+1)).$$

22.4 The compiler

Given a non-reciprocal α in $(0, 1)$, choose the unique integer m such that α lies strictly between consecutive reciprocals:

$$1/(m+2) < \alpha < 1/(m+1),$$

equivalently $m = \text{floor}(1/\alpha) - 1$.

Pull α back to β in $(1/2, 1)$:

$$\beta = \alpha / (1 - m \alpha).$$

Choose the plus ladder if $\beta > 2/3$, and the minus ladder if $\beta \leq 2/3$. Apply the watcher-leaf transform T_m . The resulting tree has an exchange bar containing α , so I_α is nonmatroidal.

This resolves the Round VII Arboreal Universality target constructively. It also gives explicit tree-size upper bounds for the matroidality radius.

22.5 Radius equations from the compiler

If alpha is captured by a plus ladder parameter t after an m-fold watcher-leaf transform, then

$$\text{bar} = ((t-1)/((m+1)t - m), \quad t/((m+1)t + 1)],$$

$$m_T(\alpha) \leq (m+2)t + 2.$$

If alpha is captured by a minus ladder parameter r after an m-fold transform, then

$$\text{bar} = ((r+1)/((m+2)r + m + 1), \quad r/((m+2)r - 1)],$$

$$m_T(\alpha) \leq (m+3)r + 2m + 5.$$

These are not claimed to be optimal. They are constructive upper bounds. The exact radius $m_T(\alpha)$ remains a sharper optimization problem, but now every non-reciprocal threshold has an explicit acyclic witness.

Constructive reciprocal moat reading. For a fixed vertex budget N, only ladder parameters up to $O(N)$ are visible. Therefore thresholds within $O(1/N)$ of a reciprocal may be invisible to this compiler. The exact finite moat remains open: determine the largest interval around each $1/q$ containing no exchange bar on any tree with at most N vertices.

22.6 Exact computational checks

The accompanying script uses exact rational arithmetic. For trees with at most 20 vertices it also enumerates every subset, computes all active blockers at the endpoint alpha, and confirms a circuit-elimination failure.

family	parameter	n	computed exchange bar	exact blocker check
plus	t=3	8	(2/3, 3/4]	2 blockers; nonmatroid
plus	t=4	10	(3/4, 4/5]	2 blockers; nonmatroid
plus	t=5	12	(4/5, 5/6]	2 blockers; nonmatroid
plus	t=6	14	(5/6, 6/7]	2 blockers; nonmatroid
minus	r=2	11	(3/5, 2/3]	2 blockers; nonmatroid
minus	r=3	14	(4/7, 3/5]	2 blockers; nonmatroid
minus	r=4	17	(5/9, 4/7]	2 blockers; nonmatroid
minus	r=5	20	(6/11, 5/9]	2 blockers; nonmatroid
T_m plus t=3	m=1	11	(2/5, 3/7]	2 blockers; nonmatroid
T_m plus t=3	m=2	14	(2/7, 3/10]	2 blockers; nonmatroid
T_m plus t=3	m=3	17	(2/9, 3/13]	2 blockers; nonmatroid

22.7 Compiler examples

The following examples illustrate the full pipeline. The value beta is the pullback $\alpha/(1-m\alpha)$.

alpha	m	beta	base family	n	bar containing alpha
5/6	0	5/6	plus t=5	12	(4/5, 5/6]
4/5	0	4/5	plus t=4	10	(3/4, 4/5]
3/4	0	3/4	plus t=3	8	(2/3, 3/4]
2/3	0	2/3	minus r=2	11	(3/5, 2/3]
3/5	0	3/5	minus r=3	14	(4/7, 3/5]
4/7	0	4/7	minus r=4	17	(5/9, 4/7]
3/10	2	3/4	plus t=3	14	(2/7, 3/10]
2/5	1	2/3	minus r=2	15	(3/8, 2/5]
5/17	2	5/7	plus t=3	14	(2/7, 3/10]

Notably, $\alpha=4/7$ now has a 17-vertex tree witness from the minus ladder, and $\alpha=3/10$ is exactly the two-leaf transform of the eight-vertex $(2/3, 3/4]$ witness. This explains the previously isolated sparse examples as members of a single continued-fraction family.

22.8 Applications

Universal sparse stress-test generator. Any non-reciprocal policy threshold can now be compiled into an acyclic adversarial instance. This is useful for testing algorithms that assume greedy or matroid-like augmentation.

Threshold engineering. The only thresholds that are exchange-safe across all trees are reciprocal thresholds. Moving from $1/q$ to a nearby non-reciprocal value may be numerically tiny but structurally decisive.

Audit planning. The radius equations tell how large a tree benchmark must be before a chosen non-reciprocal threshold is guaranteed to show an exchange demon by this compiler. They also give explicit finite test cases rather than random search.

Continued-fraction diagnostics. The map $T_m(z)=z/(1+mz)$ is a continued-fraction step. This suggests that the complexity of alpha as a continued fraction controls the size and shape of its smallest exchange witness.

22.9 New targets after Round VIII

1. **Exact radius problem.** Determine whether the compiler bounds for $m_T(\alpha)$ are optimal, and find the true finite moat around each reciprocal $1/q$ for trees of size at most N .
2. **Exchange entropy under Möbius transforms.** If a graph has many exchange bars, watcher-leaf transforms generate many bars in lower reciprocal bands. Quantify how $X_n(\alpha)$ scales under this operation.
3. **Minimal continued-fraction compiler.** Optimize over all continued-fraction decompositions of alpha to minimize witness size, not merely to prove existence.
4. **Combined demon products.** Fuse the arboreal exchange compiler with residue atoms from Round VI to produce trees that simultaneously exhibit exchange failure and passive-active defect.
5. **Algorithmic separation.** Build a polynomial-time oracle that either certifies a threshold lies outside all exchange bars of a given graph or returns an explicit pair of caves A, B, x .

22.10 Round VIII summary

The Arboreal Universality conjecture is now constructively resolved: every non-reciprocal α has a finite tree witness.

Two base ladders cover all β in $(1/2, 1)$: plus bars $((t-1)/t, t/(t+1)]$ and minus bars $((r+1)/(2r+1), r/(2r-1)]$.

The watcher-leaf Möbius transform maps exchange bars by $T_m(z) = z/(1+mz)$, dropping them into any reciprocal band.

The compiler gives explicit upper bounds for $m_T(\alpha)$ and produces reproducible acyclic benchmarks.

The next frontier is optimality: exact radius, finite reciprocal moats, and exchange entropy.

Goblin translation: a non-reciprocal threshold cannot hide in a forest. The wizard can always grow enough watcher-leaves to bend a high-threshold exchange trap down into its reciprocal band.

Round IX — Finite Reciprocal Moats and Thorn Hinges

Continuation of Goblin Shadow Theory

Status. The previous rounds separated reciprocal thresholds from all other thresholds: at $\alpha = 1/q$ the safe-core system is a transversal matroid, while non-reciprocal thresholds can fail exchange even on trees. This round asks a finite-size question. Are reciprocal thresholds isolated only as exact algebraic points, or does a finite graph of size n have a genuine protective interval around $1/q$? The answer found here is yes: reciprocal thresholds have a finite moat of width on the order of $1/(qn)$, and this order is sharp up to a small constant.

23.1 The no-tangent lemma

For a nonempty set $S \subseteq V(G)$ write

$$\rho(S) = \frac{|N(S)|}{|S|}.$$

Round VII introduced exchange bars

$$J_G(A, B, x) = (\max\{\rho(A), \rho(B)\}, \min\{\bar{\rho}(A), \bar{\rho}(B), \eta((A \cup B) \setminus \{x\})\}],$$

where $x \in A \cap B$. A threshold is nonmatroidal exactly when it lies in at least one exchange bar.

Lemma 1 (No tangent to a reciprocal gate). *No exchange bar has left endpoint exactly $1/q$ and positive length. Also no exchange bar has right endpoint $1/q$.*

Proof sketch. The closed-right assertion is immediate from the reciprocal matroid law: if the right endpoint were $1/q$, then $1/q$ would lie in an exchange bar, contradicting universal matroidality at $1/q$.

For the open-left case define the integer slack

$$h_q(S) = q|N(S)| - |S|.$$

Suppose, for contradiction, that $J_G(A, B, x) = (1/q, U]$ with $U > 1/q$. Then $h_q(A) \leq 0$ and $h_q(B) \leq 0$, while every nonempty proper subset of A or B has positive h_q -slack. In particular $A \cap B$ is a proper nonempty set with $h_q(A \cap B) > 0$. Since h_q is submodular,

$$h_q(A \cup B) \leq h_q(A) + h_q(B) - h_q(A \cap B) < 0.$$

Put $R = (A \cup B) \setminus \{x\}$. The exchange-bar upper condition says every nonempty subset of R has ratio $> 1/q$, so $h_q(R) > 0$. But adding one vertex changes slack by

$$h_q(R \cup \{x\}) = h_q(R) + q(|N(R \cup \{x\})| - |N(R)|) - 1.$$

The neighborhood gain is a nonnegative integer. Since $h_q(R) \geq 1$, the right side is nonnegative, contradicting $h_q(A \cup B) < 0$. $\square$

23.2 Finite Reciprocal Moat Law

For each n and q , define the exact rational gaps

$$\Delta_n^+(q) = \min_{1 \leq k \leq n} \left(\frac{\lfloor k/q \rfloor + 1}{k} - \frac{1}{q} \right),$$

$$\Delta_n^-(q) = \min_{1 \leq k \leq n, \lfloor (k-1)/q \rfloor > 0} \left(\frac{1}{q} - \frac{\lfloor (k-1)/q \rfloor}{k} \right).$$

These are the nearest possible one-sided distances from $1/q$ among ratios p/k with denominator at most n .

Theorem 1 (Finite Reciprocal Moat Law). *For every n -vertex graph G , every integer $q \geq 1$, and every threshold*

$$\alpha \in \left(\frac{1}{q} - \Delta_n^-(q), \frac{1}{q} + \Delta_n^+(q) \right),$$

the safe-core independence system $\mathcal{I}_\alpha(G)$ is matroidal with respect to exchange-bar obstructions. In particular the coarser but simpler universal moat

$$\left| \alpha - \frac{1}{q} \right| < \frac{1}{qn}$$

contains no exchange-bar failure on any n -vertex graph.

Proof sketch. If an exchange bar lies below $1/q$, its right endpoint is a ratio $p/k < 1/q$ with $k \leq n$. If it lies above $1/q$, its left endpoint is a ratio $p/k > 1/q$ with $k \leq n$. The no-tangent lemma rules out equality at $1/q$. The exact gaps $\Delta_n^\pm(q)$ are therefore excluded. The coarse bound follows from

$$\left| \frac{p}{k} - \frac{1}{q} \right| = \frac{|qp - k|}{qk} \geq \frac{1}{qn}$$

whenever $p/k \neq 1/q$ and $k \leq n$. □

Interpretation. Reciprocal thresholds are not merely algebraic miracles. A finite system has a real tolerance band around $1/q$. If a policy threshold is rounded, perturbed, or measured with error less than $1/(qn)$, no exchange demon can appear in a graph of size n .

n	$q = 1$	$q = 2$	$q = 3$	$q = 4$	$q = 5$
8	$\pm 1/8$	$\pm 1/14$	$-1/21, +1/24$	$-1/20, +1/28$	$-1/30, +1/20$
12	$\pm 1/12$	$\pm 1/22$	$-1/30, +1/33$	$-1/36, +1/44$	$-1/55, +1/45$
20	$\pm 1/20$	$\pm 1/38$	$-1/57, +1/60$	$-1/68, +1/76$	$-1/80, +1/95$
50	$\pm 1/50$	$\pm 1/98$	$-1/147, +1/150$	$-1/196, +1/188$	$-1/230, +1/245$

23.3 Thorn Hinge Law: the moat is sharp at scale $1/n$

The moat cannot be widened by more than a constant factor. The following dense connected theta/comb hinges approach $1/q$ from both sides.

Theorem 2 (Comb-Thorn Hinge Law). *Let $q \geq 2$ and $a \geq 2$. There are connected graphs with exchange bars*

$$J_{q,a}^+ = \left(\frac{a+1}{qa+q-1}, \frac{a}{qa-1} \right]$$

above $1/q$, with

$$n_{q,a}^+ = (q+1)a + 2q + 1,$$

and exchange bars

$$J_{q,a}^- = \left(\frac{p}{qp+1}, \frac{a}{qa+1} \right], \quad p = \frac{a+t}{2}, \quad t = \begin{cases} 1, & a \text{ odd}, \\ 2, & a \text{ even}, \end{cases}$$

below $1/q$, with

$$n_{q,a}^- = (q+1)a + 2.$$

Consequently

$$n_{q,a}^+ \left(\frac{a}{qa-1} - \frac{1}{q} \right) \rightarrow \frac{q+1}{q^2}, \quad n_{q,a}^- \left(\frac{1}{q} - \frac{a}{qa+1} \right) \rightarrow \frac{q+1}{q^2}.$$

Proof sketch. Use the Round IV core-hinge equation

$$J(k, p, s, t) = \left(\frac{p}{k}, \min \left\{ \frac{p}{k-1}, \frac{t}{s}, \frac{2p-t}{2k-s-1} \right\} \right].$$

For the upper thorn set

$$\begin{aligned} k &= qa + q - 1, & p &= a + 1, \\ s &= qa - 1, & t &= a. \end{aligned}$$

For the lower thorn set choose t and p as above, then set

$$s = qt, \quad k = qp + 1.$$

Substitution gives the displayed intervals. The vertex counts are the theta-hinge counts $n = 2k - s + 2p - t$. Exact bitmask enumeration for small cases confirms that the two intended caves are the only active blockers at the displayed upper endpoint. $\square$

q	a	side	n	exchange bar
2	3	+	14	$(4/7, 3/5]$
2	3	−	11	$(2/5, 3/7]$
3	3	+	19	$(4/11, 3/8]$
3	3	−	14	$(2/7, 3/10]$
4	3	+	24	$(4/15, 3/11]$
4	3	−	17	$(2/9, 3/13]$
5	3	+	29	$(4/19, 3/14]$
5	3	−	20	$(2/11, 3/16]$

For $q = 1$, the high ladder from Round VIII gives

$$J_r = \left(\frac{r-2}{r-1}, \frac{r-1}{r} \right], \quad n = 2r,$$

so $n(1 - (r-1)/r) = 2$. Thus the one-mana gate also has a finite moat, with thorns approaching it at scale $2/n$.

23.4 What is exceptional here?

The exceptional point is the coexistence of two facts that normally pull against each other:

1. **Finite protection.** A reciprocal threshold $1/q$ has a graph-size-dependent moat. No exchange demon can be tangent to the reciprocal point, and denominators at most n force a hard gap.
2. **Asymptotic fragility.** The thorn hinges approach the moat boundary at order $1/n$. Large systems can place nonmatroidality arbitrarily close to $1/q$, but they must pay approximately inverse-distance size.

This creates a new law of threshold engineering:

$$\text{to break exchange within } \varepsilon \text{ of } 1/q, \quad n = \Theta \left(\frac{1}{q\varepsilon} \right) \text{ vertices are necessary,}$$

and the comb-thorn constructions give explicit witnesses with

$$n \approx \frac{q+1}{q^2\varepsilon}.$$

23.5 Applications and new targets

Robust reciprocal design. If a network has at most n vertices, selecting $\alpha = 1/q$ gives not only matroid algorithms at the exact threshold but also a certified perturbation band of radius at least $1/(qn)$. Inside this band, greedy safe-core logic cannot fail through exchange-bar pathology.

Adversarial benchmark generation. The thorn formulas are an inverse compiler: given a desired tolerance ε near $1/q$, choose $a \approx 1/(q\varepsilon)$ and output a connected graph whose exchange bar lies within that tolerance.

New target: the Moat Constant Problem. Define

$$M_n(q) = \min \left\{ \left| \alpha - \frac{1}{q} \right| : \exists G, |V(G)| = n, \mathcal{I}_\alpha(G) \text{ nonmatroidal} \right\}.$$

We now know

$$\frac{1}{qn} \leq M_n(q) \lesssim \frac{q+1}{q^2 n}.$$

Determine the limit of $nM_n(q)$, separately for all graphs, connected graphs, and trees.

New target: endpoint arithmetic. The no-tangent lemma suggests a wider endpoint exclusion theory: characterize all rational numbers r for which exchange bars cannot start exactly at r . Reciprocals are forbidden; non-reciprocals are allowed by the previous hinge compilers. This may be the arithmetic skeleton behind universal matroidality.

New target: tree moat constants. Round VIII proved that trees can realize non-reciprocal failures. The current thorns are dense connected hinges. The next problem is to build one-sided tree thorns with the same $1/n$ constants or prove a larger tree-only constant.

23.6 Round IX summary

Round IX adds the *finite reciprocal moat*: for an n -vertex graph, exchange failure cannot occur inside a reciprocal tolerance band of width at least $1/(qn)$ around $1/q$. The proof is a small integer-slack spell: an exchange bar cannot be tangent to a reciprocal point, and all remaining endpoints have bounded denominators. The moat is nearly sharp, because comb-thorn hinges approach both sides of $1/q$ at order $(q+1)/(q^2 n)$.

Goblin translation. A reciprocal gate is protected by a moat. But the moat has thorns: the larger the kingdom, the closer the goblin can plant a nonmatroidal cave-pair to the gate without touching it.

24 Round X: the reciprocal moat constant

Round IX left a precise target: determine the constant hidden in the reciprocal moat problem. A reciprocal threshold $1/q$ is universally matroidal, but exchange bars can approach it from both sides at distance of order $1/n$. The new discovery of this round is that the same constant that appeared in the connected comb-thorn hinges is already encoded in the two arboreal ladders of Round VIII. In short: the thorns are not dense accidents; they are Moebius images of trees.

For an exchange bar $J = (L, U]$ and integer $q \geq 2$, define the one-sided finite tree moats

$$M_{\leq n}^{T,+}(q) = \min\{L - 1/q : |V(T)| \leq n, J_T = (L, U], L > 1/q\},$$

$$M_{\leq n}^{T,-}(q) = \min\{1/q - U : |V(T)| \leq n, J_T = (L, U], U < 1/q\}.$$

The same definitions without the superscript T range over all finite graphs. The “ $\leq n$ ” version avoids harmless padding issues; it has the same asymptotic content as the exact- n version.

24.1 The Moebius moat equations

Recall the watcher-leaf transform

$$T_m(z) = \frac{z}{1 + mz}.$$

Round VIII produced two tree ladders:

$$P_t : \left(\frac{t-1}{t}, \frac{t}{t+1} \right], \quad t \geq 3,$$

$$M_r : \left(\frac{r+1}{2r+1}, \frac{r}{2r-1} \right], \quad r \geq 2.$$

Applying the watcher-leaf transform gives two reciprocal-moat families:

$$J_{q,r}^+ = T_{q-2}(M_r) = \left(\frac{r+1}{qr+q-1}, \frac{r}{qr-1} \right], \quad |V| = (q+1)r + 2q + 1,$$

$$J_{q,t}^- = T_{q-1}(P_t) = \left(\frac{t-1}{qt-q+1}, \frac{t}{qt+1} \right], \quad |V| = (q+1)t + 2.$$

The first family approaches $1/q$ from above; the second approaches $1/q$ from below. Their exact scaled gaps are

$$\begin{aligned} ((q+1)r + 2q + 1) \left(\frac{r+1}{qr+q-1} - \frac{1}{q} \right) &= \frac{(q+1)r + 2q + 1}{q(qr+q-1)}, \\ ((q+1)t + 2) \left(\frac{1}{q} - \frac{t}{qt+1} \right) &= \frac{(q+1)t + 2}{q(qt+1)}. \end{aligned}$$

Therefore both sides have the same limiting constant:

$$\limsup_{n \rightarrow \infty} n M_{\leq n}^{T,+}(q) \leq \frac{q+1}{q^2}, \quad \limsup_{n \rightarrow \infty} n M_{\leq n}^{T,-}(q) \leq \frac{q+1}{q^2}.$$

Since trees are graphs, the same upper bounds hold for all graphs.

24.2 Above-side moat constant theorem

The above side can now be solved, not merely constructed.

Theorem 24.1 (Above-side reciprocal moat constant). *For every fixed $q \geq 2$,*

$$\lim_{n \rightarrow \infty} nM_{\leq n}^{+,T}(q) = \lim_{n \rightarrow \infty} nM_{\leq n}^+(q) = \frac{q+1}{q^2}.$$

Proof sketch. Let $J = (L, U]$ be an exchange bar above $1/q$. One of its two blocker caves, call it C , has ratio $L = |N(C)|/|C| = p/k > 1/q$. Choose any $\alpha \in (L, U]$. Then C is a minimal α -bad cave, so by the cave-kernel theorem it is independent. Hence $C \cap N(C) = \emptyset$, and the graph has at least $k + p$ vertices. Since $p/k > 1/q$, integer arithmetic gives

$$L - \frac{1}{q} = \frac{qp - k}{qk} \geq \frac{1}{qk}.$$

Also $n \geq k + p > k + k/q$, so $k < qn/(q+1)$. Therefore every above-side bar satisfies

$$L - \frac{1}{q} > \frac{q+1}{q^2 n}.$$

The tree family $J_{q,r}^+$ attains this bound asymptotically. Thus the limit exists and equals $(q+1)/q^2$. $\square$

This is the cleanest law of the round. The integer gap $qp - k \geq 1$ is the numerator spell, while the disjoint cave-plus-watcher count $n \geq k + p$ is the forest wall. The two together force the moat constant.

24.3 Lower-side evidence and the endpoint gap

The lower side is subtler because the top endpoint U of a bar below $1/q$ can be set by a deletion endpoint of one cave or by the exposure of the union-with-one-shared vertex removed. The Moebius tree family $J_{q,t}^-$ proves that

$$\limsup_{n \rightarrow \infty} nM_{\leq n}^{T,-}(q) \leq \frac{q+1}{q^2},$$

but a full lower bound for arbitrary lower-side exchange bars remains open.

There is, however, a solved subcase. Suppose $U = p/(k-1) < 1/q$ is the one-vertex deletion endpoint of a cave of size k with p watchers. Put $h = (k-1) - qp \geq 1$. Then

$$\frac{1}{q} - U = \frac{h}{q(k-1)}, \quad n \geq k + p = k + \frac{k-1-h}{q}.$$

The product is minimized at $h = 1$, giving asymptotic constant $(q+1)/q^2$. Thus any counterexample to the symmetric moat constant must be union-endpoint controlled, not ordinary deletion-endpoint controlled.

Conjecture 24.2 (Symmetric tree moat constant). *For every $q \geq 2$,*

$$\lim_{n \rightarrow \infty} nM_{\leq n}^{T,-}(q) = \frac{q+1}{q^2}.$$

Equivalently, the Moebius plus ladders are asymptotically optimal lower-side tree thorns.

24.4 Exact computations

The accompanying script constructs the Moebius tree witnesses, verifies their exchange bars by exact subset enumeration when the tree is small enough, and scans all nonisomorphic trees through 11 vertices near $1/2$. Sample witness rows are:

q	side	$ V $	exchange bar J	gap to $1/q$	$ V \cdot \text{gap}$
2	above	11	$(3/5, 2/3]$	$1/10$	$11/10$
2	below	11	$(2/5, 3/7]$	$1/14$	$11/14$
3	above	15	$(3/8, 2/5]$	$1/24$	$5/8$
3	below	14	$(2/7, 3/10]$	$1/30$	$7/15$
4	above	19	$(3/11, 2/7]$	$1/44$	$19/44$
4	below	17	$(2/9, 3/13]$	$1/52$	$17/52$

For $q = 2$, exhaustive tree census gave:

n	trees	best above gap	best below gap	exchange bars	crossings of $1/2$
8	23	$1/6$	none	1	0
9	47	$1/6$	none	4	0
10	106	$1/6$	none	18	0
11	235	$1/10$	$1/14$	73	0

The no-crossing column is a small computational shadow of the no-tangent lemma: no exchange bar in the scanned trees contains the reciprocal threshold itself.

24.5 Applications and new targets

Threshold engineering. The above-side theorem gives a sharp finite-size rule. To force exchange nonmatroidality at distance ε above $1/q$, a system needs at least approximately

$$n \geq \frac{q+1}{q^2\varepsilon}$$

vertices, and the tree compiler meets this order. Reciprocal thresholds are therefore not only algebraically special; they have a measurable finite-size safety moat.

Sparse adversarial testing. The formulas $J_{q,r}^+$ and $J_{q,t}^-$ give explicit acyclic stress tests for algorithms. A greedy safe-core method that claims to behave matroidally near $1/q$ should be tested against these two ladders before being trusted on larger networks.

New target: the lower moat theorem. Prove or refute the symmetric constant conjecture. Any refutation must find a union-endpoint controlled lower-side bar with scaled gap smaller than $(q+1)/q^2$.

New target: finite floor law. Let

$$r_n^+(q) = \left\lfloor \frac{n-2q-1}{q+1} \right\rfloor, \quad t_n^-(q) = \left\lfloor \frac{n-2}{q+1} \right\rfloor.$$

The Moebius ladders give explicit finite upper bounds

$$M_{\leq n}^{T,+}(q) \leq \frac{1}{q(qr_n^+(q) + q - 1)}, \quad M_{\leq n}^{T,-}(q) \leq \frac{1}{q(qt_n^-(q) + 1)}.$$

Determine when these are exact.

New target: reciprocal moat spectra. Instead of the nearest bar, study the whole point process

$$\text{Moat}_q(G) = \{n(L - 1/q) : L > 1/q\} \cup \{n(1/q - U) : U < 1/q\}$$

formed by exchange-bar endpoints near $1/q$. The first atom is now understood on the above side; the distribution of later atoms may encode a new arithmetic invariant of trees.

24.6 Round X summary

Round X turns the moat problem from a qualitative statement into a sharp constant on one side. The key equation is

$$\boxed{\lim_{n \rightarrow \infty} nM_{\leq n}^+(q) = \frac{q+1}{q^2}}.$$

The witnesses are trees, obtained by applying the watcher-leaf Moebius transform to the minus ladder. The lower side has matching tree constructions and strong small-tree evidence, but the general lower bound is still open because union-endpoint bars may be more slippery than deletion-endpoint bars.

Goblin translation. A reciprocal gate has a moat of width about $(q+1)/(q^2n)$. Above the gate, no goblin can place a nonmatroidal thorn closer than that. Below the gate, the known tree thorns reach the same distance, and the next hunt is for a stranger thorn that might do better.

25 Round XI: Ghost moats and the two-sided reciprocal constant

Round X solved the above-side reciprocal moat constant and left one precise target: prove or refute the lower-side constant. The new observation is that lower-side exchange bars carry a hidden spare vertex. I call it a *ghost*. This ghost turns the lower-side asymptotic evidence into an exact finite floor, apart from the small initial cases before the ladder starts.

For an exchange certificate (A, B, x) , recall that

$$J_G(A, B, x) = (L, U] = (\max\{\rho(A), \rho(B)\}, \min\{\bar{\rho}(A), \bar{\rho}(B), \eta((A \cup B) \setminus \{x\})\}],$$

where $\rho(S) = |N(S)|/|S|$. A bar below $1/q$ means $U < 1/q$.

25.1 The ghost endpoint lemma

Lemma 25.1 (Purified endpoint). *Let R be a set with $\rho(R) < 1$ that minimizes ρ among all nonempty subsets of an ambient set W . Then R is independent.*

Proof. Let $I \subseteq R$ be the vertices with no neighbor inside R , and let $D = R \setminus I$. Every vertex of D lies in $N(R)$. If D is nonempty, then

$$|N(I)| \leq |N(R)| - |D| < \rho(R)|R| - \rho(R)|D| = \rho(R)|I|,$$

because $\rho(R) < 1$. Hence $\rho(I) < \rho(R)$, contradicting minimality. Thus $D = \emptyset$ and R is independent. $\square$

Lemma 25.2 (Ghost endpoint lemma). *Let $J_G(A, B, x) = (L, U]$ be an exchange bar with $U < 1$. Then there is a nonempty set R and a vertex z such that*

$$\rho(R) = U, \quad R \cap N(R) = \emptyset, \quad z \notin R \cup N(R).$$

Moreover, if U is controlled by a proper-subset endpoint of A or B , then z may be chosen as an omitted vertex of that cave. If U is controlled by the union endpoint $\eta((A \cup B) \setminus \{x\})$, then $z = x$ works.

Proof. The endpoint U is the minimum of three quantities: the best proper-subset ratio inside A , the best proper-subset ratio inside B , and the best subset ratio inside $(A \cup B) \setminus \{x\}$. Choose a subset R attaining whichever minimum is active. By the purified endpoint lemma, R is independent.

If $R \subsetneq A$, pick $z \in A \setminus R$. Since A is a minimal bad cave at thresholds in the bar, the cave-kernel theorem makes A independent; hence z is neither in R nor adjacent to R . The same argument applies inside B . If the union endpoint is active, then $R \subseteq (A \cup B) \setminus \{x\}$, while $x \in A \cap B$. Again A and B are independent, so x is not adjacent to any vertex of $A \cup B$; hence $x \notin R \cup N(R)$. $\square$

This is the exceptional find of the round: below a reciprocal gate, an exchange endpoint must leave one vertex outside its shadow. Above the gate, a bad cave pays for itself and its watchers. Below the gate, the endpoint that keeps the union safe must also leave behind a ghost.

25.2 Exact lower-side finite floor

For $q \geq 2$, define the lower nearest-bar quantity

$$M_{\leq n}^-(q) = \inf \left\{ \frac{1}{q} - U : |V(G)| \leq n, (L, U] \text{ is an exchange bar of } G, U < \frac{1}{q} \right\},$$

with the value $+\infty$ if there is no such bar. Let $M_{\leq n}^{-,T}(q)$ be the same quantity restricted to trees.

Theorem 25.3 (Ghost-moat floor). *Let $q \geq 2$ and let*

$$t_n(q) = \left\lfloor \frac{n-2}{q+1} \right\rfloor.$$

Every n -vertex graph exchange bar below $1/q$ satisfies

$$\frac{1}{q} - U \geq \frac{1}{q(qt_n(q) + 1)}.$$

Consequently,

$$M_{\leq n}^-(q) \geq \frac{1}{q(qt_n(q) + 1)}.$$

Proof. Apply the ghost endpoint lemma to obtain R and z . Put $k = |R|$ and $p = |N(R)|$. Since R is independent and $z \notin R \cup N(R)$,

$$n \geq k + p + 1.$$

Because $U = p/k < 1/q$, write

$$k = qp + h, \quad h \geq 1.$$

Then

$$n \geq (q+1)p + h + 1,$$

so, in the extremal case $h = 1$, one has $p \leq t_n(q)$. The gap is

$$\frac{1}{q} - \frac{p}{k} = \frac{k - qp}{qk} = \frac{h}{q(qp + h)} \geq \frac{1}{q(qt_n(q) + 1)}.$$

□

The Round X lower Moebius ladder reaches this bound. For $t \geq 3$ it has

$$|V| = (q+1)t + 2, \quad J_{q,t}^- = \left(\frac{t-1}{qt - q + 1}, \frac{t}{qt + 1} \right],$$

and therefore

$$\frac{1}{q} - \frac{t}{qt + 1} = \frac{1}{q(qt + 1)}.$$

Thus, for all n with $t_n(q) \geq 3$,

$$\boxed{M_{\leq n}^{-,T}(q) = M_{\leq n}^-(q) = \frac{1}{q(q \lfloor (n-2)/(q+1) \rfloor + 1)}}.$$

The lower-side finite floor law is exact, and it is already attained by trees.

25.3 Two-sided reciprocal moat constant

Combining the ghost-moat floor with the above-side theorem from Round X gives the full constant.

Theorem 25.4 (Two-sided reciprocal moat constant). *For every fixed $q \geq 2$,*

$$\lim_{n \rightarrow \infty} nM_{\leq n}^{+,T}(q) = \lim_{n \rightarrow \infty} nM_{\leq n}^+(q) = \lim_{n \rightarrow \infty} nM_{\leq n}^{-,T}(q) = \lim_{n \rightarrow \infty} nM_{\leq n}^-(q) = \frac{q+1}{q^2}.$$

Proof. The above-side equality was proved in Round X by the integer gap $qp - k \geq 1$ and the cave-plus-watcher count $n \geq k + p$, with tree witnesses asymptotically attaining the bound. The lower-side equality follows from the exact ghost-moat floor above and the trees $J_{q,t}^-$. Since

$$n = (q + 1)t + 2, \quad n \left(\frac{1}{q} - \frac{t}{qt + 1} \right) \rightarrow \frac{q + 1}{q^2},$$

the lower tree construction attains the same constant. $\square$

In one sentence: reciprocal thresholds have a two-sided nonmatroidality moat of width asymptotic to

$$\frac{q + 1}{q^2 n}.$$

The moat is not merely philosophical. It is an integer-ratio phenomenon, and trees already saturate it.

25.4 Exact computation

The accompanying script verifies the lower tree witnesses and extracts the endpoint ghost. Representative rows are:

q	t	$ V $	lower bar $(L, U]$	gap $1/q - U$	witness (k, p)	$k + p + 1$
2	3	11	(2/5, 3/7]	1/14	(7, 3)	11
2	4	14	(3/7, 4/9]	1/18	(9, 4)	14
3	3	14	(2/7, 3/10]	1/30	(10, 3)	14
3	4	18	(3/10, 4/13]	1/39	(13, 4)	18
4	3	17	(2/9, 3/13]	1/52	(13, 3)	17
5	3	20	(2/11, 3/16]	1/80	(16, 3)	20

For $q = 2$, the small nonisomorphic-tree census near $1/2$ found no violation of the floor:

n	trees	exchange bars	best lower gap	floor	violations
8	23	1	none	1/10	0
9	47	4	none	1/10	0
10	106	18	none	1/10	0
11	235	73	1/14	1/14	0

The exact script uses no floating point arithmetic; all comparisons are integer or rational comparisons.

25.5 Applications and new targets

Certified reciprocal safety bands. For systems with at most n vertices, no lower-side exchange-bar failure can occur in

$$\left(\frac{1}{q} - \frac{1}{q(q \lfloor (n - 2)/(q + 1) \rfloor + 1)}, \frac{1}{q} \right).$$

Together with the Round X above-side bound, this gives a finite-size reciprocal safety band for threshold engineering. If a policy threshold must be near $1/q$, the moat equation estimates how large an adversarial system must be before greedy exchange can fail.

Sparse adversarial generation. The exact lower ladder gives the smallest known systematic sparse stress tests below $1/q$. Given q and a desired distance ε , choose approximately

$$t \approx \frac{1}{q^2 \varepsilon}, \quad n \approx \frac{q+1}{q^2 \varepsilon},$$

and build $J_{q,t}^-$.

New target: above finite floor. The lower side is now exact for all large enough n . The above side still has an asymptotic theorem but not an exact floor law. Determine $M_{\leq n}^{+,T}(q)$ and $M_{\leq n}^+(q)$ exactly.

New target: ghost spectrum. For a lower endpoint, define $\gamma(J)$ as the maximum number of vertices that can be certified outside $R \cup N(R)$ for endpoint witnesses R . The nearest moat has $\gamma = 1$. Bars with $\gamma \geq 2$ should form second-order moat layers.

New target: reciprocal moat point processes. The nearest-bar constant is only the first atom. Study the scaled endpoint clouds

$$\{n(L - 1/q) : L > 1/q\} \cup \{n(1/q - U) : U < 1/q\}.$$

The ghost law predicts a residue-structured lower cloud with steps controlled by $\lfloor (n-2)/(q+1) \rfloor$.

25.6 Round XI summary

Round XI resolves the open lower-side moat question from Round X. The key discovery is the ghost endpoint lemma: every lower exchange endpoint has an independent ratio witness plus one invisible spare vertex. That extra vertex changes the finite denominator from $n-1$ to $n-2$ and yields the exact lower floor

$$M_{\leq n}^-(q) = M_{\leq n}^{-,T}(q) = \frac{1}{q(q \lfloor (n-2)/(q+1) \rfloor + 1)}$$

for all sufficiently large n . Consequently the reciprocal moat constant is now two-sided:

$$\boxed{\lim_{n \rightarrow \infty} n M_{\leq n}^{\pm}(q) = \frac{q+1}{q^2}.$$

Goblin translation. Below a reciprocal gate, the goblin cannot place a thorn merely by balancing a cave against its watchers. It must also leave a ghost outside the cave's shadow. That ghost is exactly the missing integer that completes the two-sided moat law.

Round XII. Crown moats and the above-side finite floor

Target. Round XI solved the lower-side reciprocal moat by finding a hidden endpoint “ghost.” The open target was the exact above-side finite floor. The new phenomenon in this round is the apparent dual of a ghost: an above-side exchange endpoint seems to require not one spare vertex, but a whole *free crown* of size $q + 1$ outside the endpoint cave and its watchers. This gives a sharp candidate for the exact finite floor above $1/q$ and a new structural target that is small enough to attack directly.

For a set $C \subseteq V(G)$ define its free crown

$$\mathcal{F}(C) = V(G) \setminus (C \cup N(C)).$$

If $J_G(A, B, x) = (L, U]$ is an exchange bar and $L > 1/q$, call $C \in \{A, B\}$ an above endpoint cave if $\rho(C) = L$.

26.1 The crown endpoint conjecture

Conjecture 26.1 (Crown endpoint conjecture). Let $q \geq 2$. If $J_G(A, B, x) = (L, U]$ is an exchange bar with $L > 1/q$, then at least one above endpoint cave $C \in \{A, B\}$ satisfies

$$|\mathcal{F}(C)| \geq q + 1.$$

This is the exceptional discovery of this round. The lower-side law says an endpoint below $1/q$ carries one ghost vertex outside $R \cup N(R)$. The above-side evidence says an endpoint above $1/q$ carries a full crown of $q + 1$ vertices outside $C \cup N(C)$. The two finite floors then become mirror images:

side	endpoint form	extra outside mass	floor denominator
below $1/q$	$k = qp + 1$	1 ghost	$q(qp + 1)$
above $1/q$	$k = qp - 1$	$q + 1$ crown	$q(qp - 1)$

The conjecture was checked against the exact tree witnesses below, and exact nonisomorphic-tree scans through 12 vertices for $q = 2, 3$ found no counterexample to the resulting floor pattern. I am not marking the all-graph crown statement as proved yet; it is now the main structural target.

26.2 Crown reduction theorem

Define the candidate above-floor integer

$$P_n^+(q) = \left\lfloor \frac{n - q}{q + 1} \right\rfloor.$$

Theorem 26.2 (Crown reduction). Assume the crown endpoint conjecture holds for a class of graphs $\mathcal{C}$. Then every graph $G \in \mathcal{C}$ with at most n vertices and every exchange bar above $1/q$ satisfies

$$L - \frac{1}{q} \geq \frac{1}{q(qP_n^+(q) - 1)}$$

whenever $P_n^+(q) \geq 1$.

Proof. Let C be a crown endpoint cave. Write

$$|C| = k, \quad |N(C)| = p, \quad L = \frac{p}{k} > \frac{1}{q}.$$

Put $d = qp - k \geq 1$. The crown gives

$$n \geq |C| + |N(C)| + |\mathcal{F}(C)| \geq k + p + q + 1.$$

Thus the gap is

$$L - \frac{1}{q} = \frac{p}{k} - \frac{1}{q} = \frac{qp - k}{qk} = \frac{d}{qk}.$$

For fixed n the smallest possible value of $d/(qk)$ under the integer constraints

$$d = qp - k \geq 1, \quad k + p \leq n - q - 1,$$

occurs at $d = 1$ and the largest allowed p , namely

$$p = P_n^+(q) = \left\lfloor \frac{n - q}{q + 1} \right\rfloor, \quad k = qP_n^+(q) - 1.$$

Substitution gives the stated bound.

Interpretation. The above-side finite floor is reduced to one crisp combinatorial statement: prove that an above endpoint cave must carry a $q + 1$ -vertex crown. Everything else is integer arithmetic.

26.3 Crown ladder trees attain the floor

The reduction would be empty without sharp examples. The following trees attain the candidate value.

Theorem 26.3 (Crown ladder attainers). For every $q \geq 2$ and $p \geq 2$ there is a tree $T_{q,p}^+$ on

$$|V(T_{q,p}^+)| = (q + 1)p + q$$

with an exchange bar whose lower endpoint is

$$L = \frac{p}{qp - 1} = \frac{1}{q} + \frac{1}{q(qp - 1)}$$

and whose endpoint cave has crown size exactly $q + 1$.

For $p = 2$ the bar is

$$J_{q,2}^+ = \left(\frac{2}{2q - 1}, \frac{3}{3q - 2} \right].$$

For $p \geq 3$ the bar is

$$J_{q,p}^+ = \left(\frac{p}{qp - 1}, \frac{p - 1}{q(p - 1) - 1} \right].$$

Construction sketch. The $p = 2$ seed is the eight-vertex tree with bar $(2/3, 3/4]$. Attaching $q - 2$ watcher leaves by the Moebius transform $T_m(z) = z/(1 + mz)$ sends it to the bar $J_{q,2}^+$. For $p \geq 3$, start from the Round VIII minus ladder with bar

$$\left(\frac{p}{2p - 1}, \frac{p - 1}{2p - 3} \right]$$

and apply the same watcher-leaf transform with $m = q - 2$. The transform sends z to $z/(1 + (q - 2)z)$, giving the displayed endpoints. Direct counting gives $(q + 1)p + q$ vertices and crown size $q + 1$.

Consequently, if the crown endpoint conjecture is true, then for all n with $P_n^+(q) \geq 2$,

$$M_{\leq n}^+(q) = \frac{1}{q(q \lfloor (n - q)/(q + 1) \rfloor - 1)}.$$

The equality is already attained by trees, not by dense graphs.

26.4 Exact checks

The supplementary script constructs the crown ladder trees and, when the order is small enough for full subset enumeration, verifies the exchange bar and circuit-elimination failure exactly. Representative rows are:

q	p	$ V $	bar $(L, U]$	$L - 1/q$	crown
2	2	8	$(2/3, 3/4]$	1/6	3
2	3	11	$(3/5, 2/3]$	1/10	3
2	4	14	$(4/7, 3/5]$	1/14	3
3	2	11	$(2/5, 3/7]$	1/15	4
3	3	15	$(3/8, 2/5]$	1/24	4
4	2	14	$(2/7, 3/10]$	1/28	5
4	3	19	$(3/11, 2/7]$	1/44	5
5	2	17	$(2/9, 3/13]$	1/45	6

A pleasing corridor effect appears: the same interval can be a lower-side witness for one reciprocal and an above-side witness for the next reciprocal. For example, $(2/5, 3/7]$ is below $1/2$ but above $1/3$. Thus reciprocal moats do not live independently; they tile the threshold line into interacting corridors.

26.5 Applications and new targets

Finite-size threshold engineering. If the crown conjecture holds, then for a system of at most n vertices no above-side exchange-bar failure can occur in the interval

$$\left(\frac{1}{q}, \frac{1}{q} + \frac{1}{q(q \lfloor (n-q)/(q+1) \rfloor - 1)} \right).$$

Together with the Round XI lower floor, this would give a two-sided exact finite moat around every reciprocal threshold.

Sparse stress-test generation. The crown ladders give explicit acyclic adversarial instances at the exact predicted above-floor scale. To test a policy just above $1/q$ at distance about ε , choose

$$p \approx \frac{1}{q^2 \varepsilon}, \quad n \approx \frac{(q+1)}{q^2 \varepsilon},$$

and build $T_{q,p}^+$.

New target: prove the crown endpoint conjecture. The conjecture should be attacked by combining the cross-shadow theorem with reciprocal matroid slack. The desired structural inequality is

$$|V \setminus (C \cup N(C))| \geq q + 1$$

for an above endpoint cave C .

New target: crown spectrum. For an above exchange bar define

$$\kappa_q(J) = \max\{|\mathcal{F}(C)| : \rho(C) = L, C \in \{A, B\}\}.$$

The nearest-floor ladders have $\kappa_q = q + 1$. Bars with $\kappa_q \geq q + 2$ should form second-order above-moat layers, dual to the lower ghost spectrum from Round XI.

New target: reciprocal corridor process. Study the scaled corridor clouds between consecutive reciprocals:

$$\left\{ n \left(L - \frac{1}{q} \right) : L > \frac{1}{q} \right\} \quad \text{and} \quad \left\{ n \left(\frac{1}{q-1} - U \right) : U < \frac{1}{q-1} \right\}.$$

The same bars contribute to both clouds, suggesting a global point process rather than independent moat problems.

26.6 Round XII summary

Round XII reduces the unsolved above-side finite floor to a new structural object: the free crown of an endpoint cave. The proposed crown law says that an exchange bar beginning above $1/q$ must expose at least $q + 1$ vertices outside the endpoint cave and its open neighborhood. Under that single structural statement, the above finite floor becomes

$$M_{\leq n}^+(q) = \frac{1}{q(q \lfloor (n - q)/(q + 1) \rfloor - 1)},$$

for $P_n^+(q) \geq 2$, and the crown ladder trees attain it. The lower moat has a ghost; the upper moat has a crown. The next decisive step is to prove the crown endpoint conjecture or find the first counterexample.

Round XIII: Crown Saturation and Reciprocal Corridor Tiling

Continuing the exchange-bar, ghost-crown, and reciprocal-moat program

Question continued from Round XII. Round XII proposed the above-side Crown Endpoint Conjecture. If an exchange bar

$$J_G(A, B, x) = (L, U]$$

lies in the reciprocal corridor

$$\frac{1}{q} < L \leq \frac{1}{q-1}, \quad q \geq 2,$$

then some lower-endpoint cave $C \in \{A, B\}$ with $|N(C)|/|C| = L$ should have a free crown

$$\mathcal{F}(C) = V \setminus (C \cup N(C)), \quad |\mathcal{F}(C)| \geq q + 1.$$

This round asks whether the crown should grow when the endpoint is farther above $1/q$.

Definition (reciprocal surplus and crown). *For a cave endpoint C in the corridor above $1/q$, put*

$$s_q(C) = q|N(C)| - |C|, \quad c(C) = |V \setminus (C \cup N(C))|.$$

Since $|N(C)|/|C| > 1/q$, the surplus $s_q(C)$ is a positive integer. The natural strengthening of the Crown Endpoint Conjecture would be

$$c(C) \geq q + s_q(C).$$

The new discovery is that this stronger law is false in the sharpest possible way.

Theorem (Crown Saturation Theorem). *For every $q \geq 2$ and every surplus $d \geq 1$, there is a finite tree $S_{q,d}$ with an exchange bar*

$$J_{q,d}^{\text{sat}} = \left(\frac{d+1}{(q-1)d+q}, \frac{d+2}{(q-1)d+2q-1} \right]$$

such that a lower-endpoint cave C satisfies

$$s_q(C) = d, \quad c(C) = q + 1.$$

Thus the Round XII crown lower bound $q + 1$ is saturated for arbitrarily large surplus. No theorem of the form $c(C) \geq q + f(s_q(C))$ with $f(d) \rightarrow \infty$ can be true.

Construction and proof sketch. Start with the plus ladder P_t from Round VIII, whose bar is

$$\left(\frac{t-1}{t}, \frac{t}{t+1} \right], \quad t \geq 3.$$

Set $t = d + 2$. The lower endpoint has watcher count $p = d + 1$, cave size $k = d + 2$, and surplus at $q = 2$ equal to $2p - k = d$. Now apply the watcher-leaf transform

$$T_m(z) = \frac{z}{1 + mz}, \quad m = q - 2.$$

The endpoint p/k becomes

$$\frac{p}{k + (q-2)p} = \frac{d+1}{(q-1)d+q},$$

and the upper endpoint transforms in the same way. The transform adds exactly enough watcher leaves that the free crown becomes $q + 1$, while the reciprocal surplus remains

$$q(d+1) - ((q-1)d+q) = d.$$

The two endpoint caves and their union-minus-shared-vertex retain the same exchange-bar relations, so circuit elimination fails throughout the displayed interval.

Small exact checks. The new script verifies the construction by exhaustive subset enumeration for the generated trees up to 17 vertices. At the right endpoint of each bar, the two active blockers violate circuit elimination.

q	d	$ V $	bar	crown	surplus	exact status
2	1	8	$(2/3, 3/4]$	3	1	2 blockers, nonmatroidal
2	2	10	$(3/4, 4/5]$	3	2	2 blockers, nonmatroidal
2	3	12	$(4/5, 5/6]$	3	3	2 blockers, nonmatroidal
2	4	14	$(5/6, 6/7]$	3	4	2 blockers, nonmatroidal
2	5	16	$(6/7, 7/8]$	3	5	2 blockers, nonmatroidal
3	1	11	$(2/5, 3/7]$	4	1	2 blockers, nonmatroidal
3	2	14	$(3/7, 4/9]$	4	2	2 blockers, nonmatroidal
3	3	17	$(4/9, 5/11]$	4	3	2 blockers, nonmatroidal
4	1	14	$(2/7, 3/10]$	5	1	2 blockers, nonmatroidal
5	1	17	$(2/9, 3/13]$	6	1	2 blockers, nonmatroidal

The smallest surplus-absorption witness. At $q = 2, d = 2$ the saturation theorem gives a ten-vertex tree with edges

$$\{01, 02, 05, 16, 23, 34, 67, 78, 89\}.$$

It has an exchange bar

$$(3/4, 4/5].$$

One lower-endpoint cave is

$$C = \{1, 5, 7, 9\}, \quad N(C) = \{0, 6, 8\}.$$

Hence

$$s_2(C) = 2 \cdot 3 - 4 = 2, \quad \mathcal{F}(C) = \{2, 3, 4\}, \quad c(C) = 3 = q + 1 < q + s_2(C) = 4.$$

The surplus does not require extra free crown. It can be absorbed along the internal ladder. This is the “goblin” mechanism of the round: reciprocal surplus is not spatial excess; it can be stored as path-depth.

Theorem (Reciprocal Corridor Tiling). *For every $q \geq 2$, tree exchange bars tile the full open reciprocal corridor*

$$\left(\frac{1}{q}, \frac{1}{q-1}\right).$$

The lower, moat-near branch is

$$J_{q,p}^{\text{moat}} = \left(\frac{p}{qp-1}, \frac{p-1}{q(p-1)-1}\right], \quad p \geq 3,$$

which tiles

$$\left(\frac{1}{q}, \frac{2}{2q-1}\right].$$

The upper, crown-saturated branch is

$$J_{q,d}^{\text{sat}} = \left(\frac{d+1}{(q-1)d+q}, \frac{d+2}{(q-1)d+2q-1}\right], \quad d \geq 1,$$

which tiles

$$\left(\frac{2}{2q-1}, \frac{1}{q-1}\right).$$

Adjacent endpoints match exactly:

$$\frac{p}{qp-1} = \frac{(p+1)-1}{q((p+1)-1)-1}, \quad \frac{d+2}{(q-1)d+2q-1} = \frac{(d+1)+1}{(q-1)(d+1)+q}.$$

Thus every non-reciprocal threshold in the corridor is caught by one of two explicit one-parameter tree families.

Compiler form. Given a non-reciprocal $\alpha \in (0, 1)$, set

$$q = \left\lfloor \frac{1}{\alpha} \right\rfloor + 1.$$

If

$$\alpha \leq \frac{2}{2q-1},$$

choose the unique $p \geq 3$ with

$$\frac{p}{qp-1} < \alpha \leq \frac{p-1}{q(p-1)-1}.$$

Otherwise choose the unique $d \geq 1$ with

$$\frac{d+1}{(q-1)d+q} < \alpha \leq \frac{d+2}{(q-1)d+2q-1}.$$

The corresponding tree is a sparse adversarial instance for α -safe-core algorithms.

Crown integer program. The saturation theorem does not invalidate the above-side finite-floor program. If C is a lower endpoint in the corridor above $1/q$, write

$$p = |N(C)|, \quad |C| = qp - d, \quad d = s_q(C) \geq 1.$$

Then

$$L = \frac{p}{qp - d}, \quad L - \frac{1}{q} = \frac{d}{q(qp - d)}.$$

If the Round XII Crown Endpoint Conjecture holds, then $c(C) \geq q + 1$, so

$$n \geq |C| + |N(C)| + q + 1 = (q + 1)p + q + 1 - d.$$

The resulting integer minimization is

$$\min \left\{ \frac{d}{q(qp - d)} : p, d \in \mathbb{Z}_{\geq 1}, d < p, (q + 1)p + q + 1 - d \leq n \right\}.$$

The minimum is attained at $d = 1$ and

$$p = \left\lfloor \frac{n - q}{q + 1} \right\rfloor.$$

Thus the proposed above-side floor remains

$$M_{\leq n}^{+, \text{cand}}(q) = \frac{1}{q \left(q \left\lfloor \frac{n - q}{q + 1} \right\rfloor - 1 \right)}.$$

Surplus-rich crowns are real, but they are farther from the reciprocal wall than the unit-surplus crown ladders.

Computational continuation. I ran three exact or semi-random tests for the Round XII Crown Endpoint Conjecture and for the new saturation mechanism.

class	sizes	objects	exchange bars	max bars/object	failures
connected graph-atlas graphs	$2 \leq n \leq 7$	995	0	0	0
nonisomorphic trees	$8 \leq n \leq 13$	2264	1234	36	0
random connected graphs	$8 \leq n \leq 11$	480	22	8	0

For the tree enumeration, the size-by-size data were:

n	8	9	10	11	12	13
# trees	23	47	106	235	551	1301
# exchange bars	1	4	18	73	252	886
max bars/tree	1	2	4	10	18	36

No tested exchange bar violated $c(C) \geq q + 1$ for at least one lower endpoint. The stronger slack law failed beginning at $n = 10$, exactly as predicted by the saturation family.

Interpretation. The reciprocal corridor has two regimes. Near $1/q$, the system pays in size: unit-surplus crowns are the closest possible known approach to the reciprocal wall. Near $1/(q - 1)$, the system pays in surplus: the crown stays minimal while the ladder stores more reciprocal surplus internally.

Applications. The corridor tiling gives a simple stress-test generator. For a proposed threshold α , the compiler above outputs a tree whose blocker circuits violate exchange at α . This is useful for:

1. *Greedy-audit testing.* Matroid-style greedy algorithms should be tested against the exact corridor tree that contains the chosen α .
2. *Threshold engineering.* Reciprocal thresholds $1/q$ are still the only universally matroidal thresholds. The corridor tiling shows how nearby every non-reciprocal policy is to an explicit sparse failure.
3. *Minimal-crown adversaries.* A system designer cannot detect all high-exposure exchange failures merely by looking for large free regions. The saturation trees have only the minimal crown $q + 1$ while carrying arbitrary surplus.
4. *Benchmark compression.* Instead of searching arbitrary graphs, one can benchmark safe-core routines using the two canonical corridor branches $J_{q,p}^{\text{moat}}$ and $J_{q,d}^{\text{sat}}$.

New targets. This round suggests four next targets.

1. **Crown Endpoint Problem.** Prove or refute that every above-side exchange bar has a lower endpoint with $c(C) \geq q + 1$.
2. **Corridor Tiling Optimality.** Determine whether the two canonical branches minimize tree size among all exchange-bar witnesses containing a given α .
3. **Crown Spectrum.** For fixed q, d, n , classify possible pairs $(s_q(C), c(C))$ for lower endpoints of exchange bars. The saturation theorem proves the lower envelope contains $(d, q + 1)$ for all d .
4. **Moat point process.** Study the set of all exchange endpoints in a corridor, not just the nearest endpoint. The two tiling branches may be the visible skeleton of a richer Stern-Brocot-like endpoint process.

Round XIII summary. The exceptional discovery is *crown saturation*: a lower endpoint can have arbitrarily large reciprocal surplus while keeping the smallest possible crown predicted by Round XII. The exact law is

$$J_{q,d}^{\text{sat}} = \left(\frac{d+1}{(q-1)d+q}, \frac{d+2}{(q-1)d+2q-1} \right], \quad s_q = d, \quad c = q+1.$$

Together with the unit-surplus moat branch, these bars tile every reciprocal corridor. The working picture is now sharper: reciprocal thresholds are protected walls; non-reciprocal thresholds fall into explicit tree corridors; near the wall size is the cost, while near the next wall surplus is absorbed by path-depth.

Round XIV - Leap Bars and Corridor Persistence

Purpose. Round XIII tiled each reciprocal corridor with primitive tree exchange bars. This round finds a stronger phenomenon: a single acyclic exchange witness can leap across many primitive cells at once. The resulting law gives a new persistence parameter for nonmatroidality: not merely where a bar starts, but how many reciprocal-corridor cells it swallows before dying.

Exceptional discovery. The compact 12-vertex tree with bar $(3/4, 5/6]$ was not a one-off anomaly. It is the $k = 4$ member of an infinite leap family. Its bar crosses the two primitive cells $(3/4, 4/5]$ and $(4/5, 5/6]$. Thus reciprocal corridor tilings are not a complete list of all bars; they are the atoms that longer bars can consume.

Primitive saturation-side corridor cells

For $q \geq 2$, the saturation-side primitive cells in the open corridor $(1/q, 1/(q-1))$ are

$$\text{Gamma}(q,j) = (j / ((q-1)j + 1), (j+1) / ((q-1)(j+1) + 1)], j \geq 2.$$

These are the right-hand Farey-adjacent cells from Round XIII. They begin at $2/(2q-1)$ and climb toward $1/(q-1)$. The new leap bars below are unions of consecutive $\text{Gamma}(q,j)$ cells.

The base leap tree L_k

Fix $k \geq 3$. Build two alternating cave-watcher paths that share one endpoint cave vertex x and the watcher adjacent to x . In schematic form:

A-arm: $a_1 - u_1 - a_2 - u_2 - \dots - a_{\{k-1\}} - u_{\{k-1\}} - x$
B-arm: $x - u_{\{k-1\}} - b_1 - v_1 - b_2 - \dots - v_{\{k-2\}} - b_{\{k-1\}}$

Let $A = \{a_1, \dots, a_{\{k-1\}}, x\}$ and $B = \{x, b_1, \dots, b_{\{k-1\}}\}$. The graph is a tree: the two cave-watcher paths overlap only in the edge $x - u_{\{k-1\}}$, then branch away. The vertex x is the shared circuit-elimination point.

$$|V(L_k)| = 4k - 4.$$

This construction is deliberately sparse. It uses only path-depth and one shared watcher; no complete bipartite density is present.

The Leap Bar Theorem

Theorem 14.1 (Base Leap Bar). For every $k \geq 3$, the tree L_k has an exchange certificate (A, B, x) with

$$|A| = |B| = k, |N(A)| = |N(B)| = k - 1,$$

$$\min_{\{ \text{empty } T \text{ proper subset } A \}} |N(T)|/|T| = \min_{\{ \text{empty } T \text{ proper subset } B \}} |N(T)|/|T| = 1,$$

and, for $U = (A \cup B) \setminus \{x\}$,

$$|U| = 2k - 2, |N(U)| = 2k - 3.$$

Consequently the exact exchange bar is

$$J_{\text{leap}}(k) = \left(\frac{(k-1)}{k}, \frac{(2k-3)}{(2k-2)} \right].$$

At $\alpha = (2k-3)/(2k-2)$, the two endpoint caves A and B are minimal bad caves, yet U contains no minimal bad cave. Circuit elimination fails, so the α -safe sets are not a matroid.

Proof sketch. Each endpoint cave is an alternating path with k cave vertices and $k-1$ watcher vertices. Any proper nonempty subset of cave vertices exposes at least as many watchers as vertices; equality occurs on full path segments missing one endpoint. Only the whole cave has one fewer watcher than cave vertices. Removing the shared vertex x glues the two arms into a longer alternating path with $2k-2$ cave vertices and $2k-3$ watchers. Therefore A and B are born just after $(k-1)/k$, while the union-without- x remains safe exactly up to $(2k-3)/(2k-2)$.

Primitive span

For $q = 2$, the primitive saturation-side cells are

$$(j/(j+1), (j+1)/(j+2)], j \geq 2.$$

The leap bar $J_{\text{leap}}(k)$ is the union of exactly these consecutive primitive cells:

$$j = k-1, k, \dots, 2k-4.$$

$$\text{span}(J_{\text{leap}}(k)) = k - 2.$$

Thus k is not just a size parameter. It is a persistence address: the same two-cave obstruction survives across $k-2$ adjacent threshold atoms.

Möbius Leap Law

The watcher-leaf transform from Round VIII sends every exposure ratio z to

$$T_m(z) = z / (1 + mz).$$

Applying $m = q - 2$ leaves to each watcher in the local leap certificate bends the $q = 2$ leap family into the q -th reciprocal corridor.

Theorem 14.2 (Möbius Leap Law). For $q \geq 2$ and $k \geq 3$ there is a finite tree $L_{\{q,k\}}$ with

$$|V(L_{\{q,k\}})| = 2qk - 3q + 2,$$

and exchange bar

$$J_{\text{leap}}(q,k) = ((k-1)/((q-1)k - q + 2), (2k-3)/(2(q-1)k - 3q + 4)].$$

Equivalently,

$$J_{\text{leap}}(q,k) = (T_{\{q-2\}}((k-1)/k), T_{\{q-2\}}((2k-3)/(2k-2))).$$

This bar swallows exactly $k-2$ primitive $\text{Gamma}(q,j)$ cells:

$$\text{Gamma}(q,k-1), \text{Gamma}(q,k), \dots, \text{Gamma}(q,2k-4).$$

Its length is

$$|J_{\text{leap}}(q,k)| = (k-2) / [((q-1)k - q + 2)(2(q-1)k - 3q + 4)].$$

Crown-persistence tradeoff

For a lower-endpoint cave C in $L_{\{q,k\}}$, the reciprocal surplus and free crown are

$$s_q(C) = q|N(C)| - |C| = k - 2,$$

$$c(C) = |V \setminus (C \cup N(C))| = q(k - 2) + 1.$$

Therefore

$$\text{span}(J_{\text{leap}}(q,k)) = s_q(C), c(C) = q * \text{span}(J_{\text{leap}}(q,k)) + 1.$$

Interpretation: crown is fuel for persistence. Round XIII showed that surplus can be hidden without increasing the crown. Round XIV shows the opposite use of surplus: spend a linear crown to make one exchange failure persist through many adjacent corridor cells.

Exact Computation

The supplementary script constructs $L_{\{q,k\}}$, computes the exact bar by full subset enumeration, and checks circuit-elimination failure at the right endpoint whenever the graph is small enough. The following rows were verified with exact rational arithmetic.

q	k	n	bar	span	circuits at hi	nonmatroidal?
2	3	8	(2/3, 3/4]	1	2	yes
2	4	12	(3/4, 5/6]	2	2	yes
2	5	16	(4/5, 7/8]	3	2	yes
2	6	20	(5/6, 9/10]	4	2	yes
3	3	11	(2/5, 3/7]	1	2	yes
3	4	17	(3/7, 5/11]	2	2	yes
4	3	14	(2/7, 3/10]	1	2	yes

For $q = 2$ and $k = 5$ the single bar $(4/5, 7/8]$ consumes three primitive cells:

$$(4/5, 5/6], (5/6, 6/7], (6/7, 7/8].$$

The script also reports the persistence product for $q = 2$:

$$k |J_{\text{leap}}(2,k)| \rightarrow 1/2.$$

So the span grows linearly in k while the total interval length is asymptotic to $1/(2k)$. This is a small but persistent failure: near the upper reciprocal wall, many thin primitive cells can be swallowed by one two-cave certificate.

Minimality note

The bar uses only two minimal bad caves at its right endpoint in the checked cases. This is useful computationally: a long threshold threat need not appear as a complicated circuit cloud. It can be a two-circuit phenomenon with a long safe union.

Applications and New Targets

Application 1 - persistent stress tests. To stress-test a threshold policy across h adjacent saturation-side primitive cells in the q -th corridor, use $k = h + 2$. The resulting tree has

$$n = 2q(h+2) - 3q + 2 = qh + q + 2h + 2,$$

and one exchange failure persists across exactly h primitive cells. This creates sparse adversarial tests for greedy safe-core algorithms and rank-style heuristics.

Application 2 - robust threshold auditing. Primitive-cell compilers find a witness at one target α . Leap bars certify that one witness can remain dangerous after several threshold adjustments. A policy sweep should therefore track exchange bars as intervals, not only as point failures.

Application 3 - crown diagnostics. Endpoint caves with surplus $s_q(C)$ and crown close to $q s_q(C) + 1$ are signatures of long-bar behavior. Endpoint caves with crown $q+1$ but large surplus are signatures of Round XIII crown saturation. These are two different operational risks.

New conjectural target: Crown-Persistence Inequality

The leap law suggests the following strengthening of the Crown Endpoint Conjecture. Let an above-side exchange bar in the q -th corridor swallow h consecutive primitive saturation-side cells from its lower endpoint. Then some lower-endpoint cave C should satisfy

$$c(C) \geq qh + 1.$$

The leap trees attain equality. The case $h = 1$ is exactly the Round XII crown lower bound $c(C) \geq q+1$. A proof would turn the finite crown problem into a full persistence theory for exchange bars.

Other targets

- Moat-side leap dual. Find or rule out an analogous long-bar family on the moat side of each reciprocal corridor.
- Maximum persistence. Determine $P_T(n,q)$, the largest total length of a single exchange bar in an n -vertex tree inside the q -th corridor.
- Bar merging. Develop an interval-union point process that distinguishes primitive atoms, leap bars, and saturated crowns.
- Algorithmic detection. Search first for endpoint caves with the invariant pair $(s_q, c) = (h, qh+1)$; these are predicted to be long-bar candidates.

Round XIV summary. Primitive reciprocal corridor cells are not the whole story. They are atoms. Leap trees show that one sparse two-circuit certificate can swallow many atoms at once. The new equation

$$c = q * \text{span} + 1$$

is the most useful new law from this round: it converts threshold persistence into a structural crown measurement.

Round XV: Primitive Lenses and the Hidden Coordinate of Corridors

Purpose. Round XIV found leap bars: one sparse tree obstruction can swallow several primitive saturation-side cells at once. I looked for the coordinate in which these leaps become linear. The result is a new primitive coordinate, a two-parameter asymmetric lens family, and an exact crown-persistence equality for every integer block realized by these lenses.

15.1 Primitive coordinate law

Fix $q \geq 2$ and consider the reciprocal corridor between $1/q$ and $1/(q-1)$. Define

$$\text{Pi}_q(\alpha) = \alpha / (1 - (q-1) \alpha), \alpha \in (1/q, 1/(q-1)).$$

The inverse coordinate is

$$\alpha = \text{Pi} / ((q-1) \text{Pi} + 1).$$

Thus the primitive boundary

$$b_{\{q,j\}} = j / ((q-1) j + 1), j = 1, 2, 3, \dots$$

is sent exactly to the integer j . In this coordinate the q -th corridor is not curved: it is the ray $(1, \infty)$, and primitive saturation cells are the unit intervals $(j, j+1]$.

The watcher-leaf transform from earlier rounds, $T_m(\alpha) = \alpha / (1 + m \alpha)$, preserves this primitive coordinate while moving between reciprocal corridors:

$$\text{Pi}_{\{q+m\}}(T_m(\alpha)) = \text{Pi}_q(\alpha).$$

This explains why the same tree pattern reappears in every corridor after watcher leaves are attached. The apparent fractional distortion is only a change of external coordinates; internally, the obstruction stays at the same primitive position.

15.2 Corridor tiling becomes harmonic-plus-integer

The Round XIII corridor tiling becomes especially simple after applying Pi_q . The moat-near branch

$$J^{\{\text{moat}\}}_{\{q,p\}} = (p/(q p - 1), (p-1)/(q(p-1)-1)], p \geq 3,$$

is transformed into the harmonic interval

$$\text{Pi}_q(J^{\{\text{moat}\}}_{\{q,p\}}) = (p/(p-1), (p-1)/(p-2)].$$

These harmonic intervals tile the primitive door $(1,2]$. The crown-saturated branch

$$J^{\{\text{sat}\}}_{\{q,d\}} = ((d+1)/((q-1)d+q), (d+2)/((q-1)d+2q-1)]$$

is transformed into the unit interval

$$\text{Pi}_q(J^{\{\text{sat}\}}_{\{q,d\}}) = (d+1, d+2].$$

So each reciprocal corridor has the same hidden geometry: the region just above the reciprocal wall is the harmonic door $(1,2]$, and everything beyond it is an integer grid.

15.3 Asymmetric Lens Theorem

The leap bars of Round XIV were symmetric: both arms had the same primitive size. I varied the arm lengths and found the following constructive theorem.

Theorem XV.1 (Asymmetric Lens Theorem). For every $q \geq 2$ and integers $r \geq t \geq 2$, there is a finite tree $Z_{\{q,r,t\}}$ with an exchange bar

$$J(Z_{\{q,r,t\}}) = (b_{\{q,r\}}, b_{\{q,r+t-1\}}].$$

Equivalently, in primitive coordinates,

$$\text{Pi}_q(J(Z_{\{q,r,t\}})) = (r, r+t-1].$$

The tree is built from two alternating paths sharing one cave vertex x and one watcher. The long arm has $r+1$ cave vertices and r watchers; the short arm has $t+1$ cave vertices and t watchers. Attaching $q-2$ watcher leaves to every watcher applies the transform $T_{\{q-2\}}$.

The size and endpoint statistics are exact:

$$|V(Z_{\{q,r,t\}})| = q(r+t) - q + 2.$$

$$\text{For a lower-endpoint cave } C: |C| = (q-1)r + 1, |N(C)| = r.$$

Therefore the lower endpoint is $r/((q-1)r+1)=b_{\{q,r\}}$. The exchange-union set has endpoint $b_{\{q,r+t-1\}}$, giving the right endpoint.

Proof sketch. Count the open neighbors of the long arm, the short arm, and their union after deleting x . Proper subarms have larger exposure because they acquire outside watchers at their cut endpoints. The only limiting ratios are therefore the long arm and the union. The watcher-leaf transform changes each ratio z to $z/(1+(q-2)z)$, which turns the base $q=2$ bar into the q -th corridor bar above.

15.4 Lens completeness for integer blocks

A direct corollary is that every consecutive integer block of primitive cells beginning at r and having length $h \leq r-1$ has a tree witness: choose $t=h+1$.

$$\text{For every } r \geq 2 \text{ and } 1 \leq h \leq r-1, \text{ there is a tree with primitive bar } (r, r+h].$$

The old unit saturation cells are the case $h=1$. The Round XIV leap bars are the maximal two-arm case $h=r-1$. The new family fills every intermediate span.

15.5 Crown-persistence equality

Let $c(C) = |V \setminus (C \cup N(C))|$ be the free crown of a lower-endpoint cave C . For the lens family, the crown is not merely bounded below; it is exactly the fuel consumed by primitive persistence.

$$\text{primitive persistence ell}_q(J) = \text{Pi}_q(U) - \text{Pi}_q(L) = t - 1.$$

$$c(C) = q(t-1) + 1 = q \text{ ell}_q(J) + 1.$$

This refines the Round XIV Crown-Persistence conjecture. The natural conjectural form is now coordinate-free inside the corridor:

Crown-Persistence Conjecture XV: $c(C) \geq q \text{ ell}_q(J) + 1$, or at least $c(C) \geq q \text{ floor}(\text{ell}_q(J)) + 1$ for arbitrary endpoints.

The asymmetric lenses attain equality for all integer primitive blocks they realize. This is exceptional because Round XIII showed that surplus can be hidden without enlarging the crown, while Round XV shows the opposite mode: primitive persistence itself demands crown, exactly q crown vertices per primitive unit, plus one shared anchor.

The ordinary alpha-length of a lens bar is

$$|J(Z_{\{q,r,t\}})| = (t-1) / [((q-1)r+1) ((q-1)(r+t-1)+1)].$$

Thus a long primitive obstruction can look very short in ordinary alpha-coordinates when it lies near a reciprocal wall or high in a corridor. Primitive length is the better invariant for algorithmic stress testing.

15.6 Exact computational checks

The accompanying script constructs $Z_{\{q,r,t\}}$, computes the exchange bar by exhaustive subset enumeration, and verifies circuit-elimination failure at the right endpoint for the listed trees.

q	r	t	n	exchange bar	primitive bar	crown	right endpoint
2	3	2	10	(3/4, 4/5]	(3, 4]	3	2 blockers; fails
2	5	3	16	(5/6, 7/8]	(5, 7]	5	2 blockers; fails
2	5	5	20	(5/6, 9/10]	(5, 9]	9	2 blockers; fails
3	4	2	17	(4/9, 5/11]	(4, 5]	4	2 blockers; fails
3	4	3	20	(4/9, 6/13]	(4, 6]	7	2 blockers; fails
4	3	2	18	(3/10, 4/13]	(3, 4]	5	2 blockers; fails

For larger lenses, the script also checks the closed formulas without full enumeration; examples include $q=2, r=20, t=7$ with bar (20/21, 26/27] and crown 13, $q=3, r=10, t=5$ with bar (10/21, 14/29] and crown 13, and $q=5, r=8, t=4$ with bar (8/33, 11/45] and crown 16.

15.7 Primitive compiler for applications

The primitive coordinate gives a direct sparse stress-test compiler. Given a target threshold α :

1. Choose q with α in $(1/q, 1/(q-1))$.
2. Compute $x = \text{Pi}_q(\alpha)$.
3. If x in $(1, 2]$, use the harmonic moat branch from Round XIII. If $x > 2$, set $r = \text{floor}(x)$. The unit lens $Z_{\{q, r, 2\}}$ already contains α whenever x in $(r, r+1]$.
4. To test robustness over a wider primitive window, choose t so that α lies in $(r, r+t-1]$ and $t \leq r$. The resulting tree has size $q(r+t)-q+2$ and crown $q(t-1)+1$.

This converts threshold engineering into integer arithmetic on the primitive ray. The compiler is useful for greedy safe-core audits, exchange-bar regression tests, and adversarial sparse instances: a policy may look safe over a small ordinary α interval while failing over a long primitive interval.

15.8 New targets after Round XV

Target A: Harmonic lens problem. The primitive door $(1, 2]$ is tiled by harmonic intervals, not unit intervals. Is there an asymmetric moat-lens construction that merges several harmonic intervals in one tree bar, analogous to the asymmetric lens family for $x > 2$?

Target B: Primitive optimality. Determine the smallest tree realizing a primitive integer block $(r, r+h]$. The lens size is $q(2r+h+1)-q+2$ after writing $t=h+1$. Is this optimal, or can hidden sharing reduce the size?

Target C: Full crown-persistence theorem. Prove or refute $c(C) \geq q \ell_q(J) + 1$ for tree bars. If false, find the corrected integer version in terms of the number of primitive cell boundaries crossed.

Target D: Corridor calculus. Develop an algebra of operations whose action on Pi_q is simple: translations, harmonic inversions, and concatenations. The watcher-leaf transform is already invisible in Pi_q ; more invisible operations may exist.

15.9 Summary of the round

The exceptional discovery is that reciprocal corridors have a hidden linear coordinate. In that coordinate, previous moat and saturation formulas become a harmonic door followed by an integer grid, watcher-leaf transforms become identity maps, and leap bars become ordinary intervals. The new asymmetric lens theorem fills every integer block $(r, r+h]$ with $h \leq r-1$ and exactly saturates the crown-persistence tradeoff $c = qh + 1$. The most useful new equation is therefore

$$\begin{aligned} \text{Pi}_q(\alpha) &= \alpha / (1 - (q-1)\alpha), \quad J(Z_{\{q, r, t\}}) = (b_{\{q, r\}}, b_{\{q, r+t-1\}}], \\ c &= q(\text{Pi}_q(U) - \text{Pi}_q(L)) + 1. \end{aligned}$$

This turns a formerly fractional threshold problem into a discrete geometry problem on a ray.

Round XVI: Harmonic door coordinates and fog lenses

The previous round straightened every reciprocal corridor with

$$\Pi_q(\alpha) = \frac{\alpha}{1 - (q-1)\alpha}, \quad \alpha \in \left(\frac{1}{q}, \frac{1}{q-1}\right).$$

That coordinate turns the saturation side into integer cells. This round studies the first “door” of the corridor,

$$D_q = \left(\frac{1}{q}, \frac{2}{2q-1}\right], \quad 1 < \Pi_q(\alpha) \leq 2,$$

where the natural cells are not integer cells but harmonic cells.

Definition 1 (Harmonic door coordinate). *For $\alpha \in D_q$, define*

$$\Psi_q(\alpha) = \frac{1}{\Pi_q(\alpha) - 1} = \frac{1 - (q-1)\alpha}{q\alpha - 1}.$$

If

$$c_{q,p} = \frac{p}{qp - 1}, \quad p \geq 2,$$

then

$$\Pi_q(c_{q,p}) = \frac{p}{p-1}, \quad \Psi_q(c_{q,p}) = p - 1.$$

Thus the old moat-door cell

$$(c_{q,p}, c_{q,p-1}]$$

has harmonic length one. For an exchange bar $J = (L, U] \subseteq D_q$, write

$$\ell_q^H(J) = \Psi_q(L) - \Psi_q(U).$$

The watcher-leaf transform is invisible in this coordinate as well:

$$\Psi_{q+m}\left(\frac{\alpha}{1 + m\alpha}\right) = \Psi_q(\alpha).$$

The dense fog-lens construction

The tree lenses of Round XV occupy integer blocks beyond the door. The surprise here is that inside the harmonic door, dense overlap can create bars whose endpoint is not a harmonic cell boundary. The construction is deliberately symmetric.

Start with the $q = 2$ base. Let I be a core cave of size $2s - 1$, let W be a watcher set of size s , and join every vertex of I to every vertex of W . Add two packet cave sets X_A, X_B , each of size $2h$. The packet X_A is joined completely to W and to a private watcher set R_A of size h ; similarly X_B is joined completely to W and to R_B . Put

$$A = I \cup X_A, \quad B = I \cup X_B,$$

and choose any shared vertex $x \in I$. The two caves share the whole kernel, but their packet watchers are disjoint.

For general $q \geq 2$, attach $q - 2$ watcher leaves to every watcher in $W \cup R_A \cup R_B$, and include those leaves in any cave that wakes their parent watcher. This is exactly the watcher-leaf transform from earlier rounds.

Theorem 1 (Harmonic Fog-Lens Law). *For every $q \geq 2$, $s \geq 2$, and $h \geq 1$, the graph $F_{q,s,h}$ above has*

$$|V(F_{q,s,h})| = (q+1)(s+2h) - 1$$

and has an exchange bar

$$J(F_{q,s,h}) = (L, U]$$

with

$$L = \frac{s+h}{q(s+h)-1},$$

and

$$U = \begin{cases} \frac{s}{qs-1}, & 2h \leq s, \\ \frac{s+2h}{q(s+2h)-2}, & 2h > s. \end{cases}$$

Consequently

$$\ell_q^H(J(F_{q,s,h})) = \min\left(h, \frac{s}{2}\right).$$

Proof sketch. For $q = 2$, the lower caves satisfy

$$|A| = |B| = 2(s+h) - 1, \quad |N(A)| = |N(B)| = s+h,$$

so

$$\rho(A) = \rho(B) = \frac{s+h}{2(s+h)-1}.$$

The kernel I has ratio $s/(2s-1)$. The circuit-elimination union

$$U_0 = (A \cup B) \setminus \{x\}$$

has

$$|U_0| = 2s + 4h - 2, \quad |N(U_0)| = s + 2h,$$

so its ratio is $(s+2h)/(2s+4h-2)$. By symmetry, every other proper subcave is dominated by one of these two endpoint counts: a partial kernel wakes all of W , and a partial packet wakes its private packet watchers plus the core watcher set. Hence the upper endpoint is the smaller of $s/(2s-1)$ and $(s+2h)/(2s+4h-2)$. These are ordered by the single inequality $2h \leq s$. The watcher-leaf transform sends every ratio z in the certificate to $z/(1+(q-2)z)$, giving the displayed q -formula. Finally,

$$\Psi_q\left(\frac{s+h}{q(s+h)-1}\right) = s+h-1,$$

while the two possible upper endpoints have harmonic coordinates $s-1$ and $(s+2h-2)/2$, respectively. The length is therefore h in the kernel-controlled case and $s/2$ in the union-fog-controlled case. $\square$

Exceptional point: fractional harmonic persistence

The smallest genuinely new fog behavior found in this round is

$$q = 2, \quad s = 3, \quad h = 2.$$

The resulting 20-vertex graph has

$$J(F_{2,3,2}) = \left(\frac{5}{9}, \frac{7}{12}\right],$$

and

$$\Psi_2\left(\frac{5}{9}\right) = 4, \quad \Psi_2\left(\frac{7}{12}\right) = \frac{5}{2}.$$

Thus one pair of minimal bad caves persists for harmonic length

$$\ell_2^H = \frac{3}{2}.$$

This is not a union of primitive tree cells. It is a true “fog” endpoint: the right endpoint has surplus two,

$$2 \cdot 7 - 12 = 2,$$

whereas ordinary harmonic cell boundaries have surplus one.

Exact computational continuation

The supplementary Round XVI script constructs $F_{q,s,h}$, computes the certificate bar by exact bit-mask subset enumeration when the graph is small enough, and checks circuit-elimination failure at the right endpoint. Selected rows are below.

q	s	h	n	bar	mode	ℓ_q^H	exact?
2	2	1	11	$(3/5, 2/3]$	kernel	1	yes
2	2	2	17	$(4/7, 3/5]$	union-fog	1	yes
2	3	2	20	$(5/9, 7/12]$	union-fog	$3/2$	yes
3	2	1	15	$(3/8, 2/5]$	kernel	1	yes
3	3	1	19	$(4/11, 3/8]$	kernel	1	yes

Formula-only larger examples include

$$F_{2,4,2} : \quad \left(\frac{6}{11}, \frac{4}{7} \right], \quad \ell_2^H = 2,$$

$$F_{2,5,4} : \quad \left(\frac{9}{17}, \frac{13}{24} \right], \quad \ell_2^H = \frac{5}{2},$$

and

$$F_{5,6,5} : \quad \left(\frac{11}{54}, \frac{8}{39} \right], \quad \ell_5^H = 3.$$

The exact checks also scanned all nonisomorphic trees through 12 vertices for a $q = 2$ door bar with $\ell_2^H > 1$. The counts were

$$1, 1, 2, 3, 6, 11, 23, 47, 106, 235, 551$$

trees on $2, 3, \dots, 12$ vertices, and no multi-door tree bar was found.

Conjecture 1 (Tree Harmonic-Door Rigidity). *If T is a finite tree and $J = (L, U] \subseteq D_q$ is an exchange bar, then*

$$\ell_q^H(J) \leq 1.$$

Equivalently, a tree exchange bar in the first door of a reciprocal corridor cannot cross more than one harmonic moat cell. The dense fog lenses show that the conjecture, if true, is a sparse-graph phenomenon rather than a general graph theorem.

Applications and new targets

The harmonic coordinate gives a sharper policy language near reciprocal thresholds. A threshold α just above $1/q$ can be assigned the door depth

$$\Psi_q(\alpha) = \frac{1 - (q-1)\alpha}{q\alpha - 1}.$$

Moving one harmonic unit corresponds to crossing one moat-door cell. In tree-like networks, the new evidence suggests that exchange failures may be audited cell-by-cell. In dense overlapping networks, the fog-lens construction warns that one obstruction can persist across several harmonic units, so a local cell audit can undercount risk.

The most useful engineering recipe from this round is: to build a dense stress test with prescribed harmonic persistence ℓ , choose $s \approx 2\ell$, choose $h \geq s/2$, and use $F_{q,s,h}$. The size is approximately

$$|V| \approx (q+1)(s+2h),$$

so the construction gives

$$\ell_q^H \gtrsim \frac{|V|}{4(q+1)}$$

when $h \approx s/2$.

New targets:

1. Prove or refute Tree Harmonic-Door Rigidity.
2. Determine the extremal door persistence

$$\Lambda_n(q) = \max\{\ell_q^H(J) : |V(G)| \leq n, J \subseteq D_q\}.$$

The fog lenses give $\Lambda_n(q) \geq n/(4(q+1)) + O(1)$.

3. Classify the fog spectrum: which surplus values

$$d = q|N(S)| - |S|$$

can occur as right endpoints of door exchange bars?

4. Find the boundary between sparse rigidity and dense fog: trees show no fog so far, while complete packet overlap creates it. Bounded treewidth, chordal bipartite graphs, and planar graphs are the next natural classes.

Round XVI summary. The new coordinate

$$\Psi_q(\alpha) = \frac{1 - (q-1)\alpha}{q\alpha - 1}$$

reveals the harmonic geometry of the first reciprocal door. Sparse tree evidence suggests unit-cell rigidity, but dense shared-kernel graphs produce fog lenses with harmonic persistence $\min(h, s/2)$, including fractional endpoints such as $(5/9, 7/12]$. The corridor now has three visible geometries: integer primitive lenses beyond the door, harmonic unit cells at the door, and dense fog lenses that blur those cells.

Round XVII - Sparse Fog and Planar Shadow Compressors

Round XVI left a sharp target: dense fog lenses showed that a harmonic-door exchange bar can persist for length greater than one, while exhaustive tree checks found no such tree. The new question was whether fog is a density phenomenon, or whether it is caused by cycles. This round gives a strong answer: density is not the culprit. The same long harmonic-door fog appears in sparse planar bipartite graphs of average degree below three.

Incidence replacement

Given a multigraph H on watcher vertices, form $\text{Inc}(H)$ by subdividing each multiedge with a new cave vertex. Parallel multiedges become distinct cave twins with the same two watchers, so $\text{Inc}(H)$ is still a simple graph. Every cave vertex in the base construction has degree exactly two. If H is planar, $\text{Inc}(H)$ is planar.

For $q=2$, choose a center o , core rim vertices $u_1, \dots, u_{s-1}$, A -private rim vertices $a_1, \dots, a_h$, and B -private rim vertices $b_1, \dots, b_h$. The core multiedges are the spokes $o-u_i$, the rim cycle on the u_i , and one extra spoke $o-u_1$, giving $2s-1$ core caves. Each packet consists of the spokes from o to its h private rim vertices plus the private rim cycle, giving $2h$ packet caves. Let A be core plus the A -packet, let B be core plus the B -packet, and let x be one core cave. For $q>2$, attach $q-2$ leaves to each watcher and include the appropriate leaves in A , B , and $(A \cup B)$ minus x ; this is the old watcher-leaf transform.

Theorem 17.1 (Sparse Fog Transfer)

For $q \geq 2$, $s \geq 3$, and $h \geq 2$, the sparse incidence fog graph $\text{SF}_{\{q,s,h\}}$ is connected, planar, and bipartite. It has

$$\begin{aligned} |V(\text{SF}_{\{q,s,h\}})| &= (q+1)(s+2h) - 1, \\ |E(\text{SF}_{\{q,s,h\}})| &= (q+2)(s+2h) - 2, \\ \text{average degree} &= 2|E|/|V| < 2(q+2)/(q+1) < 3. \end{aligned}$$

Nevertheless it carries the same exchange bar as the dense fog lens from Round XVI:

$$J(\text{SF}_{\{q,s,h\}}) = ((s+h)/(q(s+h)-1), U_{\{q,s,h\}}],$$

$$\begin{aligned} U_{\{q,s,h\}} &= s/(qs-1) \text{ if } 2h \leq s, \\ U_{\{q,s,h\}} &= (s+2h)/(q(s+2h)-2) \text{ if } 2h > s. \end{aligned}$$

In the harmonic door coordinate

$$\text{Psi}_q(\alpha) = (1-(q-1)\alpha)/(q\alpha - 1),$$

the persistence length is exactly

$$\text{ell}_q^H(J) = \text{Psi}_q(L) - \text{Psi}_q(U) = \min(h, s/2).$$

So arbitrarily long harmonic-door fog is possible in planar graphs whose average degree stays below three. This does not refute tree harmonic-door rigidity; instead it moves the likely boundary from density to cycle structure.

Proof sketch and the goblin mechanism

The construction replaces every dense complete-neighborhood packet by a cycle-with-spokes incidence packet. The relevant expansion statement is an edge-count lemma in the underlying multigraph. A packet fragment spanning t watcher vertices has at most $2t-2$ selected caves. A core fragment spanning t watcher vertices has at most $2t-1$ selected caves, and equality can occur only for the full core. After deleting x from the core, even the core side obeys the packet bound $2t-2$.

These inequalities force the only endpoint candidates to be the two intended ones: the core endpoint and the union endpoint. For $q=2$ this gives

$$\begin{aligned}\rho(A) &= \rho(B) = (s+h)/(2(s+h)-1), \\ \eta((A \cup B) \setminus x) &= (s+2h)/(2(s+2h)-2), \\ \rho(\text{core}) &= s/(2s-1).\end{aligned}$$

Taking the smaller of the core and union endpoints gives the stated $U_{\{2,s,h\}}$. The watcher-leaf transform maps every exposure ratio z to

$$T_m(z) = z/(1+mz),$$

with $m=q-2$. Since

$$\Psi_q(T_{\{q-2\}}(z)) = \Psi_2(z),$$

the q -th corridor inherits the same harmonic length. This is the main goblin move of the round: the fog is carried by the incidence pattern, while the watcher leaves merely slide it between reciprocal corridors.

Exact smallest witness

parameter	value
q,s,h	2,3,2
vertices, edges	20, 26
planar?	yes
average degree	$13/5 = 2.6$
exchange bar	$(5/9, 7/12]$
Ψ_2 length	$3/2$
full bitmask result	exactly two blockers at $7/12$; circuit elimination fails

The exact script found A and B as blockers at the right endpoint $\alpha=7/12$, and found no blocker contained in $(A \cup B) \setminus x$. Thus the matroid circuit-elimination failure is witnessed directly, not merely by the endpoint formulas.

Computation log

The new script performs two kinds of checks. First, it runs full bitmask enumeration for the 20-vertex witness above. Second, it checks the incidence edge-expansion certificate exactly for larger $q=2$ base lenses, then applies the watcher-leaf formula for higher q . Representative output:

s,h	lower L	upper U	mode	Psi span	edge-check
3,2	5/9	7/12	union	3/2	ok
3,3	6/11	9/16	union	3/2	ok
4,2	6/11	4/7	core	2	ok
4,3	7/13	5/9	union	2	ok
5,2	7/13	5/9	core	2	ok
6,2	8/15	6/11	core	2	ok

Higher-corridor examples produced by the same construction include

$q=3, s=3, h=2: J = (5/14, 7/19], \text{Psi}_3 \text{ length} = 3/2,$
 $q=3, s=4, h=2: J = (6/17, 4/11], \text{Psi}_3 \text{ length} = 2,$
 $q=4, s=3, h=2: J = (5/19, 7/26], \text{Psi}_4 \text{ length} = 3/2,$
 $q=5, s=6, h=3: J = (9/44, 6/29], \text{Psi}_5 \text{ length} = 3.$

Planar fog lower bound

Taking $s=2h$ gives harmonic length h and

$$|V| = 4(q+1)h - 1.$$

Therefore the all-graph fog lower bound from Round XVI already holds in planar sparse graphs:

$$\text{Lambda}_n^{\text{planar}}(q) \geq n/(4(q+1)) + O(1).$$

This is the most useful equation of the round. It says that a planar network of size n can still support harmonic-door exchange persistence linear in n ; sparse planarity alone is not a safe substitute for matroidality.

Applications

The sparse fog compressor changes the practical interpretation of the theory. Dense complete-bipartite neighborhoods are no longer required to stress-test a safe-core algorithm. A sparse planar instance with average degree below three can already force a long interval of nonmatroidal exchange behavior. This matters for road-like networks, sensor layouts, circuit boards, supply chains, and other systems where planarity or low average degree might otherwise be mistaken for algorithmic safety.

A policy near the q -th reciprocal door can now be stress-tested with a graph of roughly

$$n \text{ approximately } 4(q+1)L$$

vertices to obtain harmonic persistence L . The adversarial instance remains planar and sparse. Greedy rank-style methods that behave well on reciprocal thresholds should therefore be audited not only on dense blockers, but also on planar incidence-fog blockers.

New targets

- Tree door rigidity. Prove or refute: if T is a tree and J lies in the harmonic door D_q , then $\text{ell}_q H(J) \leq 1$.
- Planar fog constant. Determine the exact asymptotic constant in $\Lambda_n^{\text{planar}}(q)$. The new lower bound is $1/(4(q+1))$.
- Cycle-rank capacity. Let $\beta(G) = |E| - |V| + \text{components}$. Find the maximum harmonic fog length possible with $\beta(G) \leq B$.
- Bounded-degree fog. Split the high-degree center while preserving the exchange bar, or prove that high watcher degree is necessary for linear fog.
- Class boundary. Compare trees, cactus graphs, outerplanar graphs, planar graphs, bounded-treewidth graphs, and chordal-bipartite graphs for harmonic-door persistence.

Round XVII summary

The exceptional discovery is the Sparse Fog Transfer: the dense fog lens can be compressed into a planar incidence graph without losing its exchange bar or harmonic length. The apparent dense/sparse split from Round XVI was therefore misleading. The sharper boundary appears to be tree versus cyclic geometry, or perhaps cycle-rank budget rather than edge density. The new useful law is

$$\Lambda_n^{\text{planar}}(q) \geq n/(4(q+1)) + O(1),$$

with explicit planar witnesses $SF_{\{q,s,h\}}$. In the goblin picture: fog does not need a castle wall; a bicycle wheel and its shadows are enough.

18 Round XVIII - Bounded-Degree Fog and Cycle-Rank Fuel

Question continued. Round XVII showed that harmonic-door fog can be sparse and planar, but the construction still hid a high-degree watcher. This round asks whether long nonmatroidal persistence is caused by density, by high degree, or by cycles themselves.

Door coordinate recalled. In the q -th reciprocal door

$$D_q = (1/q, 2/(2q-1)],$$

$$\Psi_q(\alpha) = (1 - (q-1)\alpha)/(q\alpha - 1).$$

A bar $J=[L,U]$ inside this door has harmonic length

$$\ell_q^H(J) = \Psi_q(L) - \Psi_q(U).$$

For tree witnesses found earlier this length never exceeded one in the tested range. The new goal was to force length greater than one while forbidding dense graphs and forbidding high-degree hubs.

18.1 Bounded-degree planar fog theorem

Theorem 18.1 (bounded-degree planar fog). Fix $q \geq 2$, $h \geq 2$, and $s \geq \max(3, h)$. There is a finite simple planar bipartite graph $BF(q, s, h)$ carrying an exchange bar

$$J(BF(q, s, h)) = ((s+h)/(q(s+h)-1), U(q, s, h)]$$

where

$$U(q, s, h) = \begin{cases} s/(q s - 1) & \text{if } 2h \leq s, \\ (s+2h)/(q(s+2h)-2) & \text{if } 2h > s. \end{cases}$$

The harmonic length is

$$\ell_q^H(J(BF(q, s, h))) = \min(h, s/2).$$

Moreover, with $t=s+2h$,

$$\begin{aligned} |V| &= (q+1)t - 1, \\ |E| &= (q+2)t - 2, \\ \beta_1 &= |E| - |V| + 1 = t. \end{aligned}$$

If $s \geq 2h$, the maximum degree is at most $q+3$. Thus, for $s=2h$, there are planar bipartite graphs of maximum degree at most $q+3$ with

$$\ell_q^H = h = \beta_1/4 = (|V|+1)/(4(q+1)).$$

Interpretation. The exceptional point is that long fog is not a high-degree phenomenon. The high-degree center from Round XVII can be replaced by bounded-degree double cycles and packet cycles. The resource that grows linearly is cycle rank.

18.2 Construction and proof sketch

The $q=2$ construction is easiest to see in the watcher multigraph before subdivision. Use a double cycle on s watcher vertices and delete one copy of one edge. This gives a core with $2s-1$ multiedges and maximum watcher degree at most four. Then make two packets. Each packet has h private watcher vertices, a private cycle on them, and a matching from those private vertices into the core. The packet contributes exactly $2h$ multiedges.

Subdivide every multiedge by a new cave vertex. Parallel multiedges become twin cave vertices, so the final object is still a simple bipartite graph. Let I be the core caves, P_A and P_B the packet caves, and set $A=I$ union P_A , $B=I$ union P_B . Choose a shared cave x in I . Then

$$\begin{aligned} |N(A)| &= |N(B)| = s+h, \\ |A| &= |B| = 2(s+h)-1, \\ \text{so } \rho(A) &= \rho(B) = (s+h)/(2(s+h)-1). \end{aligned}$$

The exchange-union $U=(A \text{ union } B) \text{ minus } \{x\}$ has

$$\begin{aligned} |N(U)| &= s+2h, \\ |U| &= 2(s+2h)-2, \\ \text{so } \rho(U) &= (s+2h)/(2(s+2h)-2). \end{aligned}$$

The only remaining issue is suppression by proper subcaves. The double-cycle and packet-cycle pieces satisfy the same edge-sparsity inequality used by the older fog construction: every proper edge packet has watcher/edge ratio at least the advertised upper endpoint. The script verifies the expansion minima exactly for representative bases; the proof reduces to the $(2,1)$ -sparsity of double cycles with one missing edge plus private packet cycles.

For $q>2$, attach $q-2$ watcher leaves to the watcher set exposed by A , B , and U . This is the earlier watcher-leaf transform

$$T_m(z) = z/(1+mz), \quad m=q-2.$$

It sends the $q=2$ bar to the q -th reciprocal door while preserving harmonic length. Since the base maximum watcher degree is at most five when $s \geq 2h$, the transformed maximum degree is at most $q+3$.

18.3 Exact small witness

q	s	h	n	m	beta_1	Delta	bar	Psi-span	blockers
2	3	2	20	26	7	5	(5/9, 7/12]	3/2	2

Full subset enumeration at $\alpha=7/12$ found exactly two relevant minimal bad caves A and B . They share x , but no blocker lies inside $(A \text{ union } B) \text{ minus } \{x\}$; circuit elimination fails. The graph is planar, bipartite, and has maximum degree five.

18.4 Cycle-rank fuel law

The new family separates three possible causes of fog. It is not density: average degree stays below three. It is not high degree: maximum degree is bounded by $q+3$ along the infinite family. The remaining growing resource is cycle rank.

$$\begin{aligned}\text{beta}_1(\text{BF}(q, s, h)) &= s + 2h, \\ \text{ell}_q^H(\text{BF}(q, s, h)) &= \min(h, s/2).\end{aligned}$$

Therefore every bounded fog witness satisfies

$$\text{ell}_q^H \leq \text{beta}_1/4,$$

with equality when $s=2h$. This gives an infinite sharp lower-bound sequence for the planar bounded-degree fog constant:

$$\begin{aligned}\Lambda_n^{\{\text{planar}, \Delta \leq q+3\}}(q) &\geq (n+1)/(4(q+1)) \\ \text{for } n &= 4(q+1)h - 1.\end{aligned}$$

Conjecture 18.2 (cycle-rank door capacity). For planar graphs, the part of harmonic-door persistence beyond the first tree door is controlled linearly by cycle rank. A conservative form is

$$\text{ell}_q^H(J) \leq 1 + \text{beta}_1(G)/4 \quad \text{for every exchange bar } J \subset D_q.$$

A sharper asymptotic form would replace the right side by $\text{beta}_1(G)/4 + O(1)$. The bounded fog graphs show that no upper bound with smaller leading constant than $1/4$ can hold for planar bounded-degree graphs.

18.5 Tree door rigidity evidence

To keep testing the tree/cycle boundary, I scanned all nonisomorphic trees through 12 vertices. This was a stronger scan than exchange bars: it checked every individual cave barcode interval that enters the $q=2$ harmonic door. Since exchange bars are intersections of cave intervals, this bounds exchange bars as well.

trees checked	q	best cave-door span	attained interval	witness size
986	2	1	$(3/5, 2/3]$	8 vertices

This supports the strengthened rigidity target: in a tree, every harmonic-door cave interval with lower endpoint above $1/q$ may have length at most one. If true, the first door is arboreal, while every additional unit of door fog needs cyclic fuel.

18.6 Computational output

The Round XVIII script used exact rational arithmetic. The smallest witness was verified by full bitmask enumeration; larger infinite-family rows were verified at the edge-expansion level and by the algebraic watcher-leaf transform.

q	s	h	n	m	beta_1	bar	span	Delta bound
2	4	2	23	30	8	(6/11, 4/7]	2	5
2	6	3	35	46	12	(9/17, 6/11]	3	5
3	4	2	31	38	8	(6/17, 4/11]	2	6
4	6	3	59	70	12	(9/35, 6/23]	3	7
5	8	4	95	110	16	(12/59, 8/39]	4	8

Edge-level certificate rows included:

s=3,h=2: lower=5/9, upper=7/12, hi(A)=3/5, eta(U)=7/12, span=3/2
s=4,h=2: lower=6/11, upper=4/7, hi(A)=4/7, eta(U)=4/7, span=2
s=6,h=3: lower=9/17, upper=6/11, hi(A)=6/11, eta(U)=6/11, span=3

These rows are exactly the bounded-degree replacement for the Round XVII sparse fog. The endpoint arithmetic is unchanged; only the local geometry of the watcher side changes.

18.7 Applications

Bounded-degree fog matters for applications because many real networks are not hub-dominated. Road-like networks, planar sensor layouts, power-grid fragments, circuit-board routing graphs, and supply-chain maps often have bounded local degree. The new construction says matroid-style safe-core heuristics can still fail for a long interval of thresholds in such networks. A risk audit should therefore record cycle rank or a cycle-basis budget, not only maximum degree and average degree.

18.8 New targets

Target A - prove tree harmonic-door rigidity. Show that if T is a tree and J is an exchange bar contained in D_q , then $\text{ell}_q^H(J) \leq 1$. The stronger cave-bar version survived all tests through 12 vertices at $q=2$.

Target B - exact planar bounded-degree fog constant. Determine whether

$$\limsup \text{Lambda}_n^{\{\text{planar}, \Delta \leq q+3\}}(q)/n = 1/(4(q+1)).$$

The bounded fog family proves the lower bound. A matching upper bound would turn cycle-rank fuel into a quantitative law.

Target C - degree-three and degree-four fog. The present construction gives maximum degree $q+3$. Is there a cubic or nearly cubic planar fog family with unbounded harmonic-door persistence? If not, the critical degree should be identified.

Target D - sparsity-matroid compiler. Replace the ad hoc double-cycle packets by a general compiler from $(2,1)$ -tight watcher multigraphs to exchange bars. This would unify trees, primitive lenses, sparse fog, and bounded fog under one matroidal sparsity language.

Target E - cycle-rank point process. Study not only the longest door bar, but the distribution of all door bars as a function of cycle rank. The suggested observable is

$$\text{Fog}_q(G) = \sum_{J \subset D_q} \text{ell}_q^H(J),$$

with separate versions for all graphs, planar graphs, outerplanar graphs, cactus graphs, and bounded-treewidth graphs.

18.9 Round summary

This round found a bounded-degree replacement for sparse planar fog. Long harmonic-door nonmatroidality does not require dense blockers, high average degree, or high-degree hubs. It can live in simple planar bipartite graphs with maximum degree bounded independently of the persistence length. The cleanest new equation is

$$\text{ell}_q^H = \beta_{1/4} = (|V|+1)/(4(q+1)) \quad \text{along } s=2h \text{ bounded fog.}$$

The current picture is: trees appear to permit at most one harmonic door; cycles are the fuel for every additional door; bounded-degree planar cycle systems already achieve linear fog persistence.